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MASTER

Design and analysis of a multi-link suspension for a 4WD Formula Student racing car

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Design and Analysis of a Multi-link Suspension for a 4WD Formula Student racing car

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Master's thesis

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Brian Rensen, March 2019

Abstract

Most racing cars, and a range of premium passenger cars, are equipped with an independent double wishbone suspension. A more complex extension of the double wishbone suspension is the multi-link or 5-link suspension. This type of suspension offers more kinematic freedom and flexibility, which could result in improved packaging and setup advantages. However, due to the lack of expertise and complexity, the multi-link suspension is seldom used for racing purposes.

The goal of this research project is to design a multi-link suspension aiming to improve performance, handling and driver feedback of a Formula Student race car. Formula Student is a yearly international competition for student teams from around the world, in which they compete with self built race cars. University Racing Eindhoven is participating for the fourteenth time this season. In the past a couple of URE vehicles where equipped with a multi-link suspension design. Due to lack of proper knowledge and experience of the multi-link layout, a double wishbone is used for the last six years. The emphasis of this thesis is on the review and analyzes of a multi-link. To fulfill this objective, a multi-link suspension is designed and compared with the current double wishbone layout of the URE13 vehicle.

The performance of a vehicle depends on the suspension kinematics and characteristics. An important aspect is the kinematic behavior of the front and rear suspension when a steer or bump motion is applied. To analyze the suspension kinematics a multi-body model is created using MATLAB and SIMULINK software. For a given set of vehicle parameters, suspension hard-points and suspension layout, the relevant kinematics are determined. The kinematic model is also able to identify critical collision points for the complete steer and bump range. A second kinematic model is used to investigate the influence of mechanical free-play on the kinematics for different load cases, such as concerning, braking and bump. This model is also used to analyze the steering moment for various levels of lateral acceleration and steering angles.

The development of a new multi-link suspension is accomplished by iteratively optimizing the location of the suspension hard-points and overall layout. Based on the kinematic results a new design for the front and rear suspension is obtained. In this thesis report only the final multi-link design is reviewed, a detailed description of the iterative process is not treated. In comparison with a double wishbone suspension layout the multi-link offers more optimization possibilities. Due to the virtual steering axis of a multi-link layout, the kingpin axis can have almost any orientation and position. As result, the suspension kinematics are less restricted by the packaging of the in-wheel motors. The main advantage of a multi-link suspension is the reduction of unwanted effects due to suspension play and compliance. The change in toe and camber angle, due to this suspension play and compliance can be reduced or even eliminated. This results in better driving performance and handling. However, the complexity and extra development time can be a limiting factor for the selection of a multi-link, instead of a double wishbone suspension.

Sign conventions

For the definition of vehicle cordinates and suspension kinematics, a sign convention is required. It is beneficial to distinguish the movements of a vehicle and tire based on the degrees of freedom. In order to have a clear and explicit coordinate system the ISO 8855 [1], respectively DIN 70000, sign convention is used throughout this thesis report.

0.1 Vehicle coordinate system

Since the motion of the wheel and the suspension links are related to the motion of vehicle body, most vehicle suspensions are usually designed on the assumption that the vehicle body is rigid. The translational motions of the vehicle are defined using a vehicle fixed and right-handed orthogonal axis system. The base of the body fixed x-y-z vehicle coordinate system is located at the center of gravity of the vehicle, see Figure 1. The x-axis is parallel to the longitudinal plane of symmetry of the vehicle, pointing in the driving direction. The corresponding rotation around the x-axis is defined as the roll angle φ. The y-axis is perpendicular to the driving direction and parallel to the road plane, pointing to the left. Rotation around the y-axis is defined as the pitch angle θ. The z-axis is pointing upwards and normal to the road surface. Rotation around the z-axis is described by the yaw angle ψ.

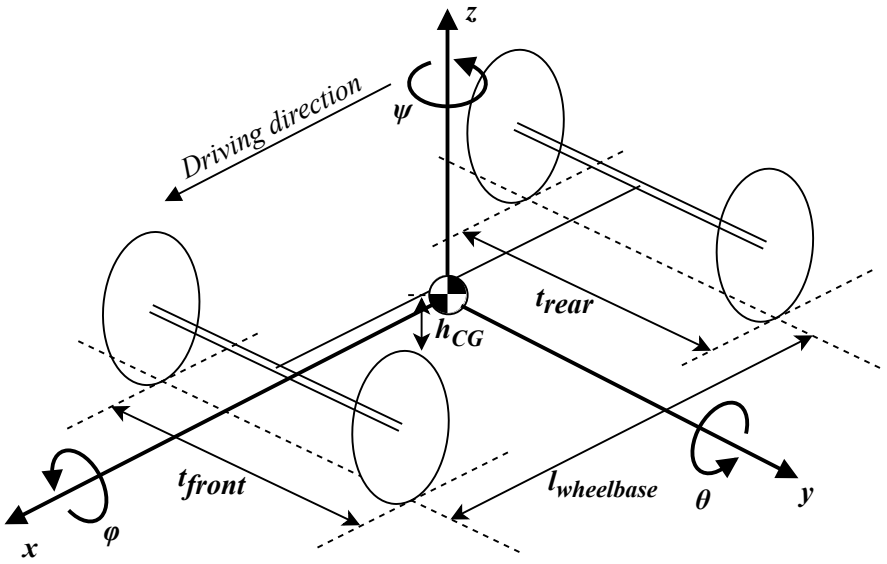


Figure 1: Vehicle coordinate system, ISO 8855 / DIN 70000 sign convention.

The point where the tire has contact with the road surface is defined as the tire contact point. The wheelbase $l_{wheelbase}$, front track-width t_{front} and rear track-width t_{rear} are prescribed by the distances between the four individual tire contact points. The wheelbase $l_{wheelbase}$ is characterized by the distance along the x-axis, between the front and rear axle. The front track-width t_{front} and rear track-width t_{rear} are prescribed by the distance in y-direction, between the centers of both front, or rear, tire contact points. The center of gravity height h_{CG} is defined as the vertical distance normal to the road plane, between the road surface and center of gravity location of the vehicle.

0.2 Tire orientation definitions

The orientation of a tire is determined by the upright geometry, vehicle setup and applied steering angle around the steering axis. In Figure 2 the tire axis systems and angle definitions are shown. The rotation of the tire around the vertical z-axis prescribes the steer or toe angle δ. The toe angle δ is defined as positive for a left-hand corner and negative for a right-hand corner. The tire inclination angle γ is defined as the rotation angle of the tire around the x-axis. The rotation angle of the tire around the spin or y-axis equals ϑ.

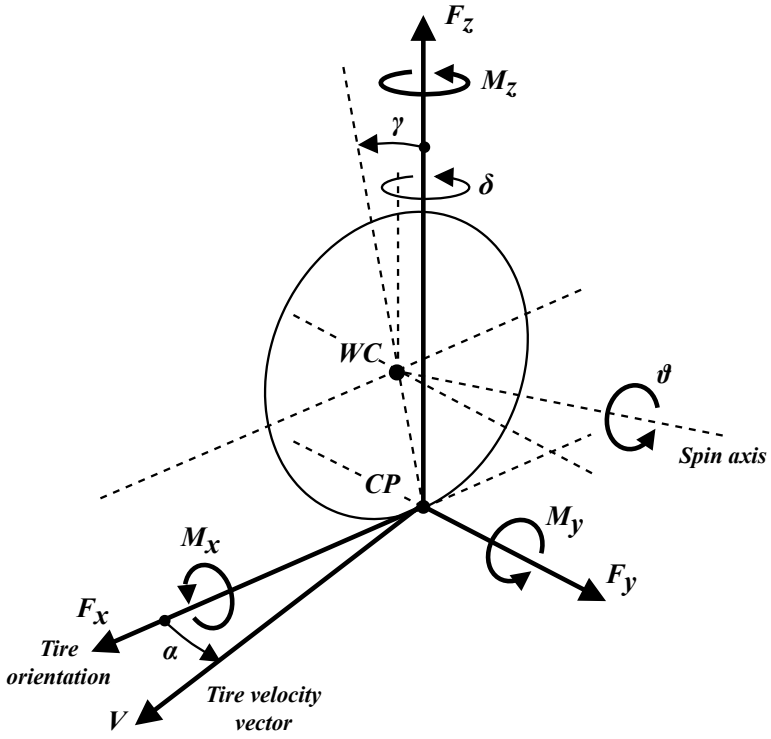


Figure 2: Tire orientation definitions, ISO 8855 / DIN 70000 sign convention.

The orientation angles of a tire are defined using a sequence of rotations. This has to be taken into account when calculating the orientation angles. The rotation sequence is described as:

Table 1: Order of rotation sequence from ground to tire axis

Order of rotation:	**Tire orientation angle:**	**Symbol:**
First	Steer or toe angle	δ
Second	Inclination or camber angle	γ
Third	Tire rotation angle	ϑ

The angle between the x-axis and the tire velocity vector V is defined as the side slip angle α.

All forces F and moments M acting on the tire are expressed with respect to the tire contact point CP. The forces and moments are defined in a right-handed orthogonal axis system, with the tire contact point being the origin.

List of symbols

Symbol	Description	Unit
α	Side slip angle tire	deg
γ	Inclination or camber angle	deg
δ	Steer or toe angle	deg
ε	Mechanical free-play ball-joint	m
θ	Pitch angle vehicle	deg
θ_{gf}	Support angle tire contact patch front (x-z-plane)	deg
θ_{gr}	Support angle tire contact patch rear (x-z-plane)	deg
ϑ	Rotation angle tire	deg
κ	Longitudinal slip ratio	$-$
λ	Combined king pin tilt angle	deg
σ	King pin inclination angle	deg
τ	Caster angle	deg
φ	Roll angle vehicle	deg
φ_{af}	Support angle front (y-z-plane)	deg
φ_{ar}	Support angle rear (y-z-plane)	deg
ψ	Yaw angle vehicle	deg
ω_{δ}	Steering velocity around steering axis	rad/s
ω_K	Angular velocity upright	rad/s
a_y	Lateral acceleration	m/s^2
A_{rod}	Cross section suspension rod	m^2
$C_{M\alpha}$	Self aligning stiffness tire	N/rad
$C_{M\gamma}$	Camber torque stiffness tire	N/rad
c_{damper}	Hard-stop damping constant mechanical free-play	Ns/m
d	Steering axis vector	$-$
d_{motor}	Diameter motor housing	m
d_{rod}	Diameter suspension rod	m
E_{rod}	Tensile modulus suspension rod	N/m^2
F_{fric}	Friction force mechanical free-play region	N
F_{rack}	Force at steering mechanism	N
F_{react}	Hard-stop reaction force	N
F_x	Longitudinal force at tire contact patch	N
F_y	Lateral force at tire contact patch	N
F_z	Vertical normal force at tire contact patch	N
f_i	Degrees of freedom spherical joint	$-$
$f_{susp.}$	Total degrees of freedom complete suspension	$-$
f_{tot}	Degrees of freedom of all individual suspension parts	$-$
g	Gravitational acceleration	m/s^2
h_{CG}	Center of gravity height	m

h_{RC}	Roll center height	m
i_{MR}	Motion ratio wheel/damper travel	$-$
K	Number of uprights	$-$
k_{spring}	Hard-stop stiffness	N/m
$k_{eq.rod}$	Axial stiffness suspension rod	N/m
L	Number of suspension links	$-$
$l_{wheelbase}$	Length of wheelbase	m
l_{mr}	Distance between motor housing and suspension rod	m
l_{rod}	Length of suspension rod	m
l_{rr}	Distance between rim and suspension rod	m
M_{steer}	Torque at steering wheel	Nm
M_{kp}	Moment acting around steering axis	Nm
M_x	Longitudinal / overturning moment at tire contact patch	Nm
M_y	Lateral / rolling resistance moment at tire contact patch	Nm
M_z	Vertical / aligning moment at tire contact patch	Nm
$m_{unsprung}$	Unsprung mass complete suspension	kg
m_v	Mass of vehicle including driver	kg
N	Number of free rotations of suspension rod	$-$
n	Mechanical trail	m
n_r	Caster offset at wheel center	m
p	Wheel load lever arm	m
p_b	Brake force distribution front and rear	$-$
p_d	Driving torque distribution front and rear	$-$
R	Rotation matrix tire	$-$
r_c	Wheel center offset	m
r_s	Scrub radius	m
$r_{steerrack}$	Pinion pitch circle radius steering mechanism	m
r_{CP}	Vector between wheel center and tire contact patch	m
r_i	Vector pointing to reference point i on upright	m
r_{tire}	Radius of undeformed tire	m
s	Screw axis	$-$
t_{front}	Track-width front	m
t_p	Pneumatic trail	m
t_{rear}	Track-width rear	m
V	Tire velocity vector	m/s
V_{CP}	Velocity vector tire contact patch	m/s
V_{WC}	Velocity vector wheel center	m/s
v_i	Velocity vector reference point i on upright	m/s
v_m	Velocity vector reference point on upright	m/s
x_{cp}	Longitudinal displacement tire contact point	m
y_{cp}	Lateral displacement tire contact point	m
z_{cp}	Vertical displacement tire contact point	m

Terminology

Term	Description
AES	Automotive Engineering Science
BF	Bottom Front
BR	Bottom Rear
CHAS	Chassis
DW	Double Wishbone
DOF	Degree of Freedom
FSAE	Formula SAE
FL	Front Left
FR	Front Right
ML	Multi-link
MP	MacPherson
RR	Rear Right
RL	Rear Left
SAE	Society of Automotive Engineers
TF	Top Front
TR	Top Rear
UPR	Upright
URE	University Racing Eindhoven
WC	Wheel Center
CP	Contact Point (tire)

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1. Introduction

1.1 Scope

In 1980, the Society of Automotive Engineers (SAE) introduced a formula car competition between universities around the world, named Formula SAE (FSAE) [2]. The FSAE program is an engineering design competition for undergraduate and graduate students. In addition to the regular university curriculum, the competition is founded to provide students with the opportunity to enhance their engineering design and project management skills by applying classroom theories in a challenging competition. The goal of the competition is to develop and construct a single-seat race car with the best overall package of design, construction, performance and cost.

The Eindhoven University of Technology is participating in the FSAE competition since 2003, the team is called University Racing Eindhoven (URE) [3]. Every year the team develops and builds a new race car in an attempt to improve the results of previous year. The team is known for its application of innovative technology. Currently, the team is building there first driverless vehicle, which eventually should drive autonomously.

The vehicle performance, handling dynamics and driver feedback are influenced by the suspension design. The majority of vehicles participating in the FSAE competition are equipped with a double wishbone suspension. An evolution of the double wishbone suspension is the multi-link or 5-link suspension. This suspension is characterized by more kinematic freedom and adjustment possibilities than a double wishbone suspension. This can result in an optimization of the suspension kinematics and may improve handling, stability and driver feedback.

1.2 Motivation

The current URE race car has a double wishbone suspension layout. In the past a couple of URE cars has been equipped with a multi-link suspension. However, the reason behind using the multi-link system is not documented properly and only a few persons are familiar with the adjustment and potential benefits of the multi-link suspension design. Also a couple of problems occurred with the first multi-link suspension design :

- Increased forces on steering wheel.
- Very sensitive to suspension play.
- Large force in rods due to design and leverage effect.

Due to these problems and the lack of proper knowledge on the multi-link suspension design, URE decided to return to a more conventional double wishbone suspension.

However, the packaging of the URE car changed substantially since then. In order to review the current suspension and investigate the new possibilities of a multi link suspension, a new study has been introduced. The following points motivate the introduction of this new study on the suspension design:

1. **In-wheel motors:** In the drive-train layout of the current URE car, the electric motors are integrated inside the wheel hubs. As a result the drive-line does not include drive shafts anymore and the suspension is more compact. This creates new design possibilities, such as: packaging, available space and more possibilities for positioning of the hard points.
2. **Identical wheel hub:** Another argument for a new study is the prospect of getting identical wheel hubs. Currently, the front and rear hubs are not identical and during this study the possibilities of identical hubs are evaluated. This will benefit the production and interchangeability of the hubs.
3. **Steering angle:** With the URE car of last year there was a problem with the maximal achievable steering angle. Due to the additional kinematic freedom and independent hard-points, using a multi-link suspension a larger steering angle could be achieved.

4. **Optimalization:** A multi-link suspension has more possibilities for optimizing vehicle handling and stability. Because of the additional kinematic freedom, the multi-link layout is more flexible and the extra hard-points could be comprehensive adjusted to fine-tune the suspension geometry. The multi-link layout offers much more options in regards to adjusting the suspension kinematics compared to double wishbones, where the kinematics are to a large extend restricted by the upright geometry. In addition, a multi-link suspension allows to locate the steering axis inside the wheel or in line with the tire contact point.

5. **Software packages:** In the past, vehicle and suspension models are created in the SIMMECHANICS generation 1 software. These models are currently outdated and not clearly archived. These models are not up to date and only revisable and adaptable by the right person. It is recommended to use the successor SIMSCAPE MULTIBODY software package, because of the improved functions and available tech-support. From MATLAB version R2018a and following, models created in SIMMECHANICS generation 1 are not supported anymore.

6. **New design:** During the last couple of years the suspension design has not changed a lot. Each year the focus is on increasing the reliability of the URE vehicle. By this, often the choice is made to not change the overall suspension layout in great detail but only concentrate on minor modifications. Performing a new study will result in an innovative new suspension design.

1.3 Objectives

The objective of this graduation project is to design a multi-link suspension, aiming to improve the performance in the context of the FSAE competition. Furthermore, a study on the advantages and disadvantages of a multi-link suspension design compared to the current double wishbone suspension of the URE13 is performed. This project serves as a preliminary study for the multi-link suspension to be used on the next URE race car. The main question is formulated as: "Which multi-link suspension design for the URE racecar will result in the best performance, in the context of the FSAE competition, and what are the advantages and disadvantages in comparison with the current double wishbone suspension design?" In order to answer this question the following goals are achieved:

1. Design of a front and rear multi-link suspension for the future URE race car. The shortcomings of the current double wishbone layout are addressed and improved by using a multi-link design.

2. Development of a suspension model in SIMSCAPE MULTIBODY, including kinematic analysis, collision check, suspension free-play and compliance.

3. Comparison of the current double wishbone suspension with the new multi-link design and verification of the advantages or disadvantages of a multi-link layout.

In order to limit the scope, the project has clear boundaries. Only a front and rear suspension are designed and analyzed, there is no redesign of the vehicle. Also a detailed design of the spring / damper assembly, steering mechanism and anti-roll system are excluded from this thesis.

1.4 Contributions

Based on the objectives, this research delivers some valuable contributions. The main contribution is providing an extensive suspension model. This suspension model can be used to analyze the kinematics and characteristics for a predefined suspension layout. Also the effect of changing suspension hard-points can be easily examined. Parallel to the kinematic analyzes, critical collision points over the complete motion range are automatically identified. The suspension model can be used by the URE team to further develop and improve the suspension design.

By using the suspension model a complete new multi-link suspension is designed. All relevant suspension kinematics and characteristics are analyzed. Based on the new multi-link layout and design evaluation, a well-grounded statement is formulated for why to use or not to use a multi-link instead of a double wishbone suspension. This will help the URE team to explain the design judges at the Formula Student event, why the choice is made for a double wishbone or multi-link design.

1.5 Report outline

In Chapter 2 a short overview of the available background information is given. A short update of the already performed research on the URE suspension design is listed. The tire used by URE is introduced and its major characteristics are briefly discussed.

In Chapter 3 a overall literature review of a suspension is discussed. The overall suspension characteristics and different suspension layouts are treated. Also the advantages and disadvantages of a multi-link suspension, in comparison with double wishbone layout are studied. Next, in Chapter 4 the design tool for analyzing the suspension kinematics, mechanical free-play and steering torque is explained. By using the suspension design tool a new multi-link suspension is created and reviewed in Chapter 5.

A comparison is made between the existing double wishbone and new multi-link suspension layout. This comparison is described in Chapter 6. The advantages and disadvantages of both suspension layouts are evaluated. Finally in Chapter 7 conclusions are drawn and recommendations for future research are given.

2. Background information

In this chapter an overview of the relevant background information is given, in order to have the correct information for a multi-link suspension design.

2.1 URE suspension history

The suspension of the URE car has evolved over time. In Table 2.1 all the URE vehicles and corresponding suspension designs are listed. Several bachelor and master projects have already been dedicated to the suspension design of an URE vehicle. Next a short summary of these projects is given.

Table 2.1: URE suspension history

Year of production:	Vehicle name:	Suspension layout:	Front push/pull:	Rear push/pull:	Remark:
2005	FSRTE01	DW	Push	Push	Never driven
2006	FSRTE02	DW	Push	Push	
2007	URE03	ML	Push	Push	
2008	URE04	ML	Pull	Push	
2009	URE05	ML	Pull	Push	
2010	URE05e	ML	Pull	Push	First electric car.
2011	URE06	ML	Pull	Push	
2012	URE07	ML	Pull	Push	
2013	URE08	DW	Pull	Push	
2014	URE09	DW	Pull	Push	
2015	URE10	DW	Push	Push	First FWD.
2016	URE11	DW	Push	Push	
2017	URE12	DW	Push	Push	
2018	URE13	DW	Push	Push	

2008
W. Lamers started the investigation of a multi-link suspension for racing applications [4]. A multi-link suspension was developed to improve handling, driver feedback and performance of the URE04. Because the tires play a crucial role in vehicle dynamics, first the tire behavior is analyzed. Based on this analyzes, the tire with the best characteristics is selected. Next, the tire characteristics are used to develop the appropriate suspension with optimal kinematics for the vehicle. The suspension kinematics have been determined using numerical optimization techniques. This method allows to adjust the position of the suspension hard-points and aims to meet the kinematic objectives as close as possible.

The most important improvements of the suspension kinematics are the harmonization of camber dependent kinematics, which result in the optimal camber angle of the tires during cornering. Also the suspension is designed to have a constant ride height during cornering which causes the suspension to operate in a specific region. The driver feedback and suspension characteristics are evaluated with a multi-body vehicle model.

2009
R. Hoeke and M. Muller continued with the optimization of the multi-link suspension geometry designed by W. Lamers [5]. In the master thesis of W. Lamers [4] the design targets and considerations were not validated properly. This was a comment of the design judges at the FSAE events. As a result, they conducted a more extensive review on the advantages and disadvantages of the multi-link suspension. Also the multi-link model of W. Lamers was optimized for better performance. The performance is validated with the dynamic simulation model. This model is also used to gain more knowledge about the vehicle dynamic behavior for different changes to the suspension.

2010
M. Hopmans performed extensive research into the properties of the tires [6]. During his study a Force and Moment test program using the TNO tire Test Trailer was conducted. Research on the influence

of tire inflation pressure and camber on the tire behavior where part of this test program. The Force and Moment measurements were conducted for three racing slick tires. Decreasing the tire inflation pressure results in an expected increase in cornering stiffness at low vertical tire forces and an increase in friction level. The cornering stiffness calculated from the tire test data was checked by performing Flatplank measurements. Using the Flatplank tire Tester located at the TU/e, the relation between the tire inflation pressure and the non-rolling tire stiffness is analyzed.

The tire data has been used to determine a Magic Formula version 6.1 tire parameter set using a fit routine in MATLAB (MF-Toolbox). Also a SIMMECHANICS vehicle simulation model was used in order to evaluate the vehicle behavior of the URE race car for the new tire data. An extensive validation process has been conducted where most of the driving maneuvers are covered. The simulation results are compared with the measured data and simulation model is a curate for almost every driving condition. All the vehicle dynamic related parts, such as the suspension kinematics and tire characteristics, have been modeled in great detail for the simulation model.

The internship of C. Ozturk covers the design and development of the rear suspension for the URE06 [7]. The kinematic characteristics of the URE05 rear suspension have been investigated and using the results of this kinematic analysis a new rear suspension is designed. The challenge is found in satisfying the different requirements for the suspension design. The main goal of the design process was to optimize the camber gain for the outer wheel during cornering. By this the structural stiffness is kept high and roll center low. After many iterations a optimized double-wishbone suspension layout is acquired, with improved camber change over body roll and steering motion.

S. van Rijk had a closer look to the damper settings of the URE05e during his internship [8]. The suspension system of the URE05e consists of a multi-link suspension that is connected to a damper using a pull-rod system. The damper is originally designed for formula 3 Euroseries cars, which is designed to use for high downforce aplications. The damping ratio at low damper velocities is therefore far above the intended region for the URE vehicle. Which resultied in a overdamped sytem whitout proper control of the body movement. This hard damping is not beneficial for roadholding performance Be this, the vehicle model of M. Hopmans [6] is used to simulate the driving behavior for specific maneuvers and a new damping curve is determined. The simulation results are used to specify the point of transition and the best damping constants for low- and high speed damping.

2012
R. van Hoek designed a completely new geometry for the suspension of the URE08 [9]. The emphasis of this new design lies on the 4WD concept for the future URE vehicle. The four wheel drive brings some complications with regard to driver feel. This problem originates from the combination of regenerative and mechanical braking and torque vectoring. By selecting a suitable combination between steering axis orientation and anti features, the driver feedback is optimized. Also camber gain was optimized by taking into account tire deformation and camber compliance during cornering. The maximum allowed roll angle was changed, which results in a smaller roll rate. This improves mechanical grip and reduces suspension travel.

The damper use was further improved by increasing the travel on the damper. This decreases the influence of friction within the damper for small movements. This required a change in the rocker mechanism and also different damper settings, because of increased damper travel and speed. A MATLAB script was developed which automatically generates coordinates for the suspension hard-points, based on the design and kinematic requirements.

2017
The bachelor end project of T. Verburg describes the validation process of the suspension model of the URE12 for longitudinal acceleration conditions [10]. The priority lies on the validation of the installation ratio, spring and damper travel, anti-effects and dynamic suspension behavior. Static measurements are conducted by prescribing a vertical motion on the rear and front axles. Dynamic measurements are done by driving various maneuvers and using different drive torque set points. The simulations are done using a MATLAB SIMMECHANICS multi-body model. However, only a small part of the suspension is validated and the results had larger deviations than expected. An additional research and improved measurement method is needed for further validation of the multi-body model.

2.2 Tire behavior and characteristics

The tires of a vehicle are the only component with a connection to the ground surface. Besides the aerodynamics, the tires are to a large extent, responsible for the forces acting on the vehicle. The forces and moments generated in contact patch of the tire are a result of tire properties, orientation, vertical load, velocity, pressure and road surface. The combination of suspension kinematics and tires characteristics will determine to a large extent the vehicle behavior. This behavior is for a large extend expressed in terms of stability, comfort, driver feedback and most important the lateral and longitudinal vehicle performance. Already numerous studies have been performed in the research on tire characteristics, sources [11, 12] are used to in the remainder of this section.

To describe the tire characteristics it is customary to use measured tire data and an advanced tire model. A widely used semi-empirical tire model for use in vehicle dynamics studies is the Magic Formula [12]. The Magic Formula tire model is used to calculate steady-state tire force and moment characteristics of a tire based on a set of input parameters: vertical force on the tire F_z, inclination angle γ, slip angle α and longitudinal slip κ. Applying a combination of these inputs will result in forces and moments: longitudinal force F_x, lateral force F_x, vertical reaction force F_z, overturning moment M_x, rolling resistance moment M_y and self aligning moment M_z. To analyze the steering feedback for the new suspension design, the self aligning moment and pneumatic trail are relevant.

Self aligning moment and pneumatic trail

The lateral forces F_y at the tire does not act at the center of the contact patch, but at a distance behind, see Figure 2.1. This distance is called the pneumatic trail t_p, it is the distance where the resultant lateral force is applied behind the geometric center of the contact patch. The pneumatic trail is caused by the progressive build-up of lateral force along the contact patch, such that lateral forces are greater towards the rear of the contact patch and this creates a moment on the tire called the self-aligning moment M_z, see Figure 2.1. The self aligning moment M_z is the moment that the tire generates as it rolls along its direction of motion. The self aligning moment M_z is depended on the side slip and inclination angle. The For small values of side slip and inclination the self aligning moment can be described by

$$M_z = -C_{M\alpha}\alpha + C_{M\gamma}\gamma. \tag{2.1}$$

where $C_{M\alpha}$ is the self aligning stiffness and $C_{M\gamma}$ is the camber torque stiffness of the tire. Using the self-aligning moment M_z and the lateral force F_y, the pneumatic trail t_p can be calculated by

$$t_p = -\frac{M_z}{F_y}. \tag{2.2}$$

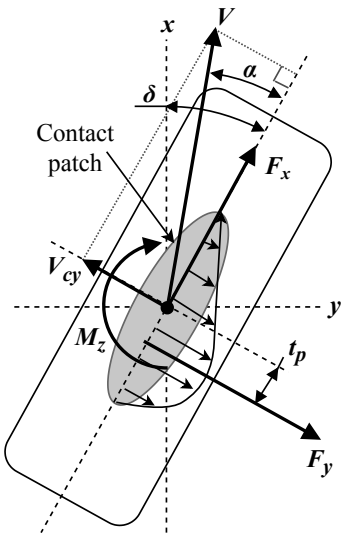
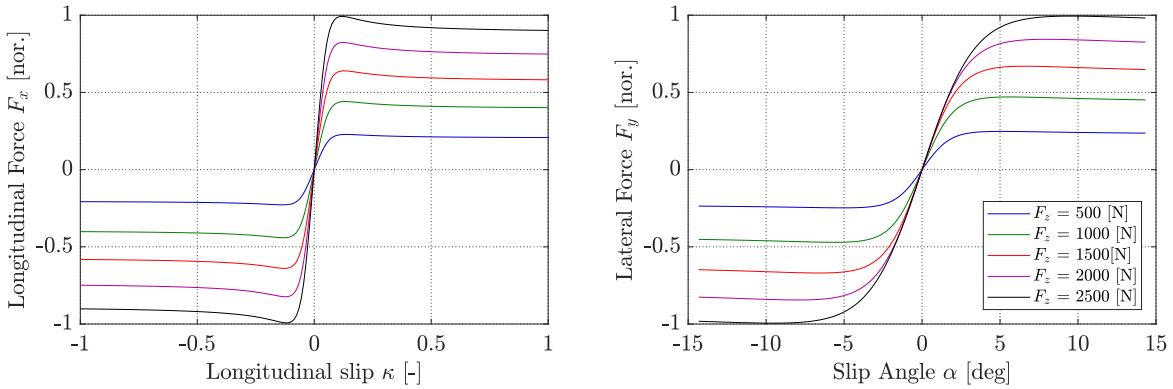


Figure 2.1: Self aligning moment M_z and pneumatic trail t_p (top view tire)

URE tire characteristics

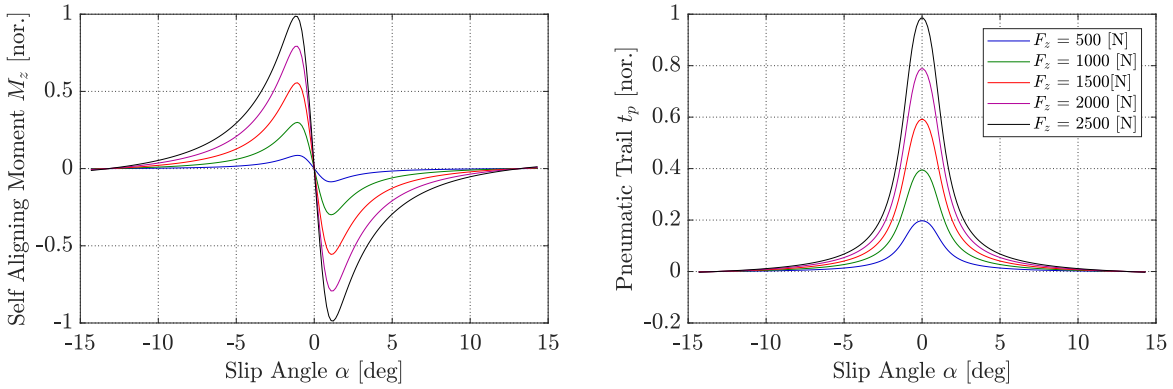
The available tire data of the current URE tire is fitted by using the Magic Formula [12]. In figure 2.2 the longitudinal and lateral force for a kappa κ and alpha α sweep of the current 205/50 R10 Apollo tire are shown. These figures are generated using the data of the Magic Formula tire model [12]. The plotted data is normalized because of confidentially.



(a) Longitudinal force **(b)** Lateral force

Figure 2.2: Kappa κ and alpha α sweep, P_{tire}= 0.8 [bar], γ= 0 [deg]

In figure 2.3a and 2.3b the self-aligning moment and pneumatic trail are plotted as function of slip angle. Pneumatic trail is at the maximum when the slip angle is zero and decreases as the slip angle increases, as can be seen in figure 2.3b.



(a) Self aligning moment **(b)** Pneumatic trail

Figure 2.3: Alpha α sweep, P_{tire}= 0.8 [bar], γ= 0 [deg]

The pneumatic trail and self-aligning moment are important parameters for analyzing the compliance and steering torque for different load cases, which will be further discussed in the remainder of this thesis.

2.3 Summary

In the literature study the history of the suspension layout of the URE car is investigated. The earlier performed research with respect to the suspension design of the URE racing car is reviewed. A summary of all performed studies is given, creating an clear overview of the research done so far. Furthermore, a general overview of the important tire data is given. This background information gives a good basis to start with the design of a multi-link suspension. Also the gained experience of the already performed research is used to further develop the suspension for the URE vehicle.

3. Literature review suspension

The ride and handling characteristics of the vehicle are to a large extend determined by the suspension. For a well designed suspension various aspects must be taken into account. In this chapter the suspension function, characteristics and different layout will be discussed. The advantages and disadvantages of a multi-link are examined in detail. Substantially research is available on the design of a multi-link suspension. However, many of these studies focus on multi-link suspension design for passenger vehicles. However, the focus of this thesis is a design for racing applications.

3.1 General function suspension

The main function of the suspension is connecting the tire and the chassis, using one or more links, resulting in one degree of freedom. This degree of freedom is a vertical directed motion of the tire relative to the chassis. The forces, acting at the tire contact patch, will be applied to the chassis trough the suspension links. Depending on the suspension geometry a specific tire contact patch path can be realized during wheel travel or steering motion. The maximum achievable performance is dependent on a well considered combination between suspension design and tires. Following characteristics should be achieved using a suspension for race car application:

- Maintain a large contact surface between tire and road surface for all driving conditions.
- Ensure an optimal wheel load distribution.
- Maintaining proper driving stability and feedback.
- Simple and fine adjustment of wheel alignment and kinematics, like camber, caster and toe, by use of minimal tools and effort.
- System reliability and rapid exchange possibility of wear parts such as: joints, bearings and tires.
- For vehicles with extensive use of aerodynamics the suspension should have no or minimal influence on the airflow through the vehicle and maintain the desired alignment with the road surface, so minimal pitch, roll and heave are desired.
- Light weight as possible.

3.2 Suspension characteristics

The suspension characteristics are described by ISO 8855: Road vehicles, Vehicle dynamics and road-holding ability vocabulary [1]. The position and alignment of the tire relative to the chassis or road surface are prescribed by the suspension geometry and kinematics. These kinematics have influence on the driving performance and driver feedback of a vehicle. The suspension characteristics can be divided into the tire alignment features and the parameters that describe the position and orientation of the steering axis. These values will vary with the vertical and steer motion of the suspension.

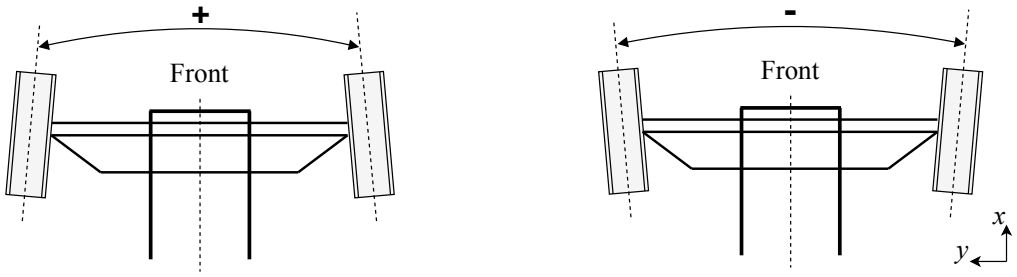
3.2.1 Tire alignment

The tire alignment characteristics describe the orientation of the tire with respect to the road. Toe and inclination angle are the two tire alignment characteristics.

Toe

Toe defines the alignment of the tire with the longitudinal axis of the vehicle. Toe is a result of a combination of static geometry, suspension setup, kinematic and compliance effects. For the left tire a positive steering angle is described as toe-out and a negative angle as toe-in. For the right tire a negative steering angle is described as toe-out and a positive angle as toe-in, see Figure 3.1. Toe settings will influence three crucial performance parameters: tire wear, straight-line stability and corner entry handling characteristics [11]. Having a to large toe-in angle will cause the tire to scrub on the outboard edges and increasing tire wear.

Excessive toe-out will cause the inboard edges to wear out and shorten the tire-life. However, the scrubbing effect can be used to increase the temperature and clean the tires of asphalt. Front wheel drive race cars are mostly set up with a small amount of toe-out, as this induces oversteer to counteract the understeer tendency of a front wheel drive car [11]. In case of a four-wheel independent suspension, also the toe must be set at the rear of the car. Toe settings at the rear have essentially the same effect on wear, stability and turn-in characteristics as they effect on the front. Setting toe-out at the rear of rear wheel drive race car will cause increased oversteer, specifically when driving torque at the rear tire is applied. The amount of toe-in or toe-out settings are highly dependent on the suspension compliance and the desired handling characteristics of the vehicle. The change in toe angle over vertical or steer motion is described by the toe gain. Toe gain is highly reliant on the steering axis orientation.



(a) Toe in (+) **(b)** Toe out (-)

Figure 3.1: Toe angle front suspension [13]

Inclination angle

The inclination or camber angle γ is the angle between the vertical axis of the tire and the vertical axis normal to the road, see Figure 3.2. Positive camber is when the top of the tire is pointing away from the car. When cornering, having negative camber on the outer tire and positive camber on the inner tire will improve grip [12]. For straight-line acceleration or braking, the highest traction is achieved when the camber angle is equal to zero [11]. However, the camber influences on longitudinal forces are generally much smaller than on lateral forces [12]. A cambered tire will also generate lateral forces during straight-line acceleration [11, 12]. During cornering a negative camber can also compensate for the deformation of the tire due to lateral forces and the tire contact patch area will increase. However, like toe settings, if the static camber is to large, tire wear will increase. Also for a large camber angle it is possible that the tire will primarily roll on its inner or outer edge during straight-line motion. The change in inclination angle over vertical or steer motion is described by the camber gain. Camber gain is like the toe gain, highly reliant on the orientation of the steering axis and the unequal length of the upper and lower suspension arms.

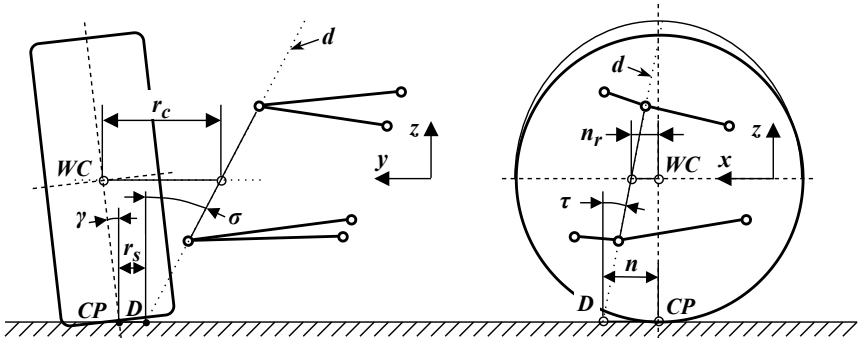


Figure 3.2: Definition of suspension parameters [14]

3.2.2 Steering axis

The location and orientation of the steering axis is mainly described by four characteristics: king pin inclination angle, scrub radius, caster angle and mechanical trail.

King pin inclination

The king pin inclination KPI σ is the angle between king pin or steering axis d and the vertical z-axis normal to the road, see Figure 3.2. The king pin or steering axis is the line around which the tire and wheel hub will rotate. It is the line that runs through the center of the upper ball joint to the center of the lower ball joint, in case of a double wishbone suspension. Combined with the caster angle τ, the king pin angle influences the variations of the tire orientation when the tire is steered. The king pin angle should be positive to have camber variation over wheel travel and proper steering feedback. However, to large king pin angle could result in increased camber gain when steering.

Scrub radius

Scrub radius r_s is defined as the distance between the intersection of the steering axis d to the center of the contact patch CP at road level, viewed in the y-z-plane, see Figure 3.2. The scrub radius is the lever arm of longitudinal forces applied at the contact patch, related to braking or traction forces when in-wheel motors are used. The scrub radius is positive as illustrated in Figure 3.2. A positive scrub radius will induce a self-centering action during steering, mainly sensitive at low speed. At high speed the self-centering action is usually minimal compared to other effects such as the caster angle effect. On the front suspension a positive scrub radius will improve stability, causing toe-out for braking and toe-in during accelerating, as result of suspension compliance. Having a small amount of scrub radius will furthermore increase the steering feedback due to longitudinal forces. In combination with torque vectoring the scrub radius should be minimized to reduce the induced torque on the steering wheel. However, when the scrub radius is zero it is conceivable that the tire will start oscillating at a specific velocity for straight line driving. At this point the scrub radius is fluctuating between positive and negative, resulting in a negative or positive moment around the steering axis, as result of the longitudinal tire forces.

Caster

Caster τ is defined as the angle between the steering axis and the vertical axis d in the x-z-plane, see Figure 3.2. In combination with the king pin angle, a positive caster angle will induce a self-centering moment of both tires and result in steering feedback during straight line driving, making the vehicle easier to drive with a better steering response and straight line stability. A positive caster will also affect the change in inclination angle when steering. A combination of positive king pin and positive caster angle will lead to a negative camber gain on the outer wheel and a positive camber gain on the inside wheel, when a steering motion is applied. The caster angle is often used in combination with the king pin angle to optimize camber gain during a steering and bump motion.

Mechanical trail

The mechanical trail n is the distance between the intersection of the steering axis to the center of the contact patch CP at road level, viewed in the x-z-plane, see Figure 3.2. The mechanical trail is defined positive as illustrated in Figure 3.2. For positive mechanical trail the tire is pulled by the vehicle comparable to a wheel of a shopping trolley. This causes a self-aligning effect for straight line driving and increase straight-line stability. A negative mechanical trail will increase the tendency of the tire to show shimmy effect. Shimmy is described as an oscillating motion of the tire. The mechanical trail in combination with the pneumatic trail, will transfer the lateral forces applied at the contact patch to a steering moment felt by the driver. The mechanical trail is mostly used to fine tune the steering torque during cornering.

Wheel load lever arm

The wheel load lever arm p is defined as the arm which transfers a vertical tire load to a torque around the steering axis. The arm is influenced by the KPI σ, the caster τ, mechanical trail n and scrub radius r_s [14]. The wheel load lever arm result in self-centering action of the steering during straight line driving or cornering. The gradient of the wheel load lever arm describes the restoring forces due to the weight of the vehicle. This can be related to the moment which is needed to steer a vehicle during stand-still. For a suspension with the fixed kingpin axis to the upright, the wheel load lever arm p can be calculated by:

$$p = r_s \tan(\tau) + n \tan(\sigma) \tag{3.1}$$

3.3 Suspension types

The most used suspension layout for racing aplications, and also current suspension layout of the URE13, is a double wishbone suspension [3, 14]. A few high performance vehicles are equipped with a multi-link layout, for example the lately introduced AMG project one [15]. In this section the double wishbone and multi-link suspension layout are discussed and a comparison is conducted.

3.3.1 Double Wishbone

A double wishbone suspension is an independent suspension layout using two A-shaped arms to locate the upright, see Figure 3.3. Mostly these arms are parallel, and are called the upper and lower arm. Each wishbone or A-arm has two spherical mounting points to the chassis and one spherical joint with the upright. The upper and lower spherical joint on the upright define the steering axis d, see Figure 3.4. Often the lower A-arm is used as support for the shock absorber.

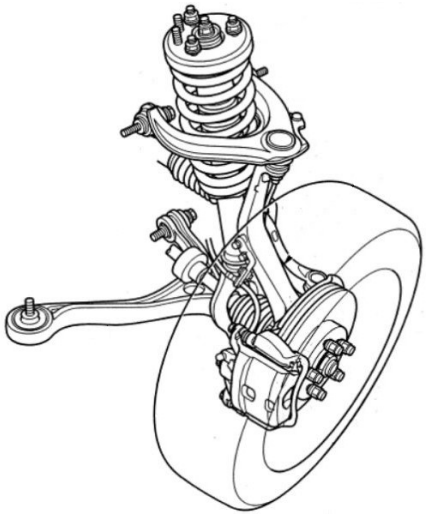


Figure 3.3: Double wishbone front suspension layout [16]

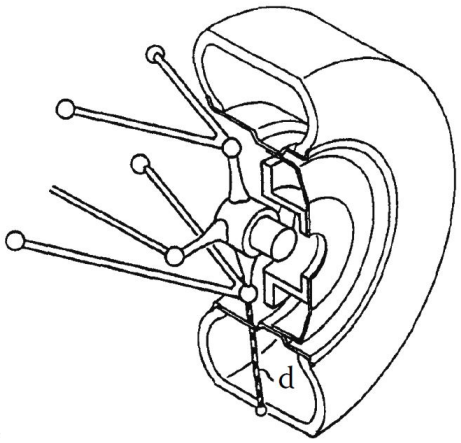


Figure 3.4: Double wishbone suspension schematic representation [14]

Generally, the upper arm is shorter to induce negative camber as the suspension is compressed. This type of double wishbone suspension, with unequal arm length, is also known as the short-long arm (SLA), type suspension. As result, body roll during cornering result in a positive camber gain on the inside wheel, whereas the outer wheel gains negative camber. In the case of the front axle, the upright is also connected to the steering rack by the steering link. Which has spherical joints at both ends. The steering link is intended to control or adjust the rotation around the steering axes. For this the connection point on the steering rack moves in lateral direction to the chassis. This lateral motion is transposed to a steering motion of the tire by the suspension mechanism and steering link. The rear suspensions contain a similar link, called toe or tie link, to control the static toe angle.

3.3.2 Multi-link

The multi-link suspension is similar in construction to the double wishbone. However, the upper and lower A-arms are now opened up into four separate links, result in a 5-link connection between the upright and chassis, see Figure 3.5. The multi-link suspension layout was first introduced on a production vehicle in 1982 on the Mercedes-Benz 190E W201 serie. The main reason for introducing the multi-link suspension was to give compact models exactly the same suspension characteristics as larger Mercedes-Benz saloons. Furthermore, Mercedes-Benz want use suspension compliance in an effective way, to improve stability and ride comfort. This is achieved by proper positioning of the independent suspension links and effectively usage of the bushing compliance.

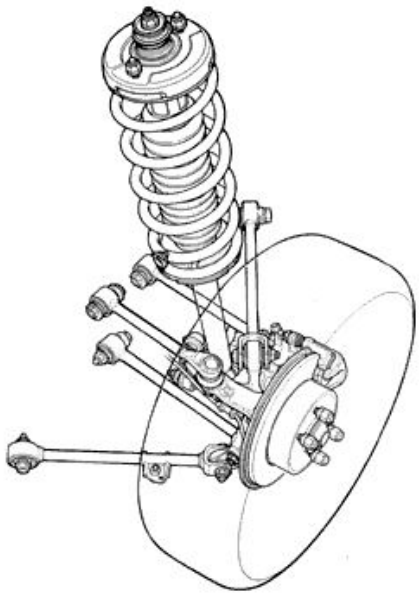


Figure 3.5: Multi-link rear suspension layout [17]

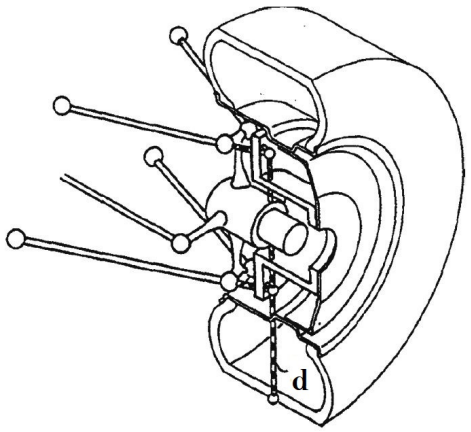


Figure 3.6: Multi-link suspension schematic representation [14]

Derived from the double wishbone, the multi-link suspension uses three or more lateral links and one or more longitudinal links, which do not have to be of equal length and can be angled in different direction. The configuration is characterized by spherical joint connection at the end of each link. By this, the suspension links are only exposed to tension and compression forces. The steering axis d for a multi-link suspension is virtual and is defined as the imaginary axis where the extend of the four individual suspension links intersect with one virtual vector in space d, see Figure 3.6. Likewise for the double wishbone, the multi-link also consist of a steering or toe link which connects the upright to the steering rack respectively chassis.

The multi-link suspensions can be divided into a 4-link or 5-link group. For a 5-link suspension layout all links are rods with spherical-joints at each end. In case of a 4-link suspension layout, the upper or lower suspension links are actually an A-arm. Thus the 4-link suspension is a combination between the double wishbone and 5-link suspension layout.

3.4 Multi-link characteristics

A multi-link will have some benefits over other suspension layouts. In this section a literature review on the advantages and disadvantages is presented.

3.4.1 Advantages

kinematic freedom

The first advantage is the increased design freedom, resulting in more possibilities for optimization of the suspension kinematics. Because of the additional kinematic freedom, the multi-link layout is more flexible and the extra hard-points could be comprehensive adjusted to fine-tune the suspension geometry. The multi-link layout offers much more options for adjusting the suspension kinematics compared to double wishbones, whereas the kinematics are to a large extend restricted by the upright geometry and in-wheel packaging. Also the steering axis is imaginary and therefore it is feasible to locate this axis anywhere with relation to the tire. The steering axis is usually determined by upper and lower spherical joint postion on the upright, in case of a double wishbone suspension. For a multi-link type suspension, a imaginary steering axis exists [18]. This imaginary steering axis is determined by the two virtual rotation points, located on the intersections of the extension lines of upper respectively lower links, see Figure 3.7 and 3.8.

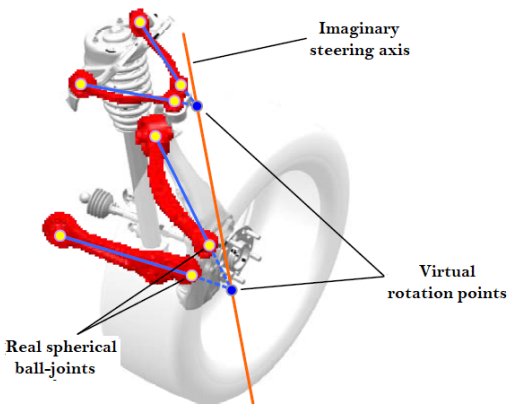


Figure 3.7: Imaginary steering axis [18]

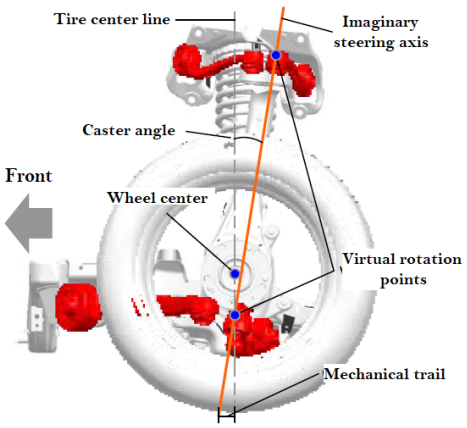


Figure 3.8: Imaginary steering axis side view [18]

During a steering motion the orientation of the imaginary steering axis varies, due to the movement of the upper and lower virtual rotation points. The steer related characteristics of a front suspension such as the caster, toe and camber angles will change depending on the applied steering angle [4, 19], see Figure 3.9.

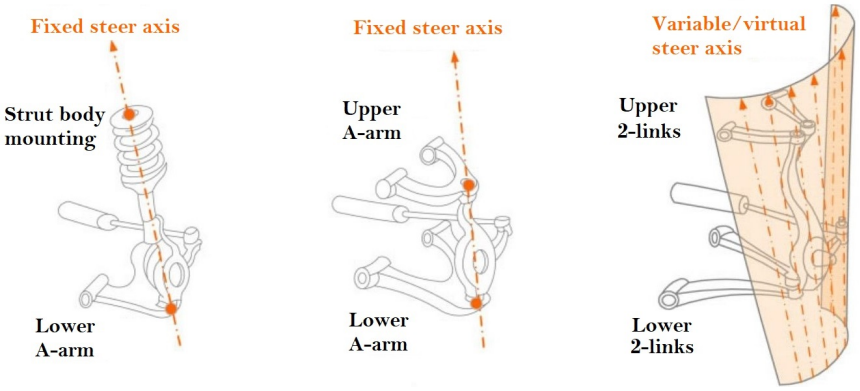
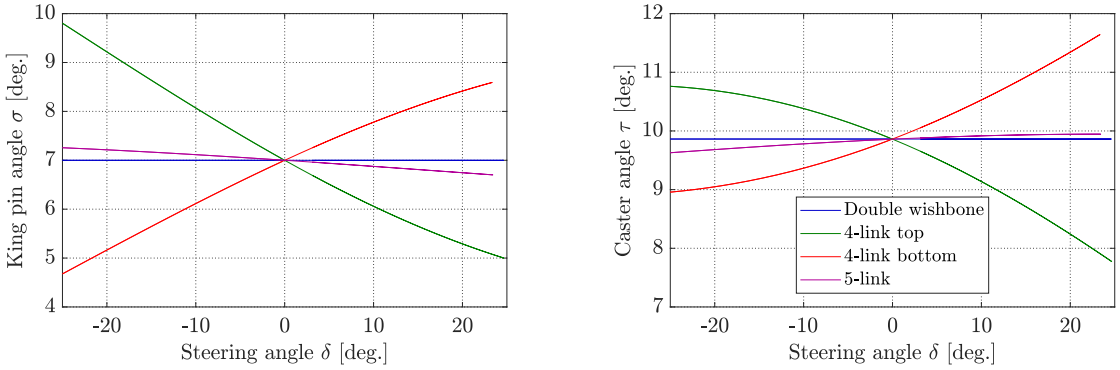


Figure 3.9: Steer axis for various suspension layouts (Mcpherson, Double wishbone & Multi-link) [20]

In Figure 3.10 the kingpin and caster angle for a double wishbone, 4-link and 5-link suspension with the same initial steer axis are plotted. For this comparison the current rear suspension of the URE13 is used. The 4-link and 5-link are created by splitting the spherical ball-joints of the upper and/or lower A-arms. Looking at kingpin and caster angle over the steering angle, a five-link suspension shows a more linear behavior. This is mainly caused by the fact the a 4-link suspension has a fixed upper or lower spherical ball-joints on the upright around which the steered tire rotates. For the 5-link, both the upper and lower virtual rotation points change with the steering angle δ.



(a) Kingpin angle

(b) Caster angle

Figure 3.10: Variation steering axis for double wishbone and multi-link suspension

Compliance

The second advantage of a multi-link suspension is the positive effect on compliance and stability. The overall compliance and deformation of the multi-link suspension due to longitudinal and lateral forces is larger compared to a fixed structure suspension, such as a double wishbone layout suspension. Therefore, the positions of the links slightly change due to compliance of the bushings combined with external forces. This will result in toe-in of the front wheels during braking, when using the right geometry and layout [18], see Figure 3.11. During braking the suspension will move backward, in relation to the vehicle body. Because of the geometry and compliance, the wheel will steer inwards or outwards, resulting in toe-in or toe-out. Effecting the oversteer and understeer behavior and improved stability during braking [21, 22].

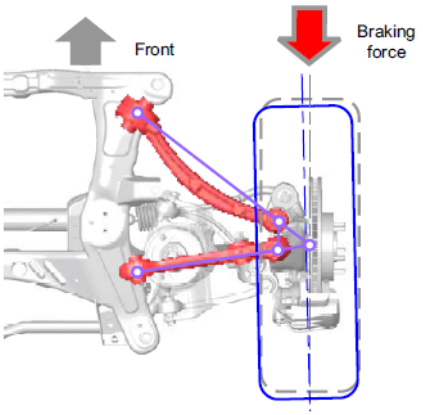


Figure 3.11: Front Toe-in at Braking [18]

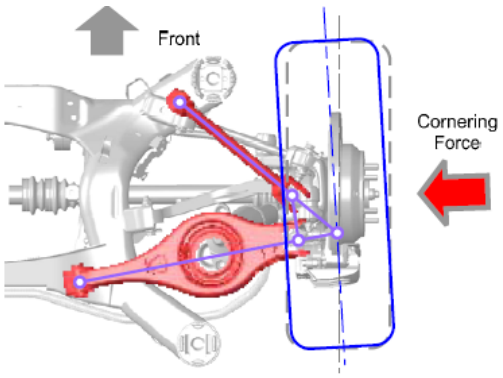


Figure 3.12: Rear Toe-in at Cornering [18]

Also the rear wheel experiences toe-in during cornering as result of thre lateral force. This will increase the understeer tendency in a cornering maneuver. Figure 3.12 shows that the instantaneous center is located rearward from the tire contact patch center, therefore the wheel has a toe-in tendency when a lateral cornering force is applied, this is required for cornering stability[18]. Likewise the rear wheel will show toe-in when a braking force is applied, because of the geometry and position of suspension links. During a braking maneuver, the front suspension will compress and the rear suspension will extend [12]. Using this effect and having toe change over wheel travel, the total change in toe angle can be restricted or minimized.

Steering angle

The third advantage of using a multi-link suspension is the larger maximal steering angle which result in a smaller turning radius. Using a multi-link suspension the instant center of the front wheel moves during a steering motion, [4] [23]. Due to the relocation of the instant center, or imaginary steering axis, and movement of the links it is feasible to achieve larger steering angles, without having collision between the inner-rim radius and suspension links, as shown in Figure 3.13.

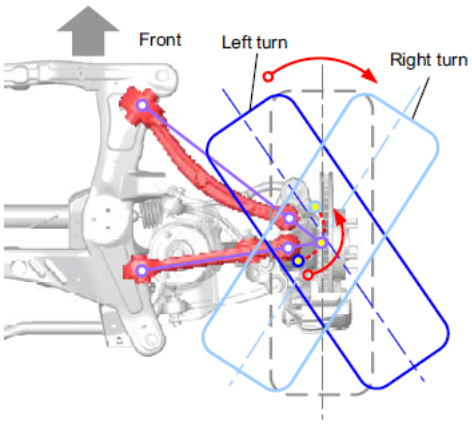


Figure 3.13: Motion of the instant center during steering [18]

Jacking

The fourth advantage is reduction of the jacking effect during a steering motion and cornering. Jacking is described by vertical movement of the vehicle body during lateral acceleration. Using a multi-link suspension design the effect of jacking could be reduced by decreasing or optimizing the vertical movement of the tire contact patch during a steering motion. A double wishbone design often shows a large jacking effect during cornering [4]. This is a disadvantage in terms of the slightly increasing ride height and change in the vertical position of the center of gravity. As result the maximal achievable lateral acceleration will decrease.

Wheel center offset

The last advantage of using a multi-link suspension is the reduction of vibrations induced by the wheel. The imaginary steering axis and its offset from the wheel center can be reduced, as shown in Figure 3.14. Due to the reduced distance between the steering axis and wheel center, the vibration of a wheel caused by the asymmetry of wheels or the road disturbance can be reduced [18]. A double wishbone and McPherson strut type suspension are restricted by their geometrical layout and in-wheel packaging, resulting in a steering axis relative far located from the wheel center. However, changing the wheel center offset will also effect the scrub radius.

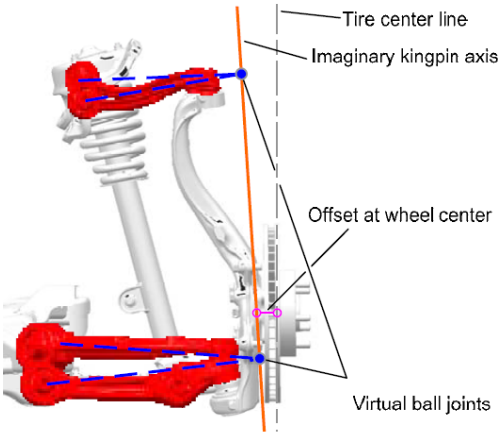


Figure 3.14: Kingpin axis and its offset at wheel center [18]

3.4.2 Disadvantages

Complexity and weight

The biggest disadvantage of a multi-link suspension in comparison to a double wishbone suspension is the complexity and possible increased weight [4]. An additional disadvantage of the increased complexity is the larger number of joints and mounting points, which may result in increased suspension play. Also the multi-link layout provides less suitable mounting positions for directly connecting a stabilizer bar to the suspension links [23].

Reduction rotational stiffness

A second disadvantage is the rotational stiffness around the longitudinal and lateral axis. For a race car application it is desirable to have a stiff suspension, so the kinematics will be less effected by the result of suspension forces. The overall stiffness depends on the compliance of the bushings that connect the links to the vehicle body and by the length of the arms. Placing these connection points further apart from each other and longer arms will result in a higher stiffness [4]. However, due to the space limitations and desired kinematics, the resulting design will have a reduced torsional stiffness compared to a double wishbone suspension.

Extra care hard-point position and tolerances

A third disadvantage is the possibility of having increased forces in the suspension links. Due to de decoupling of the hardpoints and incorrect design could result in leverage effect. This will increase the forces in the links and ball-joints [19, 24]. Also the multi-link suspension is sensitive to collapsing effect during steering motion. In Figure 3.15 an example of this collapsing effect is illustrated. Collapsing occur during steering motion when hard-point 2 moves over center with respect to the line connecting point 1 and 3 . At this point the suspension will collapse due to the applied lateral forces F_y. All this result that the multi-link suspension layout is much more sensitive to exact postioning of the hard-points and production tolerances.

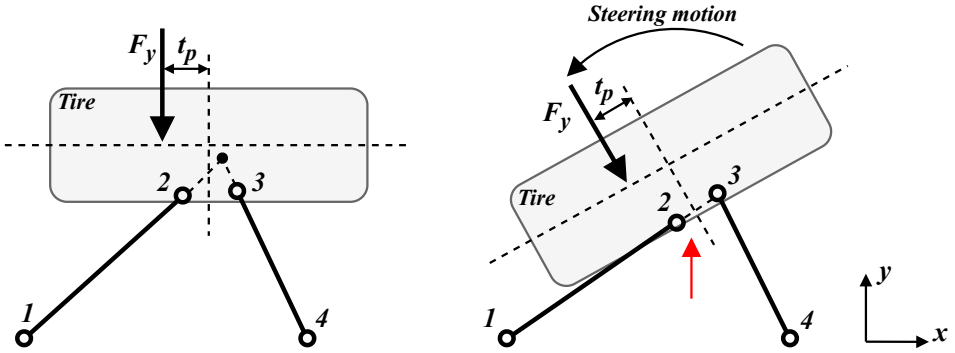


Figure 3.15: Collapsing effect by improper hard-point position for a multi-link suspension

Simulation software

The forth disadvantage is that analyzing the kinematics and tuning the suspension is nearly impossible to do by hand and requires 3D CAE analysis software, such as ADAMS or SIMMECHANICS [24, 25]. Using CAE analysis software the suspension kinematics can be determined and optimized. Also the complete motion of a multi-link is harder to analyze without having visual simulations and kinematic graphs.

3.5 Design process

The main goal for the suspension design of the URE race car is the best possible tire-road contact. This will improve tire performance and thereby the overall vehicle performance and handling of the car. Because of FSAE regulations and vehicle layout, some components and parameters are already fixed, this result in sub optimal design. Based on the design targets, load cases and packaging a suspension geometry is defined, in order to gain maximal performance. Determining the final suspension geometry is an iterative process. This iterative process consist of: kinematic, collision check, compliance and driver feedback analyses. Also the choice between double wishbone or multi-link layout is one of the considerations during the design process. In Figure 3.16 a flowchart for the suspension design process is given. The scope of this project is illustrated with the red boxed frame. The suspension design process is divided into various steps, only the most relevant will be shortly discussed.

Determining overall vehicle parameters, packaging and design targets
In the first stage the overall vehicle and design targets are stated. These targets are mainly based on vehicle dynamic considerations, for example: ride frequencies, roll stiffness, lateral load distribution, ride height, wheel travel range, under or over steer gradient, Ackermann settings, and turning radius. Once the vehicle targets are identified, the more detailed targets for the suspension can be determined.

Suspension layout
Based on the vehicle targets and packaging, an initial suspension layout can be selected. In most cases a choice between double wishbone or multi-link suspension layout has to be made. Furthermore, a choice will be made for the spring/damper system. This can be direct acting or using a rocker system with a push or pull rod. Also the type and position of the steering system need to be determined. The type can be classified in a rack and pinion or recirculating ball system. The position of the steering system can be in front or behind of the wheel center, and in plane with the top or bottom suspension links, or in between.

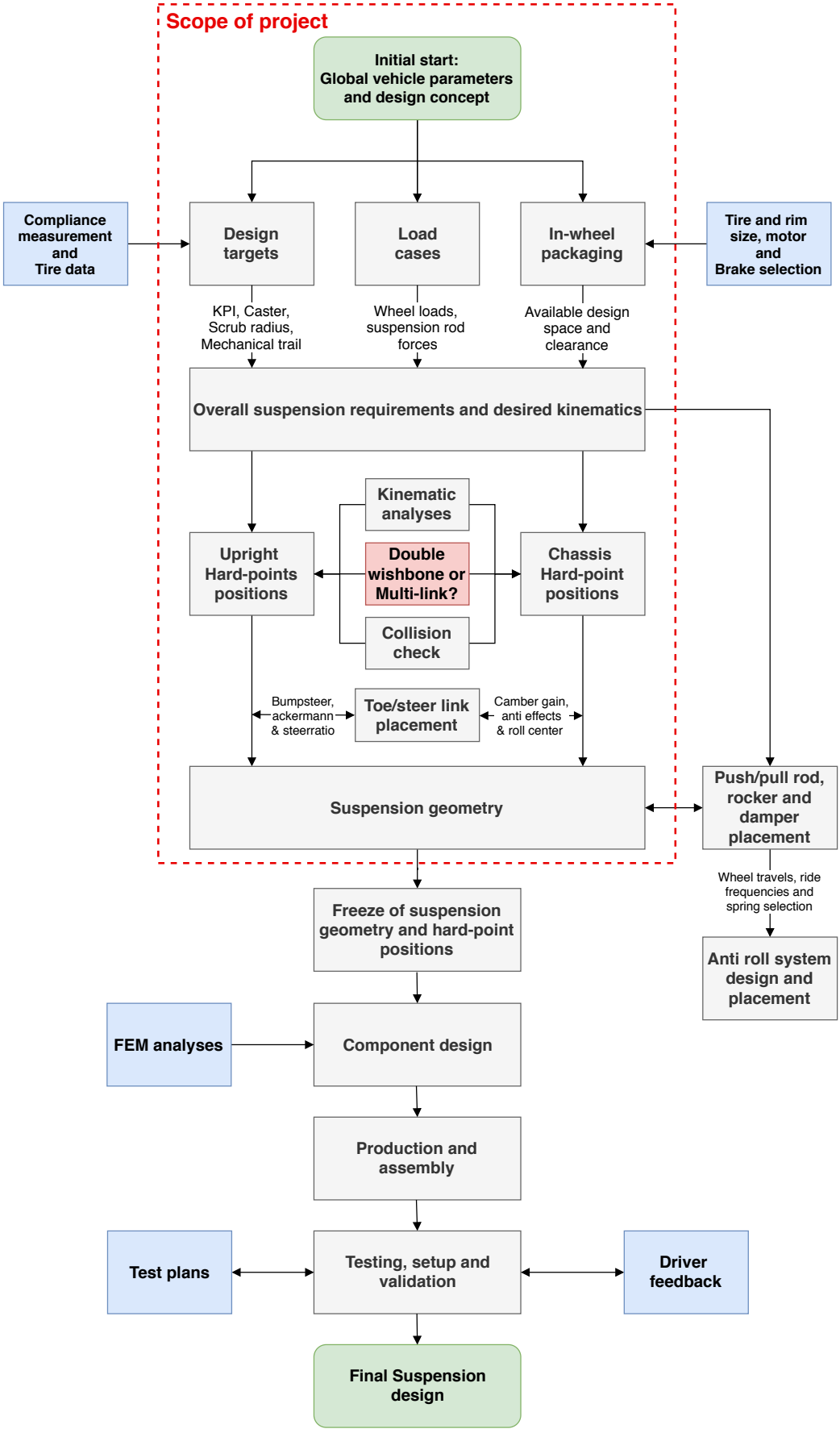


Figure 3.16: Suspension design flowchart.

Kinematic analysis, determining hard-point positions and collision check
The position of the hard-points determines the kinematic characteristics of the suspension. The analysis of the kinematics can be done graphically, using CAD package, or by the use of dedicated suspension software packages. The kinematics can be divided into the steer and bump kinematics.

For the steer kinematics the following properties are considered:

- Toe, camber and caster angles
- Kingpin inclination angle
- Mechanical trail and scrub radius
- Ackermann change with steering angle

The bump kinematics properties are:

- Toe, camber and caster gain
- Roll center height
- Instant centers and axis
- Anti features

Based on the wheel motions and hard-point location a collision check can be performed in order to identify potential collision between the individual suspension components. If a collision is present the hard-points should be relocated without affecting the overall suspension kinematics.

Compliance analysis
The compliance of the complete suspension will affect the kinematics during different driving maneuvers. It is difficult to quantify and improve the compliance of a suspension. The compliance consists of mechanical free-play in the ball-joints and elasticity of individual components. The resulting compliance should be taken into account for a dynamic analysis. In general, free-play and compliance should be minimized in order to have precise control of the vehicle and to have predictable suspension kinematics.

Suspension loads
Once the geometry and hard-points positions are established, the load in each suspension component can be calculated. This can be performed by using a multi-body suspension simulation model and applying different load cases at the contact patch in order to determine the forces in the various suspension components. These load cases can be related to acceleration, braking, cornering and driving over a bump. Often, loads that have been measured on a similar suspension, for example older URE vehicles, are used instead. It is possible to determine the suspension load cases for specific driving maneuvers using a multi-body vehicle simulation model.

3.6 Summary

The suspension kinematics are essential to establish proper driving performance and handling of a vehicle. The suspension characteristics can be divided into tire alignment parameters, and the steering axis orientation and position. The tire alignment parameters influence the driving stability of the vehicle. Whereas, the steering axis placement is to a large extend responsible for the driver feedback. The suspension kinematics are determined by the hard-points and suspension layout. The most used suspension type in the racing industry and FSAE competition is a double wishbone suspension. In comparison to a double wishbone suspension layout, multi-link can offer more design freedom. Due to the virtual steering axis of a multi-link suspension, the kingpin axis can have almost any orientation and position. Because of the extra kinematic freedom, the grip performance of a tire can be maximized with an optimized wheel trajectory using a multi-link suspension layout. Furthermore, proper position of the hard-points will result in using suspension play and compliance in an advantage way. However, the complexity and need for a proper design tool can be a restraining factor to choose a multi-link instead of double wishbone suspension layout. In conclusion, a multi-link suspension is less restricted by the upright geometry and in-wheel packaging. By this the multi-link has more design freedom and the suspension characteristics can extensively be tuned to optimize tire and vehicle performance.

4. Suspension Design Tool

For analyzing the suspension characteristics a suspension simulation model is used. Because most commercial software packages have some shortcomings, as discussed in Appendix A, a complete new suspension simulation model is created. Based on the current knowledge and the fact that other URE team members should be able to use this suspension model, MATLAB is be used. Using MATLAB the suspension model can be modeled and adapted to the personal preferences. Developing a suspension multi-body model from scratch, ensures good insight in the definitions of the suspension characteristics and mathematical equations involved. It is possible to freely enhance the model and implement additional calculations. In addition, the developed suspension model could be used in combination with other models created in MATLAB. This means that the suspension model is flexible and can be easily extended. For example, the suspension model can be combined with a multi-body vehicle model for analyzing the vehicle dynamics.

The developed suspension design tool is able to calculate and plot all needed suspension characteristics for a given suspension layout, using a set of hard-points and setup parameters. The suspension design tool consist of different modules in order to fulfill the project goal and design a multi-link suspension for the future URE race-car. The first module is the multibody model, used to investigate the kinematic behavior of a suspension layout. This model is created in SIMSCAPE, which is the updated multibody toolbox of MATLAB SIMMECHANICS. A more detailed description and comparison between SIMSCAPE and SIMMECHANICS can be found in Appendix B. The post processing of the simulation results is done using a set of mathematical equations, which calculate the relevant suspension kinematics. The second module consists of collision check and visualization of the suspension movement. Mathematical equations are used to determine the potential of suspension collisions. The kinematic multibody model is extended by including suspension play. This module is used to investigate the effects of suspension play, compliance and determine the suspension loads for different load-cases. The last module is a simplified suspension model with the capability of applying forces at the tire contact patch. Based on the applied forces, the resulting steering torque can be determined. In this chapter the suspension design tool and included modules will be discussed in detail.

4.1 Kinematic suspension model

To analyze the suspension kinematics a multibody model is used. In Figure 4.1, a screen shot of this suspension model is given. The chassis is fixed to the world, whereas the wheel platforms can be separately actuated in vertical direction. Based on the simulation results, the front and rear suspension characteristics are analyzed. A detailed description of the this SIMSCAPE model is given in Appendix C.

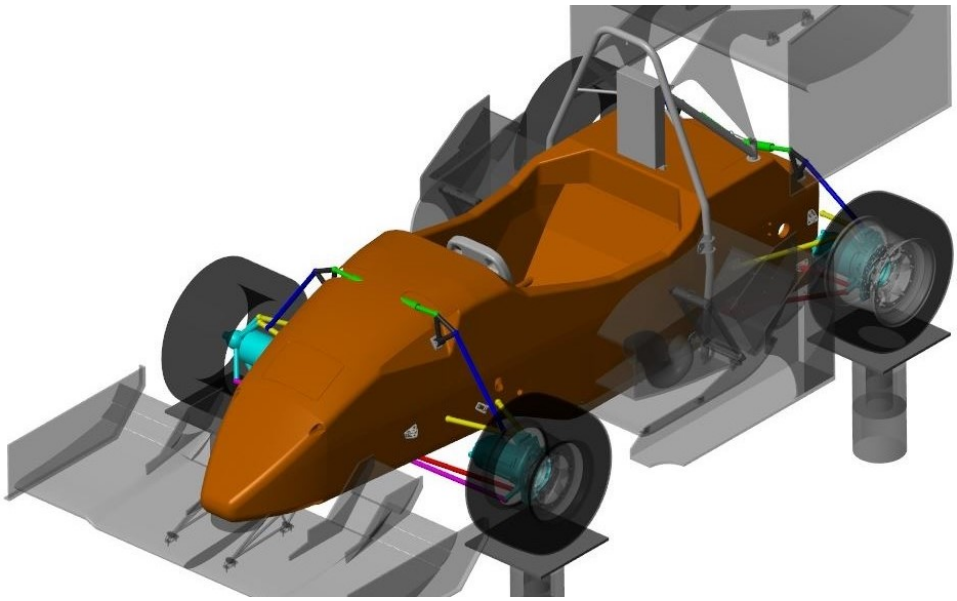


Figure 4.1: Suspension simulation model in SIMSCAPE.

The multi-link suspension mechanism consists of an upright, which is connected to the chassis, suspension box and steering rack with six connection links and twelve ball joints. The six links can be divided into: four construction rods, one pull/push rod, and one steer rod or toe rod for the front respectively rear. The combination of these components constrain the movement of the upright. In Figure 4.2 a schematic representation of multi-link suspension is shown. The chassis is fixed in space and can not move. Whereas the position of the upright and wheel center is determined by position and length of the six individual connection links, see Figure 4.3.

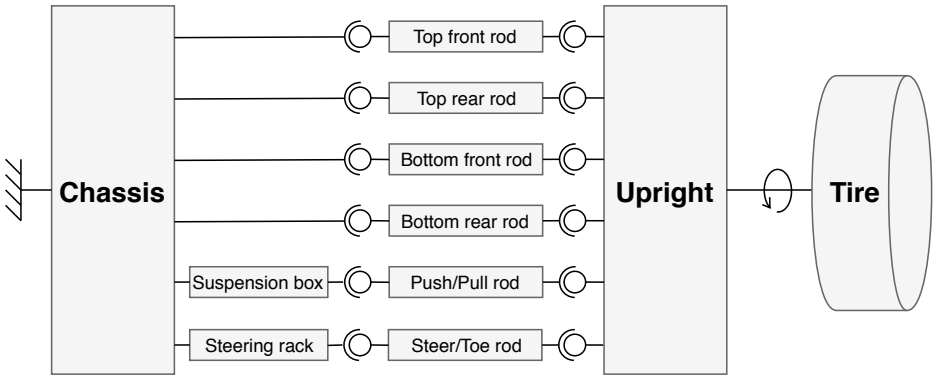


Figure 4.2: Schematic representation of a multi-link suspension.

The number of degrees of freedom of the complete suspension is

$$f_{susp.} = 6(K + L) - N + \sum f_i = 6(1 + 6) - 6 - (12 * 3) = 0, \tag{4.1}$$

where K is the number of uprights, L is the number of links, N is the number of free rotations of all links, and f_i is the total degree of freedom of each joint. The suspension consists of seven bodies; one upright, four suspension links, one steer/toe link, and one push/pull rod. Each individual body has six degrees of freedom, three translation and three rotational degrees. The total degrees of freedom therefore equals $f_{tot} = 7 \cdot 6 = 42$. Each link is able of rotate around there individual axial axis, resulting in $N = 6$. In total there are twelve spherical joints, each reducing the total degrees of freedom by three.

To determine the position of the tire contact point, the lowest point on the wheel is placed on a wheel pad. The wheel pad has the freedom to move in the longitudinal, lateral and yaw direction. Furthermore, it is assumed that the tire cannot slide sideways with respect to the wheel pad and can have a inclination angle. The distance between the wheel center and the tire contact point on the wheel pad surface remains constant and equals the tire radius r_{tire}.

Using a combination of bodies and joints in the multi-body model, the wheel pad is connected to the ground. In Figure 4.4 a graphical representation of these constraints are shown. Starting from the ground plane, first the tire contact point is allowed to translate in x and y-direction, presented by the lateral and longitudinal degree of freedom. Next, the tire is allowed to steer around the world z-axis and move in vertical direction, the steering and vertical DOF. Finally, the tire is allowed to incline around the longitudinal axis of the steered tire, the camber DOF. The tire has five degrees of freedom in total, two rotational DOFs and three translational DOFs. The rotation around the y-axis or axle of the tire is suppressed, meaning that the tire cannot roll forward or backwards with respect to the wheel pad.

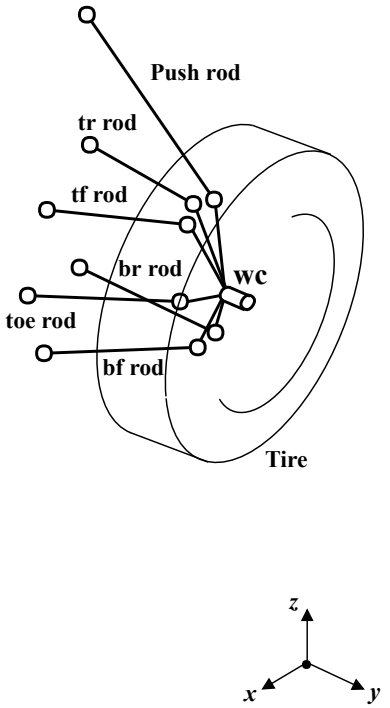


Figure 4.3: Graphical representation of suspension layout

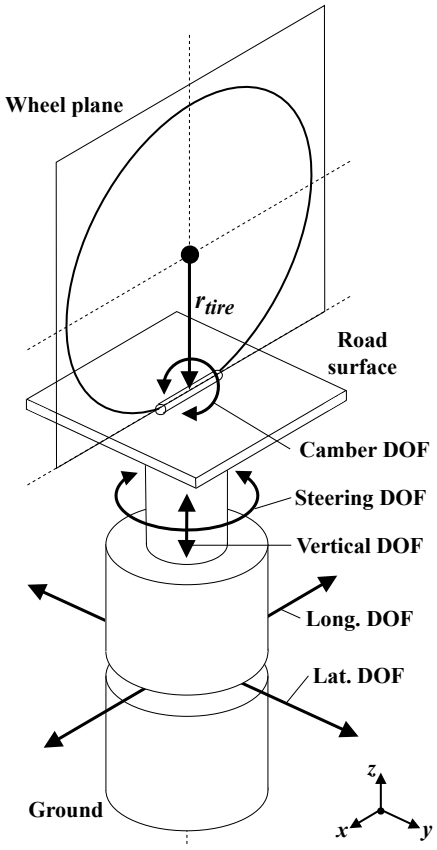


Figure 4.4: Graphical representation of wheel pad construction

4.1.1 Suspension kinematics

In order to analyze the suspension kinematics, the multi-body model has two inputs. Both are used to prescribe the motion of the suspension system. The first one is steering motion introduced by the steering rod and the second one is vertical motion of the tire contact point. Using both inputs, the suspension kinematics and characteristics are determined. Depending on the applied motion, two instantaneous axis can be defined. The first instantaneous axis is the steer axis, which is linked to the steering degree of freedom of the upright. The second instantaneous axis is linked to the bump movement of the upright. Both axis are instantaneous, meaning that they migrate during the motion of the suspension. The pair of instantaneous virtual axis is used to determine the suspension steer and bump characteristics.

For a double wishbone the both upper and lower A-arm span a plane. The intersection line of these two planes is described by the instant axis, see Figure 4.5. Around this axis the tire will rotate during vertical motion. This axis will intersect with the vertical plane through the contact patch of both the front and rear wheels. The intersection point is called the side view instant center. Also the axis will intersect with the vertical plane through the front two wheels, this intersection point is described by the front view instant center.

The steering and bump characteristics of a spatial suspension system, such as a multi-link, can not be calculated by the geometrical projection of the suspension linkage on the cross sectional plane in side and rear views, as in the case for a double wishbone suspension. The reason for this is that the motion of the upright, relative to the chassis, is represented by a screw motion, see Figure 4.6. The upright will rotate around the instantaneous screw axis s in space. This motion is characterized by a rotation around the instantaneous axis together with an applied axial motion along that axis. The screw axis of a spatial multi-link suspension is not suitable as reference axis to calculate moments and forces or to establish static force balance. Since the screw axis shows an axial motion coupled with a rotation. Forces that are not applied perpendicular to the screw axis will generate a torque, even if they intersect with that axis.

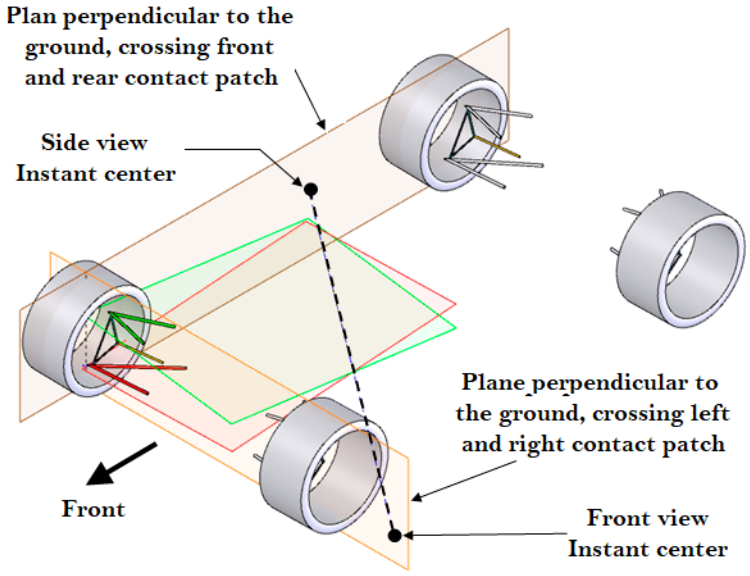


Figure 4.5: Instant axis and centers of the front right double wishbone suspension [26].

The proposed methods in [14] and [26] are used to determine the suspension steer and bump kinematics, employing the assumption of an instantaneous screw axis. A method for the kinematic analysis of the suspension mechanism is given in Figure 4.7. The angular velocity vector ω_K of the upright and the velocity vector v_M of a reference point M on the upright are used to calculate the steer and bump characteristics. The motion of the upright K is characterized by the velocity vector v_M of the reference point M and the angular velocity vector ω_K of the upright. For simplicity, the wheel center is used as reference point M. The velocity vector v_i of any point i at the upright can be described by:

$$v_i = v_M + \omega_K \times r_i \tag{4.2}$$

With r_i as the vector pointing from the reference point M to point i. Equation (4.2) will be used to further analyze and calculate all suspension kinematics. This approach is applied to calculate the instantaneous axis for both steering and bump, and will be further discussed in the remainder of this section.

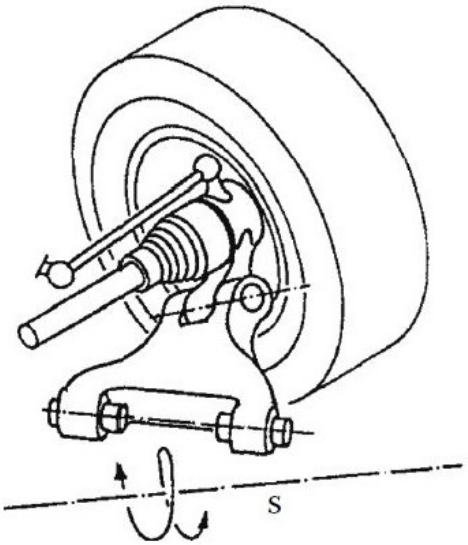


Figure 4.6: Screw axis s of a spatial Mechanism [14].

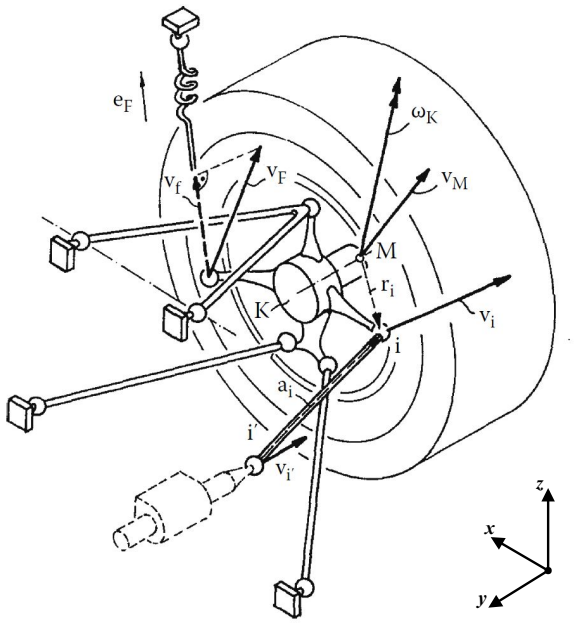


Figure 4.7: Motion analysis of a 4-link suspension [14].

Orientation of the tire

The orientation of the tire is defined by the rotation sequence discussed in section 0.2. The rotation of the tire, with respect to the world coordinate frame, is represented by a rotation matrix R, given in (4.3). The rotation matrix R describes the transformation from ground to wheel center. This rotation matrix for the tire is directly extracted from the multi-body model, during kinematic analyses for bump or steer motion.

$$\begin{aligned} R &= R^{43}(\vartheta)R^{32}(\gamma)R^{21}(\delta) \\ &= \begin{bmatrix} \cos(\vartheta) & 0 & -\sin(\vartheta) \\ 0 & 1 & 0 \\ \sin(\vartheta) & 0 & \cos(\vartheta) \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\gamma) & -\sin(\gamma) \\ 0 & \sin(\gamma) & \cos(\gamma) \end{bmatrix} \begin{bmatrix} \cos(\delta) & -\sin(\delta) & 0 \\ \sin(\delta) & \cos(\delta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} c(\delta)c(\vartheta) - s(\delta)s(\gamma)s(\vartheta) & s(\delta)c(\vartheta) - c(\delta)s(\gamma)s(\vartheta) & -c(\gamma)s(\vartheta) \\ -s(\delta)c(\gamma) & c(\delta)c(\gamma) & s(\gamma) \\ c(\delta)s(\vartheta) + s(\delta)s(\gamma)c(\vartheta) & s(\delta)s(\vartheta) - c(\delta)s(\gamma)c(\vartheta) & c(\gamma)c(\vartheta) \end{bmatrix} \\ &= \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix} \end{aligned} \tag{4.3}$$

Using the tire rotation matrix R the different angles, toe δ, camber γ and rotation ϑ, are determined by using:

$$\delta = \tan^{-1}\left(\frac{-R_{21}}{R_{22}}\right), \tag{4.4}$$

$$\gamma = -\sin^{-1}(R_{23}), \tag{4.5}$$

$$\vartheta = -\sin^{-1}(R_{31}). \tag{4.6}$$

Steering Characteristics

In order to analyze the steering characteristics, the steering rod between the chassis and upright is actuated, which results in a steering motion around the steering axis of the upright. The steering rod is actuated by a motion of the steering rack in y-direction. The steering rod transfers the lateral motion of the steering rack, to a steering motion of the upright. During the analyzes of the steering characteristics, it is assumed that the spring/damper is incompressible and the tire contact patch can freely move in lateral, longitudinal and vertical direction.

For calculating the steering characteristics, the steering velocity, ω_δ, around the tire contact point CP in x-y-plane is used, see Figure 4.8. The following equations are taken from the proposed method in [14]. This steering velocity is calculated by:

$$\omega_\delta = -\omega_{K_x}\tan(\gamma)\sin(\delta) + \omega_{K_y}\tan(\gamma)\cos(\delta) + \omega_{K_z} \tag{4.7}$$

Where, ω_{K_x} is the angular velocity of the upright around the x-axis. ω_{K_y} and ω_{K_z} are the angular velocity of the upright around the y-axis respectively z-axis. The velocity of the tire contact point V_{CP} is used to calculate the steering characteristics. This velocity vector is calculated by:

$$V_{CP} = V_{WC} + \omega_K \times r_{CP} \tag{4.8}$$

Where, r_{CP} is the vector pointing from the wheel center WC to the tire contact point CP.
The kingpin inclination σ and caster angle τ, see Chapter 3.2, are calculated using the angular velocity of the upright ω_K in combination with the following equations:

$$\sigma = -\arctan(\frac{\omega_{Ky}}{\omega_{Kz}}) \tag{4.9}$$

$$\tau = -\arctan(\frac{\omega_{Kx}}{\omega_{Kz}}) \tag{4.10}$$

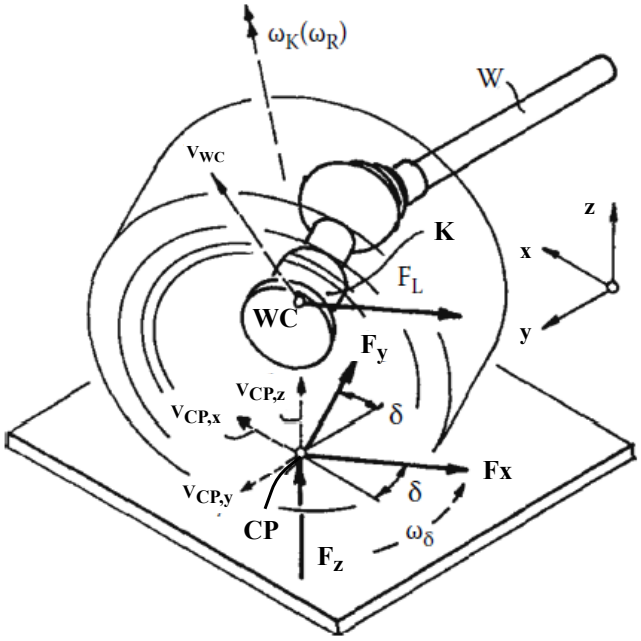


Figure 4.8: Forces and velocities acting on an steered wheel [14].

Using the steering velocity around the tire contact patch ω_δ, the velocity of the wheel center V_{WC} and the velocity of the tire contact point V_{CP}, the remaining steering characteristics are calculated. The scrub radius r_S is derived by

$$r_s = \frac{-(V_{CP,x}\cos(\delta) + V_{CP,y}\sin(\delta))}{\omega_\delta}. \tag{4.11}$$

The wheel-center offset r_C,

$$r_c = \frac{-(V_{WC,x}\cos(\delta) + V_{WC,y}\sin(\delta))}{\omega_\delta}. \tag{4.12}$$

The mechanical trail n,

$$n = \frac{-(V_{CP,x}\sin(\delta) + V_{CP,y}\cos(\delta))}{\omega_\delta}. \tag{4.13}$$

The castor offset at wheel center n_τ,

$$n_\tau = \frac{-(V_{WC,x}\sin(\delta) + V_{WC,y}\cos(\delta))}{\omega_\delta}. \tag{4.14}$$

The wheel-load lever arm p followed by

$$p = -\frac{V_{WC,z}}{\omega_\delta}. \tag{4.15}$$

Bump characteristics

In order to analyze the bump characteristics, the chassis is fixed to the world and the upright is actuated in vertical direction, by prescribing a vertical motion to the wheel pad. It is assumed that the shock absorber can be compressed or extended, which results in one DOF of the upright in vertical direction.

Roll center

The first characteristic which is determined with a vertical movement of the suspension, is the roll center height. The roll center is the point around which the chassis rolls and is determined by the position of the instant centers IC an support angle φ, see Figure 4.9. The front and rear suspensions normally have different roll centers. The imaginary line between the front and rear roll center is known as the roll axis. The roll angle of the chassis while cornering depends on the position of the roll axis. The lateral forces induced a the vehicles center of gravity will be transmitted to a roll motion of the chassis.

By reduction of the distance between the roll axis and the center of gravity, the less the chassis will roll in a corner. The roll center height can be calculated using the following equation:

$$h_{RC} = \left(\frac{t}{2}\right)\left(\frac{V_{CP,y}}{V_{CP,z}}\right) \tag{4.16}$$

with t for the track-width, $V_{CP,y}$ and $V_{CP,z}$ for the velocity of the tire contact patch in y and z-direction. It should be noted that the track-width for front and rear can vary from each other.

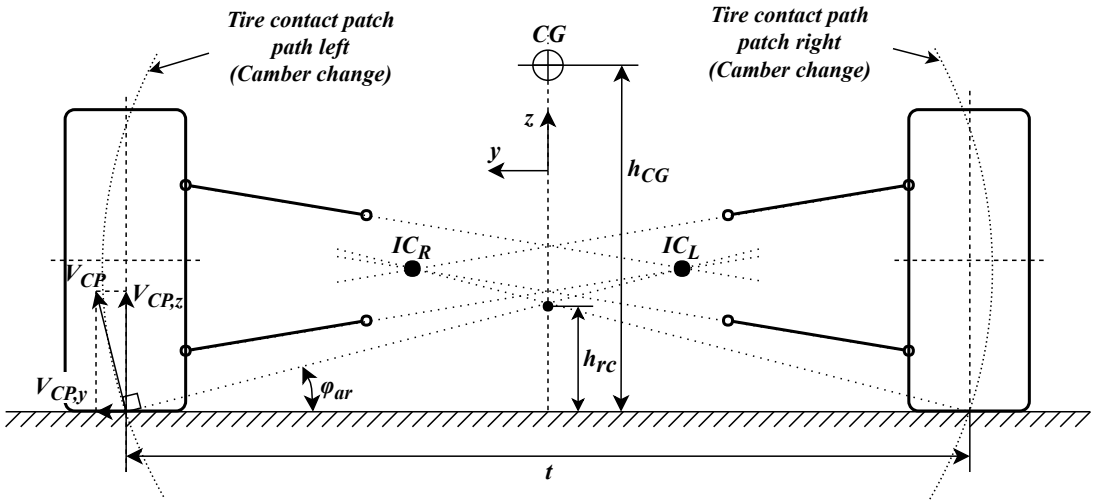


Figure 4.9: Roll-center height and front view support angles.

Support angles

The second characteristics are the support angles. The front view support angle controls the roll motion of the vehicle during cornering and is closely related to the roll center height. The support angle is calculated using the front view instant center IC, see Figure 4.9. The front view support angle can be calculated using the following equation:

$$\tan(\varphi_{ar}) = \frac{dy_{cp}}{dz_{cp}} = \frac{dy_{cp}/dt}{dz_{cp}/dt} = \frac{\dot{y}_{cp}}{\dot{z}_{cp}} = \frac{V_{CP,y}}{V_{CP,z}} \tag{4.17}$$

The side view support angle are determined by the side view instant centers IC_f and IC_f, see figure 4.10. These control the pitch motion of the chassis when accelerating or braking. The instant center IC is also known as the instantaneous center of zero velocity. For a specific motion, the upright appears to rotate around this point. The front side view support angle is calculated by:

$$\tan(\theta_{gf}) = \frac{dx_{cp}}{dz_{cp}} = \frac{dx_{cp}/dt}{dz_{cp}/dt} = \frac{\dot{x}_{cp}}{\dot{z}_{cp}} = \frac{V_{CP,xf}}{V_{CP,zf}} \tag{4.18}$$

And the rear side view support angle by:

$$\tan(\theta_{gr}) = \frac{V_{CP,xr}}{V_{CP,zr}} \tag{4.19}$$

Anti-features

The third characteristic are the anti-features. When a vehicle brakes or accelerates, load transfer will take place. Load transfer is described as the change in the vertical load of the tires. As result of this load transfer, the suspensions will be actuated. During braking the front suspension will be compressed whereas the rear will extend. By this the pitch angle of the vehicle will change. The chassis pitches forward during during braking and pitches backward during acceleration. The force acting at the tire contact patch, during acceleration or braking, are transmitted to the suspension links and push/pull rod. The part which is transmitted trough the push/pull rod is called the elastic part.

The remaining forces are transmitted trough the upper and lower suspension links. Because the springs and dampers are not actuated this is called the direct part. The amount of forces which are transmitted to the suspension links are described by the anti-features [11, 14]. The values for the anti-features are expressed in terms of percentages. A value of 100% means that the longitudinal tire forces are totally transmitted by the suspension links, and not through the spring and damper system. The description of the anti-feature also holds for lateral acceleration of the vehicle.

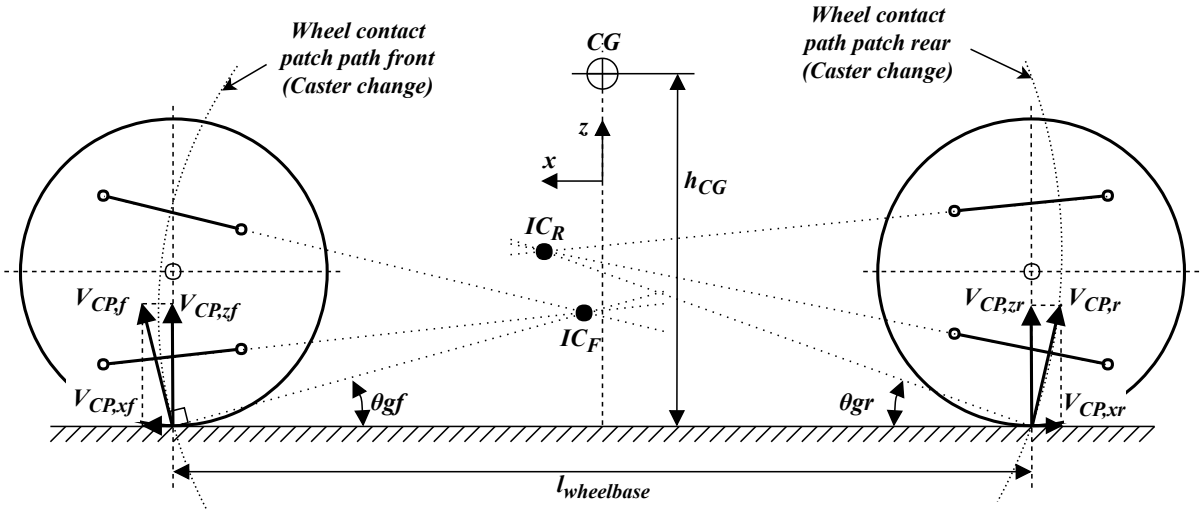


Figure 4.10: Side view support angles.

Using the support angles the anti-features such as anti-roll, anti-dive, anti-rise, anti-squat, anti-lift are determined. The front view support angle φ_{ar} is used to calculate the anti-roll percentage, which relates the amount of lateral force at the tire contact point that is transmitted to the chassis to generate a roll moment. Anti-roll describes the percentage a body will roll during cornering. 100% anti-roll means no roll during cornering. Anti-roll is calculated by

$$anti - roll = \frac{\tan(\varphi_{ar})}{h_{CG}/t} \cdot 100\%. \tag{4.20}$$

with h_{CG} as the vertical distance between the center of gravity of the vehicle and the road surface.

Anti-dive is defined as the amount of longitudinal force directed trough the suspension links of the front suspension during braking, and describes the amount that the front suspension will be compressed. The anti-dive percentage at the front is calculated by:

$$anti - dive = \frac{\tan(\theta_{gf})}{h_{CG}/p_b l_{wheelbase}} * 100\% \tag{4.21}$$

where $l_{wheelbase}$ is described as the wheelbase of the vehicle and p_b the brake force distribution between the front and rear axle.

Anti-rise is defined as the amount of longitudinal force that is transmitted through the links of the rear suspension during braking, and describes the amount that the rear suspension will be extended. The anti-rise on the rear is calculated according by:

$$anti - rise = \frac{\tan(\theta_{gr})}{h_{CG}/(1-p_b) l_{wheelbase}} \cdot 100\% \tag{4.22}$$

A four wheel driven vehicle, like the current URE vehicle, also has an amount of anti-lift at the front and anti-squat at the rear during accelerating. Anti-lift is defined as the amount of longitudinal force that is transmitted through the links of the front suspension during acceleration. Also it describes the amount the front suspension will extend during acceleration. The anti-lift percentage at the front is calculated by:

$$anti - lift = \frac{\tan(\theta_{gf})}{h_{CG}/p_d l_{wheelbase}} \cdot 100\% \tag{4.23}$$

Where p_d the driving torque distribution between the front and rear axle. Anti-squat is defined as the amount of longitudinal force that is transmitted through the links of the rear suspension during acceleration. Also it describes the amount the rear suspension will compress during acceleration. The anti-squat percentage at the rear is calculated by:

$$anti - squat = \frac{\tan(\theta_{gr})}{h_{CG}/(1 - p_d)l_{wheelbase}} \cdot 100\% \tag{4.24}$$

Camber and toe increase

The camber and toe change rates over wheel travel are determined by using the wheel rotation matrix R of (4.3). For the rear suspension only the camber and toe change with vertical motion are crucial. Whereas, at the front also the camber gain over steering angle is taking into consideration.

Tire contact path trajectory

The tire contact path trajectory in x, y and z-direction during vertical suspension travel or steering, are measured in the kinematic suspension model by using a body sensor. The body sensor is connected the point where the tire makes contact with the wheel pad surface.

4.2 Collision check

During the complete motion range of the suspension, the suspension rods should remain clear from collisions. Currently the collision check is done by a visual inspection using the CAD assembly in Siemens NX. However, this is a very time consuming task. Using Siemens NX, every possible position of the suspension needs to be checked by hand. During the kinematic analysis, discussed in Section 4.1, already every possible position of the suspension is simulated. A collision check is implemented in the simulation model and runs together with the kinematic analysis. The most critical parts during collision are the outer case of the motor and the inner rim radius. In figure 4.11 and 4.12 these occurrences of a collision are shown. The exact implementation method will be explained in more detail in the remainder of this section.

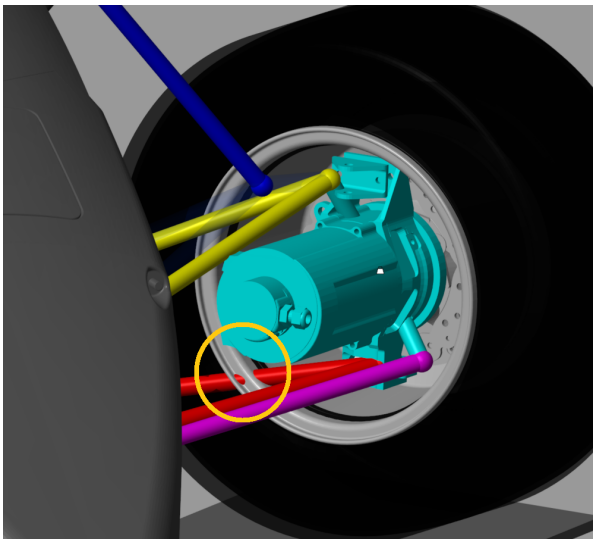


Figure 4.11: Collision with rim (Indicated with yellow circle).

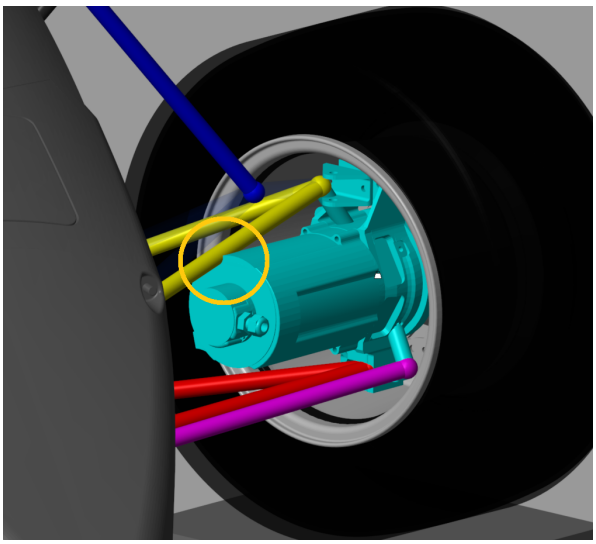


Figure 4.12: Collision with motor (Indicated with yellow circle).

Collision with motor housing

The possibility of having collision with the motor housing is determined by calculating the distance between the middle point of the back-end of the motor V_0 and the closest point on the suspension rod I.

The collision point is the position where the vector, between start and end point of the suspension rods, crosses the reference plane parallel with the back of the motor housing. This distance is determined by using different vectors and the position of the suspension hard-points, see Figure 4.13.

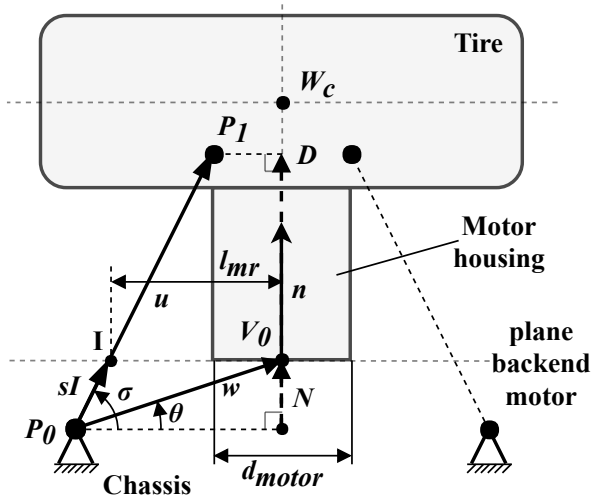


Figure 4.13: Vector notation collision motor.

The vector normal to intersection plane, pointing at wheel center, with V_0 as base is described by:

$$\overline{n} = W_c - V_0 \tag{4.25}$$

where, W_C is the center position of the tire. V_0 is the center point on the back-end of the motor housing. The vector $\overline{u}$ is described as the vector of a suspension link, pointing from ball-joint to ball-joint and is noted by:

$$\overline{u} = P_1 - P_0 \tag{4.26}$$

where, P_0 is related to the hard-point on the chassis and P_1 to the hard-point on the upright. The vector pointing from the chassis hard-point the back-end of the motor is described by:

$$\overline{w} = V_0 - P_0 \tag{4.27}$$

Both vectors $\overline{w}$ and $\overline{u}$ are projected to the normal vector $\overline{n}$. Vector $\overline{w}$ is projected on the normal vector $\overline{n}$, resulting in vector $\overline{N}$, calculated by:

$$\overline{N} = \overline{w} \cdot \overline{n} = \mid \overline{w} \mid cos(\theta) \tag{4.28}$$

Also vector $\overline{u}$ is projected on the normal vector, resulting in vector $\overline{D}$, calculated by:

$$\overline{D} = \overline{u} \cdot \overline{n} = \mid \overline{w} \mid cos(\sigma) \tag{4.29}$$

Using vector $\overline{u}$ and the fraction between vector $\overline{N}$ and $\overline{D}$, the vector $\overline{sI}$ pointing from P_0 to point I is determined. Point I is the position where the vector $\overline{u}$ intersects with the reference plane, parallel with the back of the motor housing. Vector $\overline{sI}$ is calculated according to:

$$\overline{sI} = \frac{\overline{N}}{\overline{D}}\overline{u} \tag{4.30}$$

The position of the intersection point I of the suspension rod with the crossing plane is described by:

$$I = P_0 + \overline{sI} \tag{4.31}$$

Finally, the distance from motor housing to suspension rods is determined by:

$$l_{mr} = \mid V_0 - I \mid \tag{4.32}$$

This distance should not be smaller then the the sum of the motor housing diameter d_{motor} and the suspension rods thickness d_{rod} divided by two. No collision between the back-end of the motor housing and the suspension rods occurs if the following condition is satisfied:

$$l_{mr} \geq \frac{d_{motor} + d_{rod}}{2} \tag{4.33}$$

Collision with rim

For determination of the possibility of having collision with inner side of the rim, the same method as described above is used. But now by calculating the distance between the wheel center, at the inner side of the rim V_0, and the virtual collision point at the suspension rods I. The collision point is the position where the vector, between start and end point of the suspension rods, crosses the reference plane parallel with the inner side of the rim.

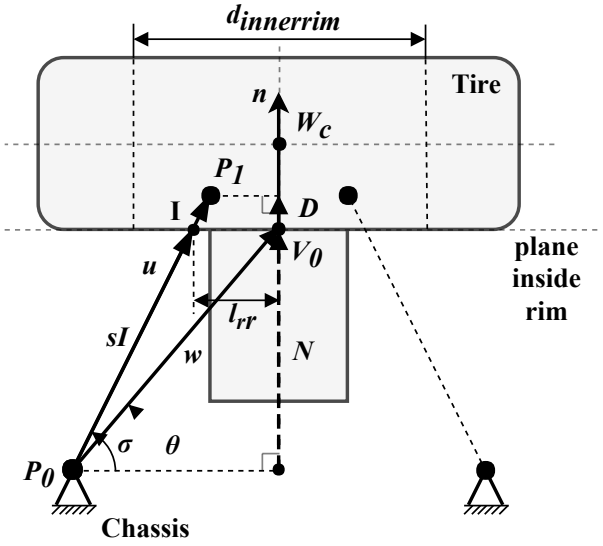


Figure 4.14: Vector notation collision rim.

The distance from the inner side center of the rim to suspension rods is calculated by:

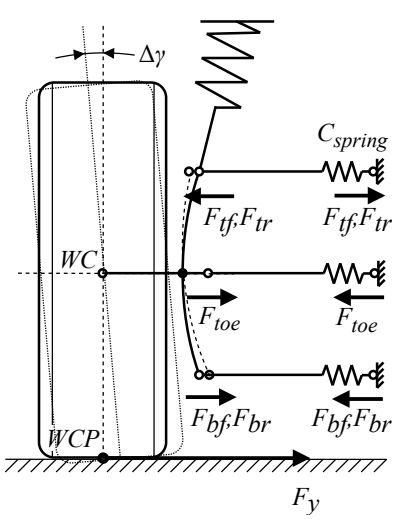
$$l_{rr} = \mid V_0 - I \mid \tag{4.34}$$

This distance should not be larger then the the sum of the inner rim diameter d_{rim} and the suspension rods thickness d_{rod} divided by two. No collision with the inner side of the rim and the suspension rods occurs if the following condition is met:

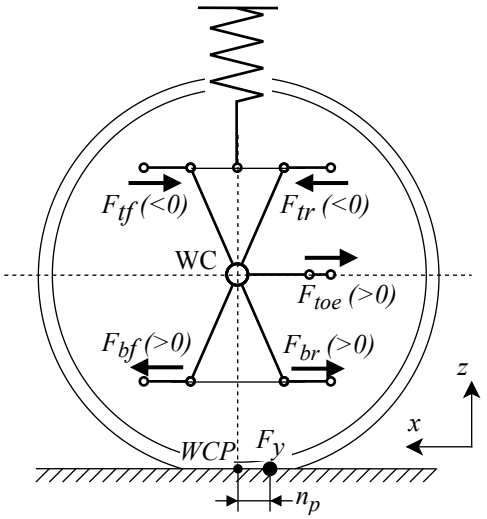
$$l_{rr} \leq \frac{d_{innerrim} + d_{rod}}{2} \tag{4.35}$$

4.3 Mechanical free-play and compliance

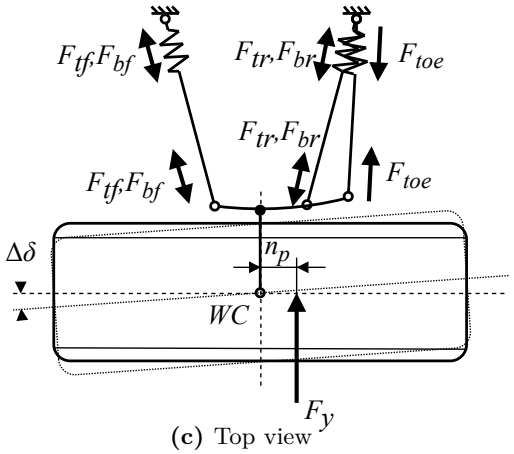
From the literature study of section 3.4, the conclusion is drawn that a the multi-link suspension layout can use compliance in an advantages way. Also in the first multi-link suspension of URE there where some problems with high suspension rod forces due to suspension play. Using this knowledge, the suspension simulation model is extend by adding compliance. The compliance is introduced by mechanical free-play of the ball-joints and an axial stiffness of the suspension rods. In Figure 4.15 a simplified multi-link suspension is loaded with a lateral force. This force is applied at the tire contact patch with a offset in y-direction due to the pneumatic trail t_p. The applied force will be transmitted via the tire and upright to the suspension rods. The suspension rods may be loaded with a tension or compression force. As a result of these forces, mechanical free-play and compliance, the length of the suspension rods will change and suspension kinematics like toe and camber will be affected.



(a) Rear view



(b) Left view



(c) Top view

Figure 4.15: Suspension compliance due to lateral force F_y.

In the example of Figure 4.15 the toe rod, lower front and lower rear suspension rods are loaded with a compression force: F_{toe}, F_{bf} and F_{br}. Whereas, the upper front and rear suspension rods are loaded with a tension force: F_{tf} and F_{tr}. This causes a change in camber angle $\Delta\gamma$. Furthermore, the lower rear link is loaded by a considerably greater reaction force F_{br} than the lower front link F_{bf}, which results in a toe change $\Delta\delta$.

4.3.1 Mechanical free-play

The free-play is implemented by decoupling the translational degree of freedom, between the ball-joint and suspension link, in the z-axis of the suspension rod. In case the displacement is larger than the determined free-play region, a reaction force $F_{react.}$ is applied in the axial direction of the suspension links.

The suspension simulation model consist of three possible models for the mechanical free-play. However, only the last model can be used for all type of compliance measurement, whereas the first two only can be used for very small free-play values and forces. Only the last model is used and will be discussed in this section. The explanation of the other two models can be found in Appendix D. The model which is currently implemented is shown in the Figure 4.16. In this model it is assumed that the ball-joint center can only move in axial-direction of the suspension rod.

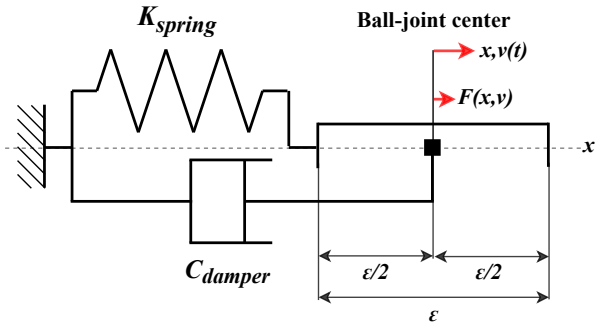


Figure 4.16: Schematic representation of the free-play model.

In the region from $-\varepsilon/2$ till $\varepsilon/2$ there is no contact between the ball-joint center and hard stops. In this region only a reaction force is modeled by the damping of the ball-joint. This reaction forces is depended on the velocity. When the suspension is in motion the ball-joint center will travel to one of the two hard stops. At the point of hitting one of the hard stop, a second reaction force is applied. This force is modeled by a spring, which is related to the axial stiffness of the suspension rods and ball-joint. Combination of these states and forces, the reaction force $F_{react.}$ is described by:

$$F_{react.}(x,t) = \begin{cases} K_{spring} * (x - \varepsilon/2) + C_{damper} * v & \text{if } x \leq \varepsilon/2 \\ C_{damper} * v & \text{if } \varepsilon/2 < x < \varepsilon/2 \\ K_{spring} * (x + \varepsilon) + C_{damper} * v & \text{if } x \geq \varepsilon/2 \end{cases}$$

where, ε describes the mechanical free-play region in the ball-joint, x is the displacement of the ball-joint center, v is the velocity of the ball-joint center, D_{damper} corresponds to the damper constant of the ball-joint, and K_{spring} corresponds to the stiffness of the suspension rod and ball-joint.

The advantage of placing the damper over the total motion region is that the forces will not increase instantaneously by hitting the hard-stop, but slowly increase until the force is in equilibrium. At this point the reaction forces is equal to the force in the suspension rods. In Appendix E the results are plotted for different hard-stop stiffness and damping coefficients. Using the different plots it can be noticed that increasing the damper coefficient will result in longer simulation times. Eventually, the suspension motion is damped out to its equilibrium position, as result of the change in length of the suspension rods due to the free-play.

Based on the simulation results and the simulation time, a damping coefficient is selected. By having a to small damping coefficient, the reaction forces are insufficient to guide the suspension to a steady state position. Also the actual time to completely damp out the suspension increases. By increasing the damping coefficient, the time it takes for the suspension to move in its steady state reduces. However, this leads to increased simulation times. By performing different simulation and analyzing the actual simulation time and resulting time it takes for steady state position, the selected damping coefficient is:

$$C_{damper} = 15e4[Ns/m] \tag{4.36}$$

The free-play has already been investigated in the past by P. van Amelsfort [27]. The free-play of different ball-joints has been measured. The ball-joint where exposed to various sinusoidal forces. This force was used to simulate the loads on a ball-joint and analyze the mechanical free-play before and after the load cycle. Based on the measurements and the fact that still the same ball-joints are used today, the the mechanical free-play of a ball-joint equals:

$$\varepsilon_{balljoint} = 0.5e^{-3}[m] \tag{4.37}$$

4.3.2 Axial stiffness suspension rods

The stiffness of the rods in axial direction are implemented in the simulation model. The axial stiffness of a rod can be described as a linear spring, as illustrated in Figure 4.17.

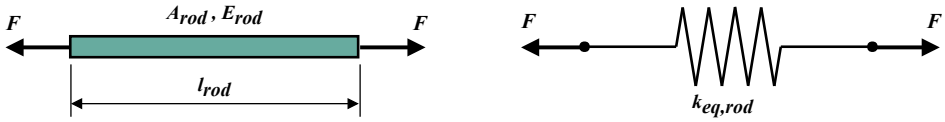


Figure 4.17: Schematic representation of modeling a rod as spring.

The axial stiffness of a rod is defined as:

$$k_{eq,rod} = \frac{A_{rod} \cdot E_{rod}}{l_{rod}} = \pm 1.20e^{8}[N/m] \tag{4.38}$$

where, A_{rod} is the cross-sectional area $[m^2]$, E_{rod} is the tensile modulus $[N/m^2]$, and l_{rod} the length of the rod $[m]$. Because the suspension rods are made of carbon tubes and aluminum inserts at the ends, the cross-sectional area and tensile modulus of carbon rod are used.

4.4 Steering torque

The steering torque at the handwheel is derived by converting the moments acting about the steering axis d to a resulting torque at the steering wheel axis. The method proposed in [14] is used to to calculate the moment around the king pin axis M_{kp}. The moment around the kingpin axis is a result from the moment around the vertical world axis at point D, where the steering axis intersect with the ground plane. The moment around the king pin axis is calculated by:

$$M_{kp} = M_D \cdot cos(\lambda) \tag{4.39}$$

where λ is noted as the total king pin tilt angle, by combination of the king pin axis σ and caster angle τ. The total king pin tilt is derived according to:

$$\lambda = \tan^{-1}\left(\sqrt{\tan(\sigma)^2 + \tan(\tau)^2}\right) \tag{4.40}$$

In Figure 4.18 the definitions of steering system and torque related variables are illustrated. The contribution to the steering torque as a result of longitudinal forces, at the tire contact patch, is dependent on the scrub radius r_s and is described by:

$$M_{r_s} = -cos(\lambda) \cdot F_x \cdot r_s \tag{4.41}$$

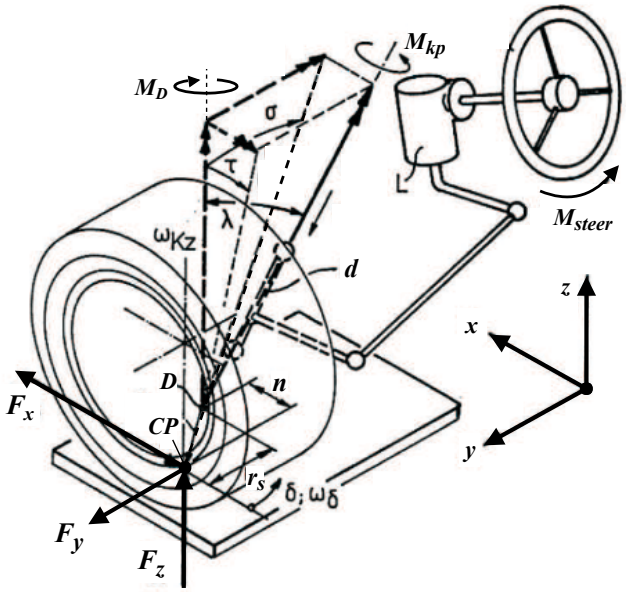


Figure 4.18: Definitions of steering system and torque related variables [14].

The contribution to the steering torque due to a lateral force is dependent on the mechanical n and pneumatic trail t_p, defined as

$$M_n = -cos(\lambda) \cdot F_y \cdot (n + t_p). \tag{4.42}$$

Also the vertical force will generate a moment around the steering axis For this the wheel load lever arm p is used. The wheel load lever arm transfers the vertical tire load to a moment around the steering axis, this moment is derived by:

$$M_p = -F_{z,l} \cdot p. \tag{4.43}$$

The total steering moment about the steering or king pin axis is the sum of the above moments and is calculated by:

$$M_{kp} = M_{r_s} + M_n + M_p \tag{4.44}$$

Using (4.44) the total moment about the steering axis is a function of the suspension kinematic properties and the applied force at the tire contact point. The steering moment at the steering wheel is determined using the method proposed in [28]. In Figure 4.19 the lay-out of the steering system is illustrated.

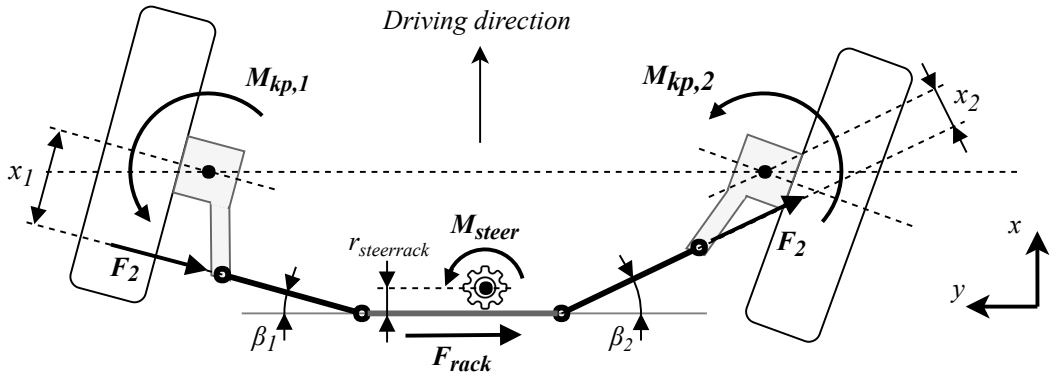


Figure 4.19: Steering system linkage arrangement

The resulting force at the steering rack mechanism F_{rack}, as indicated in Figure 4.19, can be calculated by:

$$F_{rack} = F_1 cos(\beta_1) + F_2 cos(\beta_2) = \frac{M_{kp,1}}{x_1} cos(\beta_1) + \frac{M_{kp,2}}{x_2} cos(\beta_2), \tag{4.45}$$

where F_1 and F_2 are the axial forces in the two steering rods. $M_{kp,1}$ and $M_{kp,2}$ are the moment around the left and right steering axis. The shortest distance between the steering axis and the extended steering rod link is defined by x_1 and x_2. The angle between the steering rack and the steering rods are β_1 and β_2.

Finally, the force acting on the steering rack is transformed into to a moment felt at steering wheel. The steering wheel moment M_{steer} is calculated by:

$$M_{steer} = F_{rack} \cdot r_{steerrack}, \tag{4.46}$$

where, $r_{steerrack}$ equals the pinion pitch circle radius of the steering mechanism. By substitution of (4.44) into (4.45) and (4.46) the steer wheel moment can be calculated for a specific load case consisting of applied forces at the wheel contact patches. The steering moment will be further evaluated in section 6.5.

4.5 Summary

Since a multi-link suspension is a spatial mechanism and the appearance of a virtual steer axis, it is impossible to analyze the kinematics of such a suspension layout with simple mathematical equations or by manual design iterations. For this purpose a multibody model in MATLAB has been developed. The developed suspension model is able of calculating and plotting all required suspension kinematic characteristics for a given suspension layout and geometry. The kinematics are determined by the input parameters, such as initial hard-points and setup parameters, like toe and camber. A built-in collision check is implemented, which can be used to detect critical collision points for the complete motion range of the suspension. Additionally, the suspension simulation model includes mechanical free-play and compliance. This is used to analyze the change in toe and camber angle for different load cases. Finally, a model for calculating the steering torque is developed. Based on the forces acting on the tire contact patch, as result of lateral acceleration, the needed steering torque can be calculated.

5. Suspension design considerations

In this chapter the new multi-link suspension and its design considerations will be discussed. The evaluation of the kinematic characteristics and comparison with the existing double wishbone suspension, will be discussed in Chapter 6.

5.1 Requirements

The suspension should satisfy certain requirements. These requirements will limit the boundary conditions and result in a suspension design with the desired characteristics. Based on these requirements a new suspension is designed.

5.1.1 Vehicle parameters

The suspension is designed for a specific vehicle layout. This vehicle layout is based on the current URE13 main dimensions. The vehicle parameters can be found in Table 5.1. These parameters will determine the overall dimensions of the multi-link suspension.

Table 5.1: Vehicle parameter requirements

Weight	195 kg (without driver)
Weight distribution front/rear	50/50
CoG height	0.28 m
Wheelbase	1.553 m
Track front	1.180 m
Track rear	1.139 m
Drive-line	4WD in-wheel electric motor
Tire size	205/50R10
Rim size	10 $inch$
Motor dimensions	Length from wheel center 145 mm and diameter 102 mm

5.1.2 Competition restrictions

The suspension design should also satisfy with the rules given by Formula SAE [29]. The suspension design is partially restricted by these rules. Neglecting these rules means disqualification and no participation in the Formula Student events. The most important FSAE rules relevant to the suspension design are listed below.

- The vehicle must be equipped with a fully operational front and rear suspension system including shock absorbers and a usable wheel travel of at least 50 mm with the driver seated (25 mm jounce and 25 mm rebound).
- The minimum static ground clearance of any portion of the vehicle, other than the tires, including a driver, must be a minimum of 30 mm.
- All suspension mounting points must be visible for technical inspection, either by direct view or by removing covers.
- The steering rod must be mechanically connected to the wheels. Steering systems using cables or belts for actuation are prohibited.
- The steering system must have positive steering stops that prevent the steering linkages from locking up. The stops must be placed on the rack and must prevent the tires and rims from contacting any other parts.
- Allowable steering system free play is limited to a total of 7 $deg.$ measured at the steering wheel.

- Rear wheel steering, which can be electrically actuated, is permitted if mechanical stops limit the steering range of the rear wheels to a maximum of 6 *deg.*. This must be demonstrated with a driver in the vehicle and the team must provide the equipment to determine the steering angle range to be verified at technical inspection.

- The vehicle must have a wheelbase of at least 1525 *mm*.

5.1.3 Suspension kinematics

To have a clear comparison between the current double wishbone and the new multi-link suspension design, the suspension kinematics are primarily kept identical unless specified otherwise. The following kinematic characteristics will differ for new current suspension.

Toe
The toe angle effects the performance of the vehicle during different driving maneuvers. Due to the mechanical free-play and compliance the toe angle will change. Therefore, it is important to take suspension deformation into consideration to optimize the toe change under different driving maneuvers, such as braking or cornering. One problem of the current suspension, is the short arm between the upright and steering rod or toe rod. The combination of this short arm and the mechanical free-play in ball-joints results in large toe change as result of longitudinal and lateral forces. The total toe change should be minimized to improve vehicle handling.

Scrub radius
The scrub radius should be minimized at the front axle in order to reduce the effect of torque-vectoring on steering torque. Torque vectoring will introduce different longitudinal forces at the four wheels of the vehicle. The steering torque is induced by the difference in driving torque, applied at the left and right tire. By reducing the scrub radius r_s , the resulting moment at steering wheel will be minimized, see (4.41).

King pin inclination angle
The current king pin inclination angle at the front is 13 *deg.*, which is large compared to other formula student vehicles. Due to in-wheel motor packaging constraints and current scrub radius of 4 *mm*, the king pin angle can not be reduced for the current double wishbone suspension layout. The king pin inclination angle should be reduced to have a prescribed camber gain, reduction of the distance from steering axis to wheel center and result in optimization of jacking effects. The king pin angle should be reduced to around 6-7 *deg.* for the new multi-link suspension.

Camber
Based on the tire measurements and the thesis reports of W. Lamers [4] and R. van Hoek [9] it can be concluded that the camber angle on the outside wheel needs to be minimized during cornering. By reducing the camber angle of the heavily loaded outside tire the lateral performance will increase during cornering.

Bump steer
Bump steer is the change of the toe angle due to suspension travel. Bump steer can in some cases have positive effect on vehicle stability. However, the effects and optimal amount of bump steer are hard to determine. The choice is made to eliminate or minimize bump steer over the entire vertical motion range of the suspension.

Steering angle
Because the track-width and wheelbase will not change the maximum steering angle is determined on forehand. The steering angle of the inner wheel should be 25 *deg.* minimum and the outer wheel 19 *deg.* minimum, in order to achieve a corner radius of 4.5 *m*. These angles should be possible without any suspension collision.

5.1.4 Setup requirements

In order to allow different suspension setups for testing, the suspension has some setup requirements. These requirements are based on the current suspension adjustment possibilities.

- The desired static camber angle should be -3 *deg.* at the front and -2 *deg.* at the rear. It should be adjustable between 0 and -4 *deg.* for the front axle and between 0 and -3 for the rear axle .
- The desired toe range should be approximately 1 *deg.* out front and 0.5 *deg.* in rear. It should be adjustable between -2 and 2 *deg.* for both front and rear axle.

5.2 Design process

A schematic representation of this kinematic analyses and design process is given in Figure 5.1. During the design process the kinematics are analyzed first. This is done by prescribing a steer and bump motion to the kinematic model. Based on the steering motion the steer kinematics are determined and the change over steering range is analyzed: KPI, caster, scrub radius, mechanical trail and wheel center offset. The rear axle will not be steered, however for calculating the steer kinematics also a small steering motion for the rear suspension is simulated. Subsequently, the bump kinematics are checked. These bump kinematics are camber and toe gain, anti effects, roll center height and inclination angle over body roll.

During the kinematic analysis the complete motion range of the suspension is simulated and a collision check is performed for these conditions. In case of unsatisfactory kinematic behavior or the occurrence of a collision, the initial position of the hard-points and layout should be adjusted and the analysis is performed again.

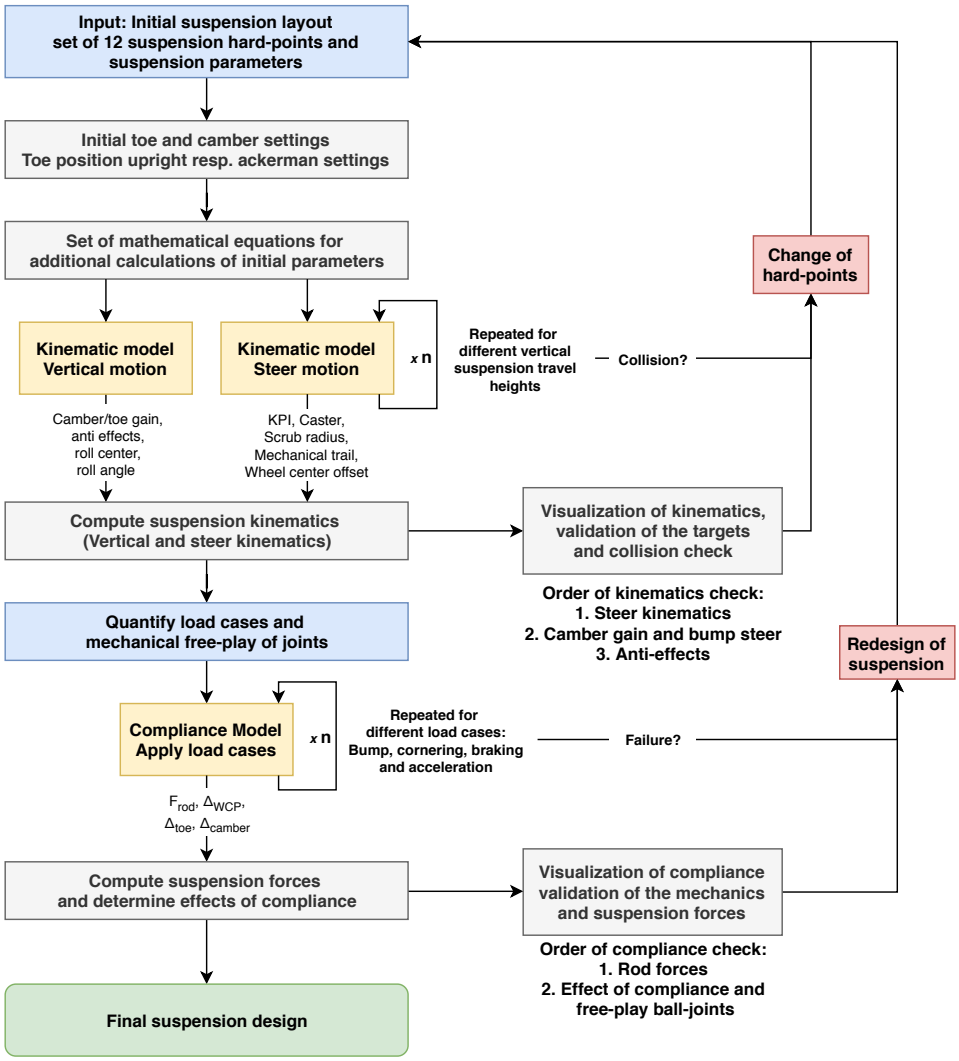


Figure 5.1: Design process flowchart using the kinematic suspension model

In a second step, after the kinematics targets are met, the influence of compliance and mechanical free-play as result of applying forces at the tire contact patch is analyzed. A force in x, y and z-axis is applied and the resulting camber and toe changes are evaluated. During this suspension compliance simulation, the spring/damper is taken as incompressible. Furthermore the wheel is fixed to the upright, so no rotational degree of freedom in roll motion. Also the suspension height is at initial position, thus zero suspension travel. For the compliance simulations three load cases defined: bump, braking and cornering.

The forces are applied at the tire contact point. However, the lateral forces in y-direction of the tire do not act at the center of the contact patch, but at a distance behind the nominal contact patch as result of the pneumatic trail t_p. Based on the compliance analysis the choice can be made to change the hard-points positions or layout. However, due to the change in hard-points also the overall suspension kinematics will change. The complete design process should be performed from the start and the kinematics should again be analyzed.

5.3 Rear suspension design

The design of the rear suspension assumes that the current monocoque will be used again. Using the current layout of the monocoque the possible hard-point positions on the chassis are determined. Most hard-points on the chassis can only be moved in x and z-direction, because the y-axis is limited by the side walls of the monocoque and the height of the suspension brackets. The position of hard-points on the upright are almost endless, although the position of the rim, brake disk, brake caliper and motor are fixed and constrain the available space for positioning of the hard-points.

The final design, as result of the iterative process, is given in Figure 5.3. In the figure also the suspension brackets, representing the current hard-points are shown, in order to visualize the change in hard-point positions.

The changes to the current double wishbone layout are:

1. The upper and lower hard-points on the upright are separated in order to achieve a multi-link layout. The distance between these two new points needs to be adequate, for compliance and ease of assembly. See figure 5.2.

2. The front hard-points on the upright are placed more inside the rim, compared to the rear hard-points, this is done in order to reduce the possibility of collapsing effect as result of suspension compliance or toe settings. Collapsing will occur when a hard-point moves over center, with respect to other suspension links.

3. The front bottom suspension rod 5 is extended to the front, resulting in a large angle between the suspension rod and monocoque. This larger angle will result in increased rotation stiffness of the upright and rod forces will be reduced. Also the rear upper suspension rod 2 is placed under an angle in order to reduce suspension forces and compliance effects.

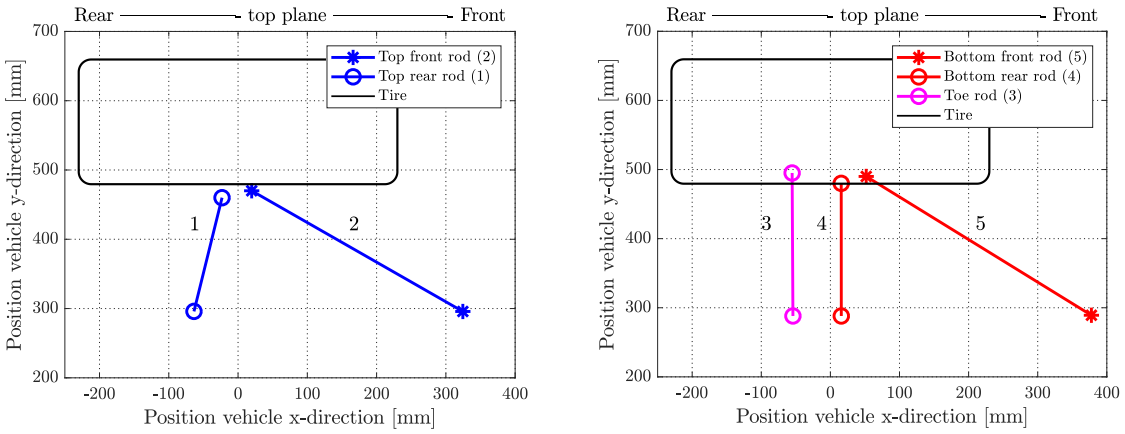


Figure 5.2: Top view rear suspension, top and bottom plane

4. The toe rod 3 is precisely orientation in order to reduce bump steer and toe compliance. For this the toe rod is placed almost parallel to the bottom rear suspension rod 4, as shown in figure 5.2. A parallelogram is created and toe compliance is almost reduced to zero, this will be discussed further in Chapter 6.

5. The mechanical trail is increased. This results in better in-wheel packing for the toe link position on the upright. The trailing arm between the steering axis and toe link is increased. By increasing this distance the change in toe angle as result of mechanical free-play and compliance are reduced. Increasing the mechanical trail will also have positive effect on the force distribution in the individual suspension rods.

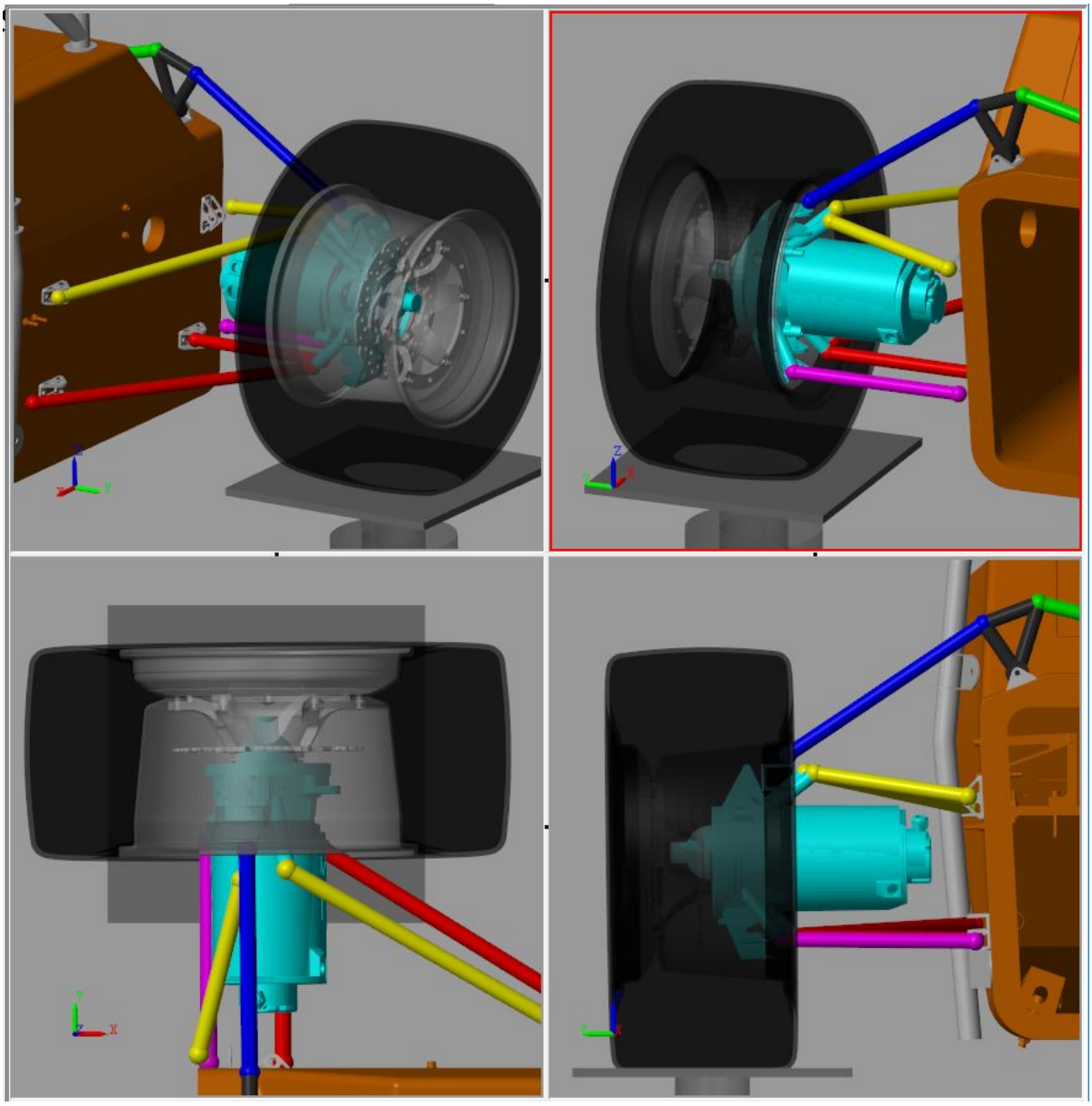


Figure 5.3: Final rear multi-link suspension design.

5.4 Front suspension design

For the front suspension the location of the hard-points is based on the current monocoque. Again, most hard-points on the chassis can only be moved in one plane. The upper two hard-points can be moved along the x and z-axis, because the y-axis is limited by the side walls of the monocoque. The bottom two hard-points can be moved in x and y-axis, because the z-axis is limited by the bottom wall of the monocoque. The position of the steering rack can not be changed because of the space available and competition regulations. Again, the hard-point positions on the upright are almost endless. Likewise on the rear, the rim, brake disk, brake caliper and motor dimensions are given and constrain the possible space for the hard-point position. One important aspect to consider is the steering motion of the front wheels. Due to the steering motion, the available space for positioning of hard-points will be limited because of the possible collision between the suspension rods, motor housing and inner rim radius.

The same kinematic characteristics of the current vehicle is aimed for. The only main difference is related to the kingpin inclination angle. For URE13 the kingpin inclination angle is 13 *deg*.. Using the multi-link layout a king pin angle of 5.9 *deg*. is achieved.

The final design is a result of an iterative process, shown in Figure 5.5. Again the brackets of current hard-points are showed, in order to visualize the change in hard-point location.

The changed compared to the current front double wishbone layout are:

1. The upper and lower hard-points on the upright of the URE13 are separated in order to achieve a multi-link layout. The distance between these two new points need to be adequate for compliance and assembly reasons, as shown in Figure 5.4.

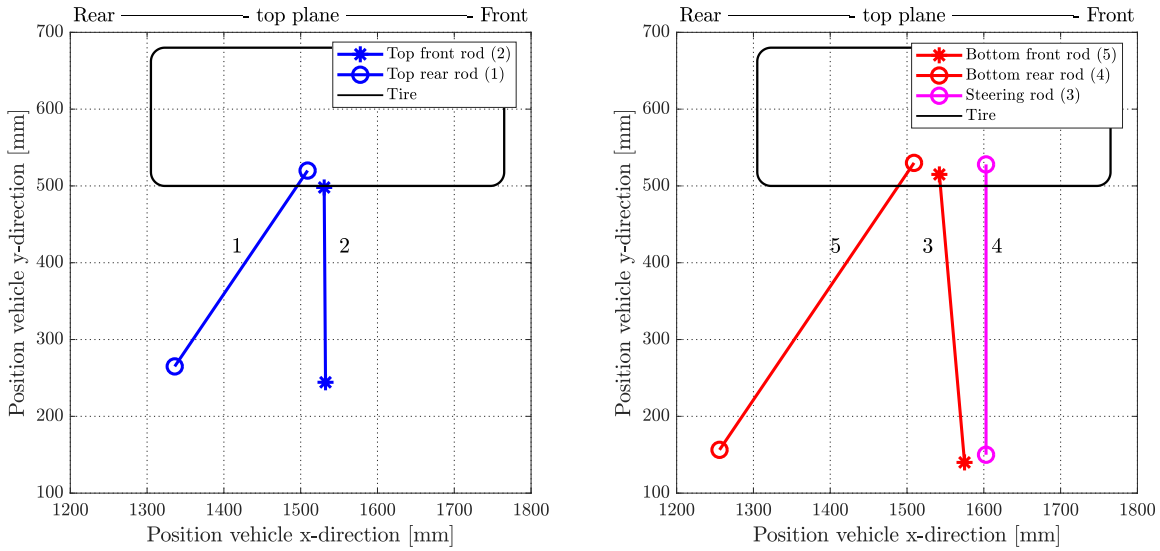


Figure 5.4: Top view front suspension, top and bottom plane

2. The rear hard-points on the upright are placed more inside the wheel, this is done in order to reduce compliance due to ball-joint free-play, variation of the steering axis over steering motion and the possibility of collapsing effect.

3. The rear top and bottom suspension rods 1 and 4, are extended more to the rear, resulting in an increase angle between the suspension rod and monocoque. This will result in increased rotation stiffness of the upright with respect to the chassis and rod forces will be reduced. The brackets are shifted to the position where the front roll-hoop is located. At this point the monocoque has the highest stiffness and overall suspension compliance will be reduced.

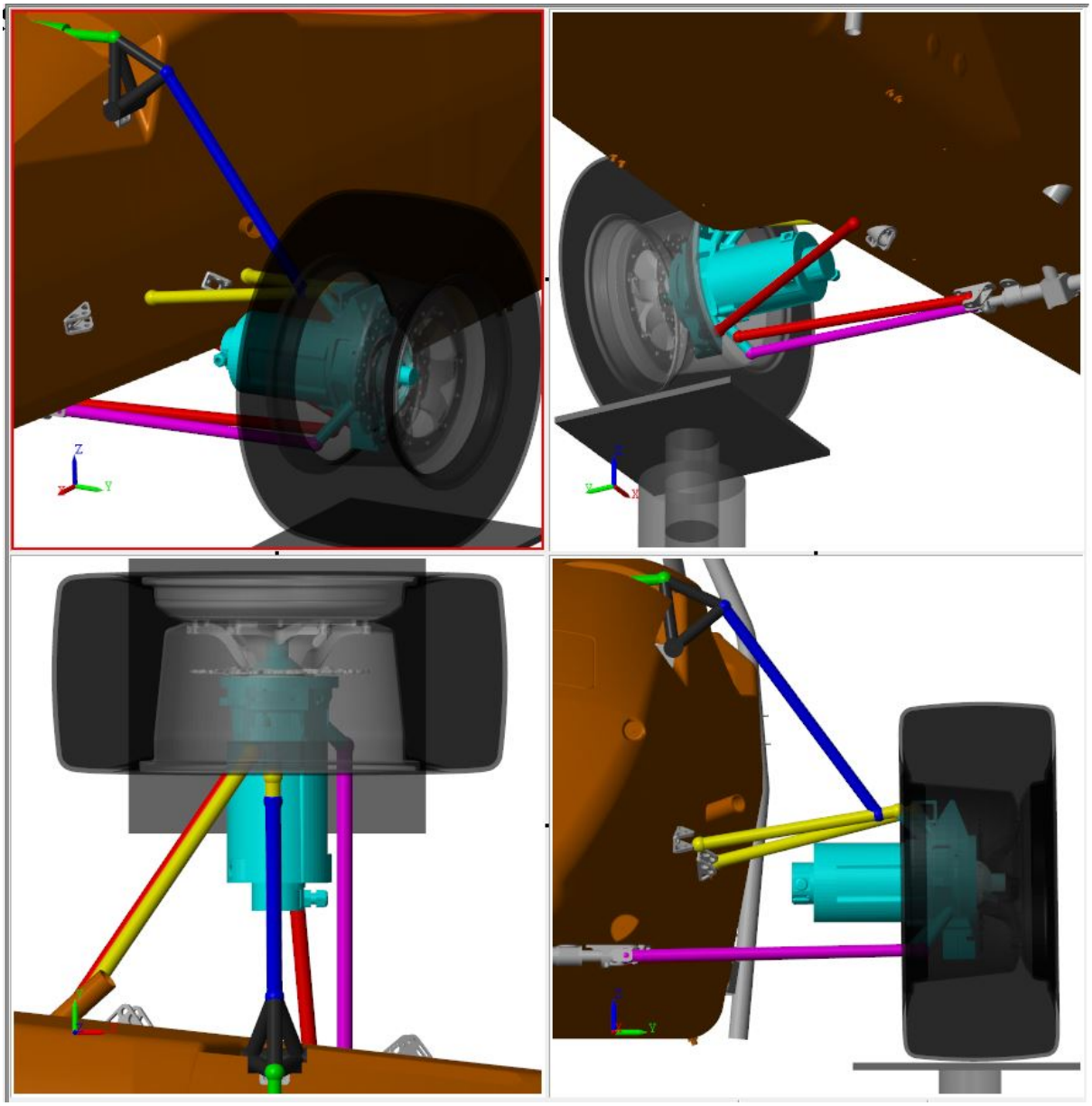


Figure 5.5: Final front multi-link suspension design.

4. The steering rod 3 is placed perpendicular with the monocoque in order to reduce toe compliance. Also the toe hard-point on the upright is moved to the front and the distance between steering axes and toe hard-point is increased by a factor of two. For a double wishbone this can not be achieve, because of collision with the inner rim. Using a multi-link the toe hard-point on the upright can be placed more outside of the rim without affecting the kinematics and toe gain. As result, a larger steering angle can be reached without having collisions.

The position of the push-rod needed some special considerations, because mounting the push-rod directly on the upright will introduce some unwanted effects on the steering torque. The axial forces of the push-rod will be transformed to a moment around the steering axis. To solve this, the push-rod is connected to the top front suspension rod, resulting in no unwanted and non-linear steering torques induced by the push-rod.

A second advantage of mounting the push-rod on the top front suspension rod is avoiding collisions. The push-rod needs to be placed in line with the vertical steering axis in order to reduce the induced steering torque by the push-rod during steering. Based on the current suspension layout and kinematics it is not possible to position the push-rod on the upright in line with the virtual steering axis, without any collision with the rim.

The push-rod is connected on the top front suspension rod. However, for this design an extra bracket or construction is needed. This extra construction is needed to mount the push-rod on the suspension rod and withstand the bending moment. The connection of the push-rod should be constructed like a fork connection. An example is shown in Figure 5.6.

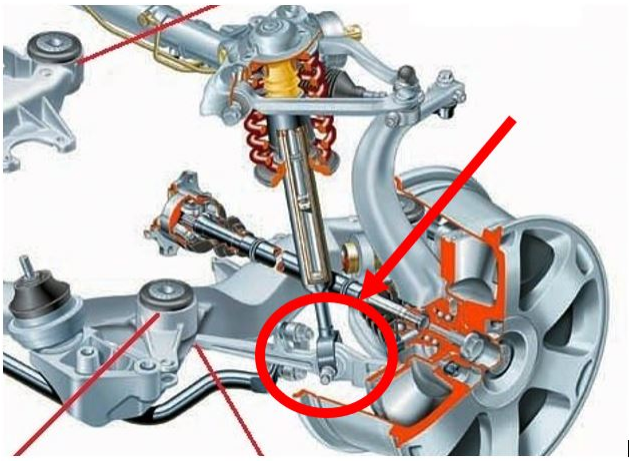


Figure 5.6: Push-rod connection concept front (AUDI S6) [30]

A small disadvantage of mounting the push-rod on the top front suspension rod, is that the connection point on the push-rod will slightly vary in longitudinal direction during steering. This will result in the push-rod not being exactly in plane with the rocker system, resulting in some bending moments on the rocker mechanism. The rocket system should be redesigned in order to withstand these side forces.

5.5 Suspension adjustments possibilities

Camber and toe adjustment

It is required to adjust some suspension settings for tire alignment purposes and to change the vehicle behavior. The allowable adjustments are the static camber angle and the static toe angle. The static camber angle is changed by increasing the distance between the upper two connection hard-points and the upright. This is done by placing shims between brackets and upright construction. For the top two connection points a proper bracket must be designed, on which both hard-points are fixed. By placing shims between the upright and bracket, the camber angle can be adjusted without affecting the kingpin inclination angle.

The static toe angle can be adjusted in the same way as it is done for the URE13, by changing the length of the toe or steering rod. This length is also changed by use of shims. Adjusting this setting will have minimal influence on the other suspension kinematics.

An iterative loop has been implemented in the suspension model for creating a setup table. Based on changes in the number of shims, the static toe and camber angle are determined. The resulting static camber and toe angle for a set of shims are listed in a table. In Appendix F the setup tables for the rear suspension are illustrated and also the way of using them is described. Adjustments of camber and toe are slightly depended of each other. By using the table a quick and easy overview of the effects on number of shims is given.

Ride height adjustment

The ride height can easily be changed by changing the length of the push-rod. In the new multi-link design and current double wishbone suspension this is achieved by adding or removing shims.

Anti-effects adjustment

The anti-effects can be changed by moving the hard-point on the chassis of the upper rear suspension rod for the front and the upper front suspension rod for the rear. By moving this hard-point, the plane resulting by the two upper suspension rods will be tilted. By tilting this plane the anti-effects will change. To achieve this, a bracket will be designed with multiple holes for selecting different anti-effects. This bracket need to be designed in such a way that it will not change other kinematics or only in a negligible amount. This can be controlled by checking the change in kinematic characteristics, for the adjusted hard-point position, using the suspension simulation model.

5.6 Summary

Based on the requirements a multi-link design for the front and rear suspension has been created. The design of the multi-link suspension is obtained by an iterative optimization process. For every change in hard-points location or changes in the overall layout the resulting kinematics are analyzed until the requirements are satisfied. All considerations and improvements compared to the current double wishbone suspension are described. Only the final design is evaluated, the in between iterative steps are not treated. Furthermore the adjustment possibilities of the multi-link design are discussed. The suspension properties which can be adjusted are initial toe and camber angles, ride height, and anti-effects.

6. Design evaluation

The multi-link suspension is compared to the old double wishbone in order to highlight the performance improvements and substantiate the choice for a multi-link suspension. Based on the kinematic analyses for vertical wheel travel, steering motion and compliance, a design evaluation is given. The final kinematics characteristics of the multi-link are compared to the double wishbone of the URE13 and can be found in Appendix G and H. The most crucial changes and improvement will be discussed in this chapter. Al figures in this chapter are generated by use of the MATLAB simulation model.

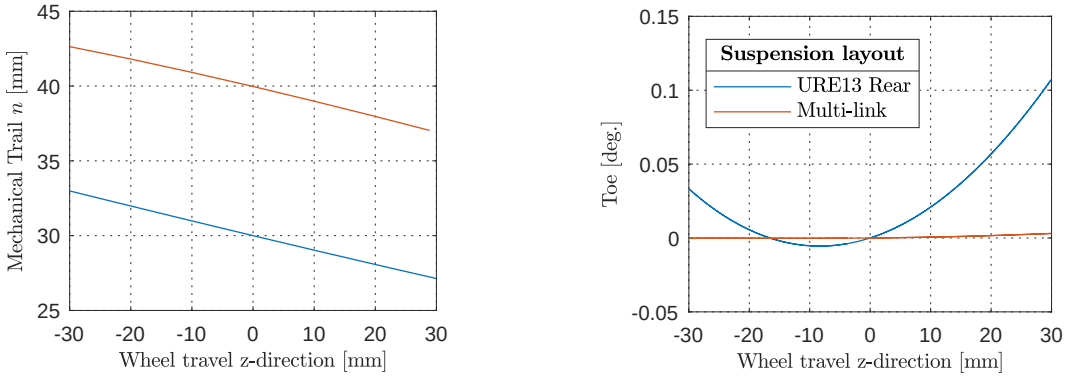
6.1 Rear suspension kinematics

In this section the changes of the kinematic characteristics of the rear suspension will be discussed. As shown in Table 6.1 most kinematics are identical, the only changes are the mechanical trail and toe change with wheel travel. For the rear suspension only the dependency on vertical motion of the suspension is analyzed.

Table 6.1: Multi-link and URE13 rear suspension characteristics

Parameter	**URE13 DW**	**Multi-link**	**unit**
Track width	1139	1139	$[mm]$
KPI angle	7.0	7.0	$[deg.]$
Scrub radius	40.0	39.9	$[mm]$
Caster angle	9.9	9.9	$[deg.]$
Mechanical trail	30.0	40.0	$[mm]$
Roll center height	55.0	54.0	$[mm]$
Wheel center offset	68.2	68.2	$[mm]$
Camber gain	0.43	0.42	$[deg./deg.$ roll$]$
Camber increase	0.44	0.44	$[deg./10\ mm$ travel$]$
Toe increase	0.03	0.0	$[deg./10\ mm$ travel$]$
Anti-rise	36.7	36.7	[%]
Anti-squat	55.0	55.0	[%]

In Figure 6.1a the mechanical trail is shown for both the new multi-link and URE13 suspension layout. The mechanical trail is increased by moving the wheel center of the tire 10 mm backwards on the upright. The main argument for increasing the mechanical trail is to create more space for positioning of the toe rod on the upright. Also by increasing the mechanical trail, the force distribution over the suspensions links will be improved. The forces as result of the lateral force will be divided more over the two bottom suspension links. In order the have the same wheelbase the complete suspension, including the damper and rocker assembly, is moved 10 mm to the front of the chassis. The wheelbase is kept identical in order to maintain the same static axle loads.



(a) Mechanical trail

(b) Bumpsteer

Figure 6.1: Kinematic changes rear suspension

The second change of the rear suspension kinematics is the reduction of bump steer or toe change with vertical travel. This reduction is obtained by careful positioning of the steer-rod mounting point on the upright. As illustrated in Figure 6.1b bump steer is almost eliminated. The effects on the resulting compliance and suspension forces for the rear suspension will be discussed in detail in section 6.4.

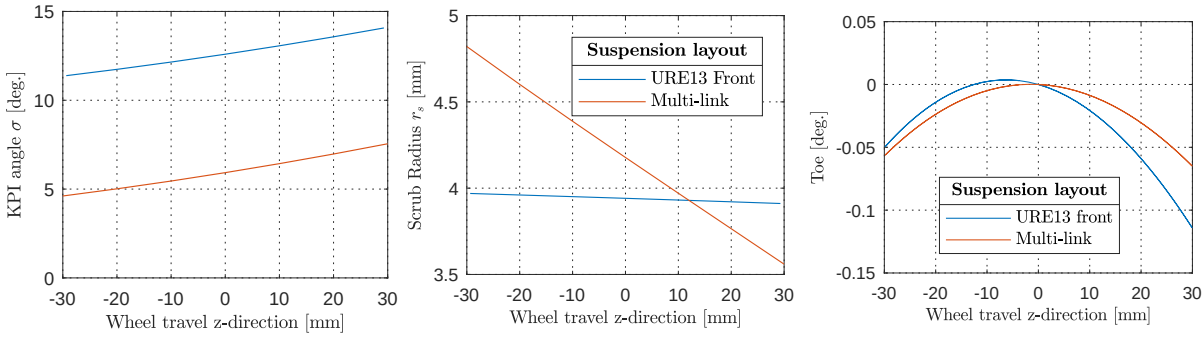
6.2 Front suspension kinematics

By evaluating the suspension kinematics of the front suspension, the steering motion should also be taken into account. The steering axis of a multi-link suspension is virtual and changes during steering motion. This will result in an increased dependency of the suspension kinematics with steering motion. In Table 6.2 the changes for the new front multi-link suspension compared to the current double wishbone are listed. Most characteristics are unchanged. Changes can be found in the KPI angle, wheel center offset and toe change/increase.

Table 6.2: Multi-link and URE13 front suspension characteristics

Parameter	URE13 DW	Multi-link	unit
Wheelbase	1180	1180	[*mm*]
KPI angle	13.0	5.9	[*deg.*]
Scrub radius	3.9	4.2	[*mm*]
Caster angle	1.76	2.1	[*deg.*]
Mechanical trail	7.0	7.2	[*mm*]
Roll center height	51.1	52.1	[*mm*]
Wheel center offset	55.5	28.0	[*mm*]
Camber gain	0.53	0.51	[*deg./deg.* roll]
Camber increase	0.53	0.51	[*deg.*/10 *mm* travel]
Toe increase	0.04	0.02	[*deg.*/10 *mm* travel]
Anti-dive	32.9	32.7	[%]
Anti-lift	49.3	49.1	[%]

As discussed, the king pin angle is reduced from 13 to 5.9 *deg*. Figure 6.2a and 6.3 show the dependency on wheel travel and steering angle. For vertical motion the king pin follows the same linear increase over wheel travel, however with an offset compared to the current double wishbone. In addition the scrub radius will slightly vary over wheel travel motion, see Figure 6.2b.



(a) King pin angle **(b)** Scrub radius **(c)** Bump steer

Figure 6.2: Kinematic change over vertical wheel travel

Based on the change in scrub radius and linear increase of the KPI angle over wheel travel, it can be concluded that the steering axis is shifting in lateral direction. The change in scrub radius should be taking into consideration for braking. During braking the suspension will be compressed and the scrub radius will reduce as result of the suspension kinematics. The scrub radius should have a sufficient positive value and not reduce to zero. When the scrub radius is zero, the actual value will fluctuate between positive and negative. This is the result of slight changes due to compliance and play of the suspension. At this point the longitudinal brake force at the tire will be transmitted to a positive or negative moment around the king pin axis. This unwanted effect could lead to vibrations of the steering

wheel during braking.

For the new multi-link design also bump steer is reduced. However, due to in-wheel packaging and possible collision, the bump steer could not be further reduced by proper positioning of the toe rod, see Figure 6.2c. The steer motion for the front suspension is further analyzed. The position and angle of the steering axes will change over a steering motion for a multi-link suspension. Based on the change in steering axis, the left and right tire kinematic characteristics are analyzed. The comparison between the old URE13 and new multi-link suspension are shown in figure 6.3.

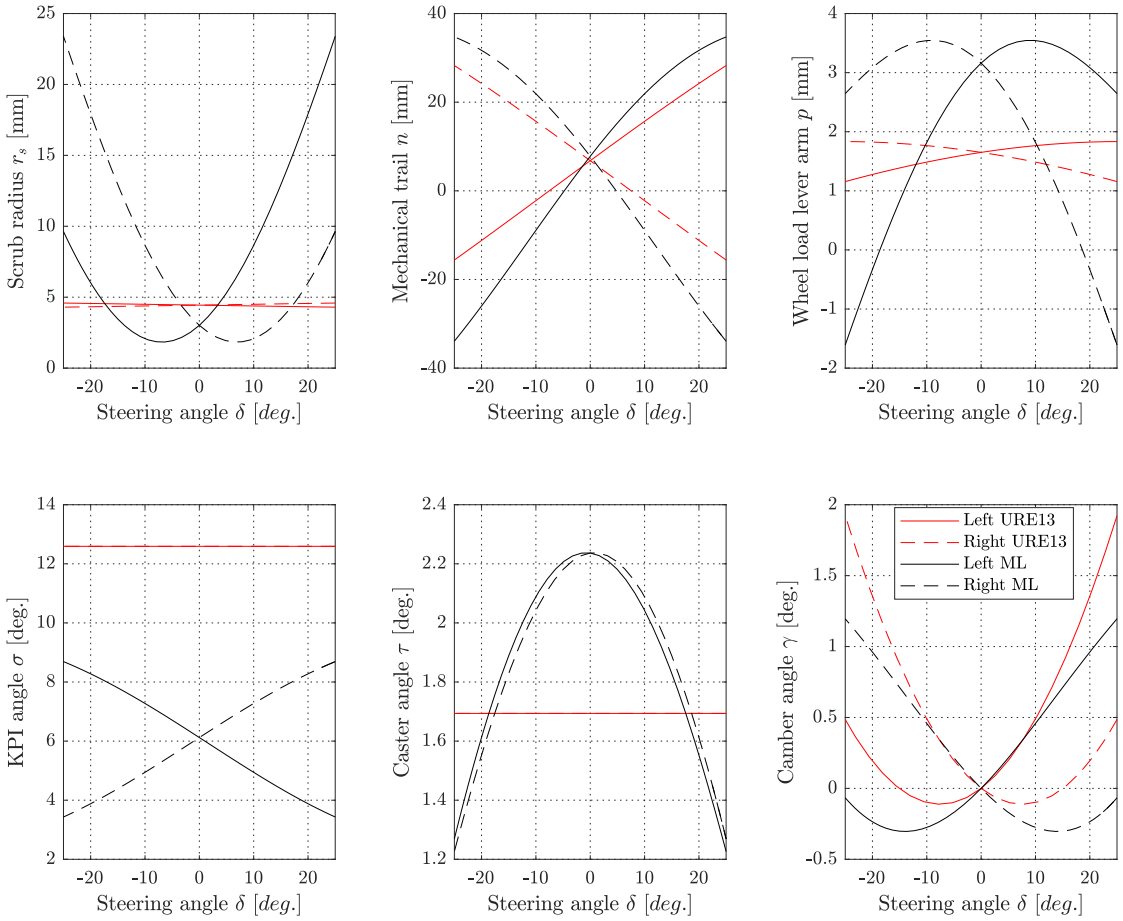


Figure 6.3: Left and right tire orientation characteristics comparison

The change in camber γ,steering angle δ and steering axis orientation are dependent on the suspension kinematics. The longitudinal and lateral tire forces are mainly influenced by the camber angle. Whereas, the difference in left and right steering angle δ will effect the cornering behavior. Combination of both will effect the maximal achievable lateral acceleration, steering feedback and vehicle balance.

The optimal camber is discussed in Section 5.1.3. The camber angle on the outside tire should be minimized. This means that the camber angle should be 0 *deg.* during cornering to achieve maximal lateral acceleration. There are three manners to influence the camber angle. The first one is having camber change with suspension travel. The camber change is influenced by the location and movement of the instant centers. The second manner is combination of a caster and kingpin inclination angles. The combination of both will influence the camber angle during steering motion. The last method of changing the camber angle is by applying a static camber angle from the neutral position of the tire.

Both the steering and vertical motion will influence the camber angle. Body roll during corner will generally generate a positive camber angle at the outside wheels and a negative camber angle at the inside wheels. For both front and rear suspension, the camber angle could be reduced by having static camber. Whereas, in case of the front suspension, the steering motion can be used to compromise this camber change.

A positive kingpin inclination contributes to a positive camber change during steering. For the heavily loaded inner tire this positive camber gain is a unwanted effect. A positive camber will reduce the lateral grip performance of a tire. Having a positive caster angle will provide negative camber change during a steering motion. Which is a desired effect for reduction of the positive camber change as result of the KPI angle and improving lateral grip during cornering.

Combination of both king pin and caster angle result in a camber change during steering motion. In the bottom right corner of Figure 6.3 the camber change over steering motion is illustrated. A positive steering is defined as a left-hand corner. It can be noticed that the outer tire will shown an increase in negative camber, while the inner tire show decrease in positive camber. This improvement of having more negative camber on the outer tire, can primarily be related to the reduction of KPI angle. Having negative camber on the heavily loaded outer tire will increase the lateral performance during cornering. Also a smaller initial setup camber angle is needed, which has positive effect on straight line driving and reduction of the tire wear. This negative camber for steering will comply for the positive camber due to body roll.

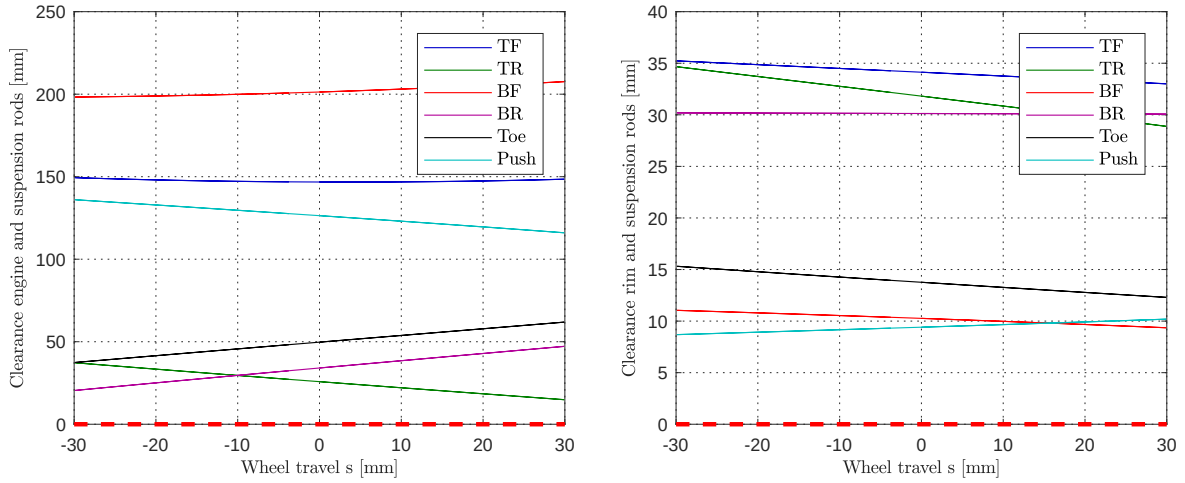
The overall change in scrub radius, mechanical trail, KPI and caster angle will influence the steering wheel torque during cornering. Because some kinematics vary with steering angle, the steering moment is further analyzed in Section 6.5.

6.3 Collision check

The collision check, as discussed in Section, 4.2 is used to analyze possible collisions of the suspension. By plotting the clearance distances between the suspension rods and rim respectively motor housing this collision check is visualized. The distances are plotted for every possible position in the complete motion range of the suspension. A negative value for the clearance will mean a collision.

Rear

For the rear suspension no steering motion is applied, only the collisions during bump motion are evaluated. In Figure 6.4a and 6.4b the clearance between the suspension rods and motor respectively inner rim radius are shown as function over wheel travel motion. All values are positive for the complete wheel travel range from -30 till 30 mm, indicating that no collisions occur. The static toe settings are neglected because they are small and will hardly influence the clearance distance.



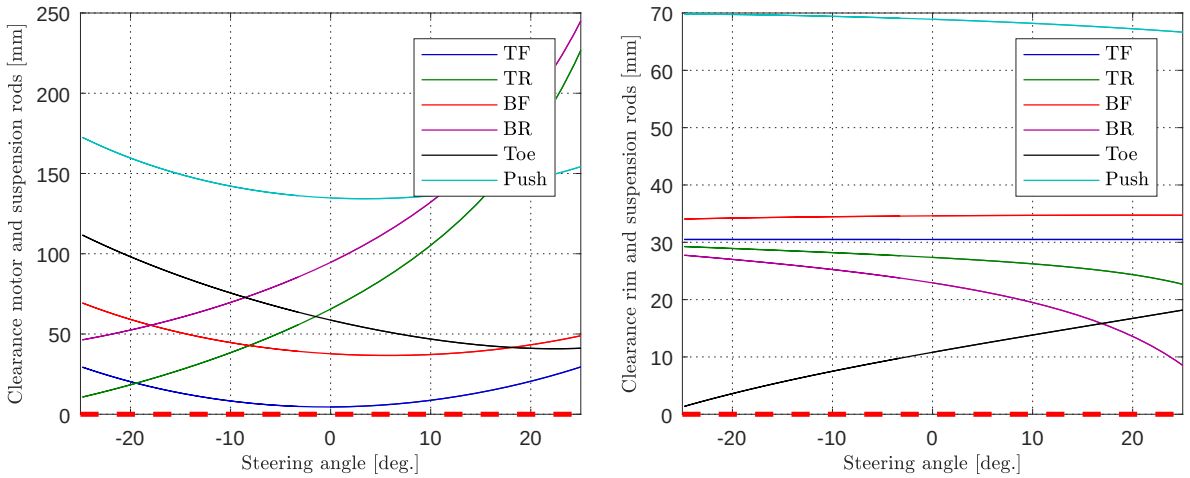
(a) Clearance distance motor and rods.

(b) Clearance distance rim and rods.

Figure 6.4: Collision data rear multi-link suspension.

Front

For the front suspension the steering motion should also be taking in consideration when evaluating possible collisions. The steering motion at minimum and maximum wheel travel are the most critical conditions. In Figure 6.5a and 6.5b the clearance between suspension rods and motor house respectively inner rim radius are shown as function of the steering motion for a fully compressed suspension.

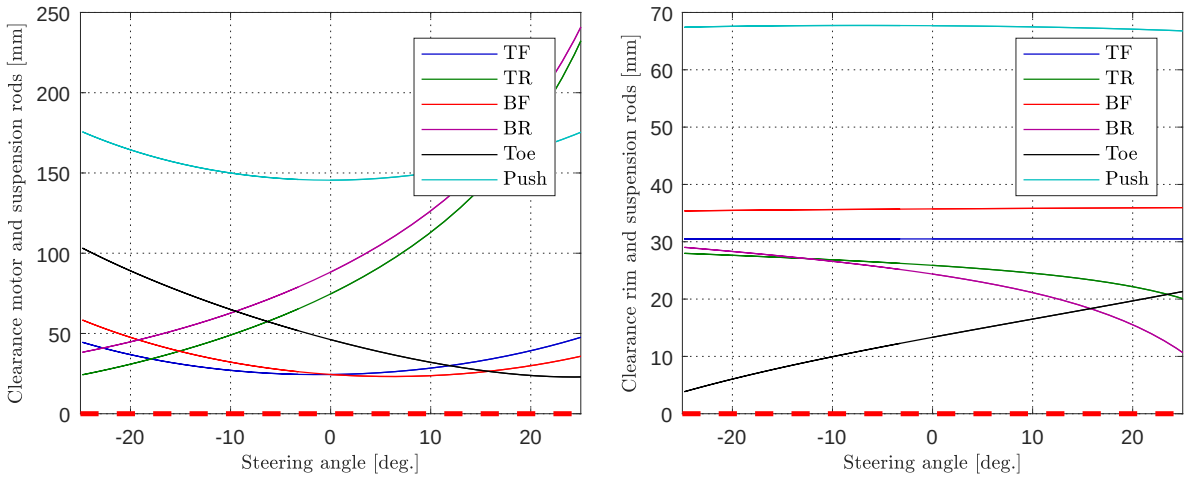


(a) Clearance distance motor and rods.

(b) Clearance distance rim and rods.

Figure 6.5: Collision data front multi-link suspension, fully compressed.

In Figure 6.6a and 6.6b the collision data for a fully extended suspension are shown.



(a) Clearance distance motor and rod.

(b) Clearance distance rim and rod.

Figure 6.6: Collision data front multi-link suspension, fully extended.

No collisions occur for the complete motion range, all values are positive. Using these plots, the critical collision points can be easily identified. This will save a lot of time compared to the current method by visual inspection using the 3D CAD models in Siemens NX.

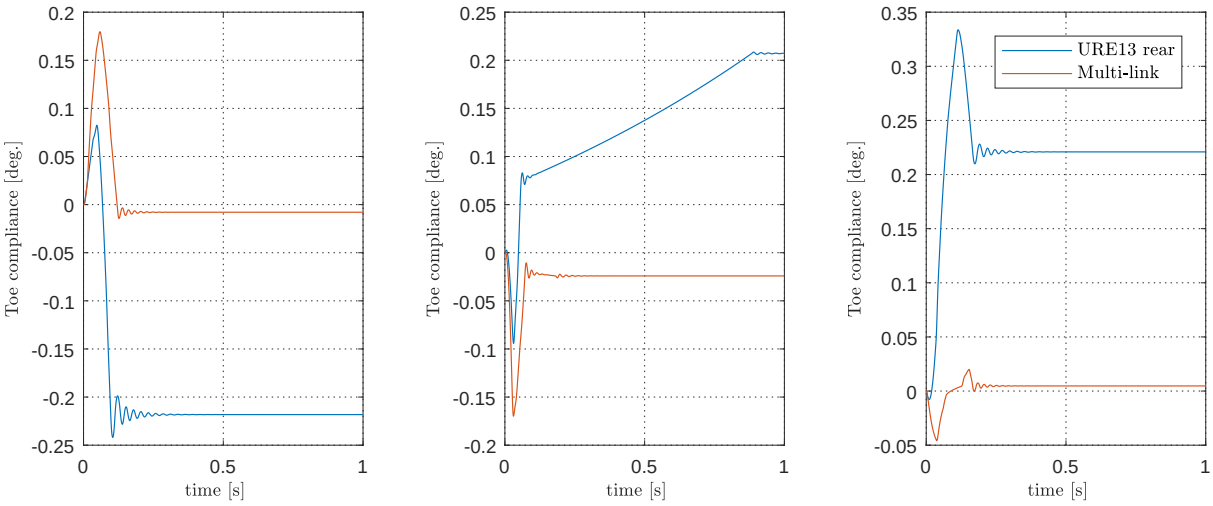
6.4 Suspension free-play and compliance

The advantage of a multi-link layout is that using the correct geometry, the toe and camber change, due to mechanical free-play and the compliance, can be used in an advantageous way. Making effective use of the hard-point positions, the change in toe and camber angle can be reduced or even eliminated. To evaluate the effects of suspension free-play, three load cases are analyzed: bump, cornering and braking. The load cases are estimated by analyzing the resulting forces from a vehicle simulation model and measured data during various driving maneuvers. In Table 6.3 the corresponding forces are listed.

Table 6.3: Load cases

Load case:		**Fx,wcp**	**Fy,wcp**	**Fz,wcp**	**unit**
Front	**Bump**	0	0	5000	$[N]$
	Cornering	0	-4000	2500	$[N]$
	Braking	-2259	0	1738	$[N]$
Rear	**Bump**	0	0	5000	$[N]$
	Cornering	0	-4000	2500	$[N]$
	Braking	-1217	0	936	$[N]$

In Appendix I the results for the compliance simulations for different load-cases are given. A fixed mechanical free-play of $0.5e^{-3}$ m is assumed for each ball-joint. In Figure 6.7 an example of the results of a compliance simulation are shown. The toe compliance as a function of time is plotted. At time zero the load case is applied at the tire contact point. After a fraction of a second the suspension is settled for its steady state position. For bump and braking this takes around 0.25 seconds and for cornering 0.8 seconds. Depending on the direction of the applied forces and the extension or shortening of the suspension links, as result of mechanical free-play, the toe or camber angle will change.



(a) Bump **(b)** Cornering **(c)** Braking

Figure 6.7: Toe compliance simulation results rear suspension

Based on multiple simulations a summary of the compliance results for three different load cases is listed in Table 6.4. The suspension play for the rear suspension is reduced using a multi-link layout, in comparison with the current double wishbone layout. The change in camber is decreased by around 30 %, whereas the change in toe due to free-play is almost eliminated for all three load cases.

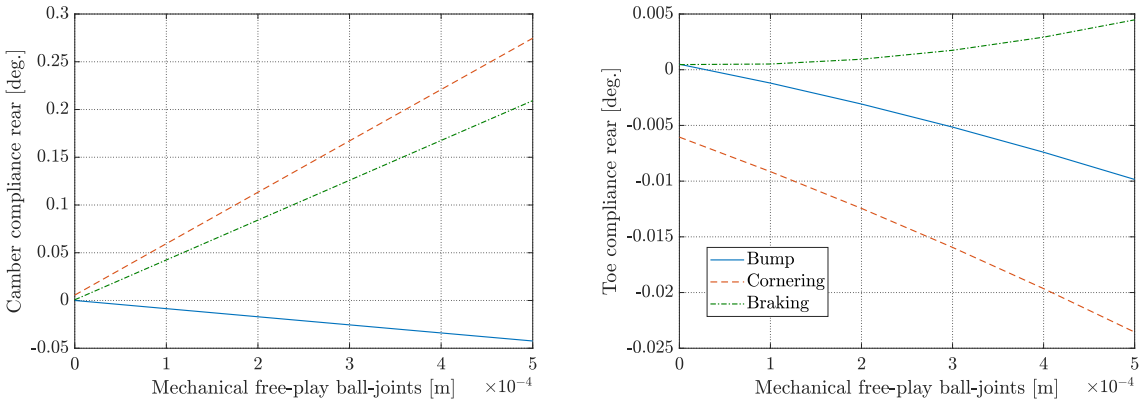
Table 6.4 shows that the change of toe angle due to free-play is quite high for the current URE13 front suspension. This is basically the effect of the small arm/distance from steering rod hard-point on upright and the steering axis. The toe compliance is reduced by increasing this distance and proper use of the hard-point position on the upright, using suspension play in an advantageous way. Using a multi-link suspension the play in toe angle of the front is decreased by around 40 % for bump and cornering, whereas the toe change during braking is again eliminated.

Table 6.4: Suspension play due to mechanical free-play in ball-joint

Load case:		URE13 DW		Multi-link		unit
		Toe	Camber	Toe	Camber	
Front	Bump	-1.43	-0.23	-0.89	-0.07	[*deg.*]
	Cornering	1.48	0.23	0.89	0.36	[*deg.*]
	Braking	-1.38	-0.13	0.11	-0.27	[*deg.*]
Rear	Bump	-0.26	-0.07	-0.01	-0.04	[*deg.*]
	Cornering	0.25	0.35	-0.02	0.28	[*deg.*]
	Braking	0.26	0.34	0.00	0.21	[*deg.*]

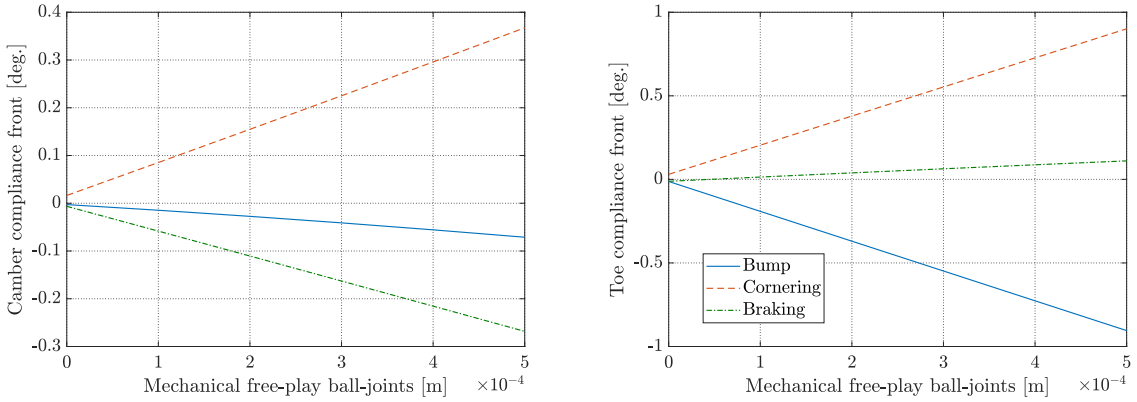
However, the camber play is slightly increased by using the multi-link design. A trade-off is made between reduction of the toe and camber change. Toe will have much more influence on driving stability and cornering performance [11], therefore the choice is made to focus on reduction of the toe free-play as much as possible and keep the camber free-play in a feasible range.

It is hard to determine the actual mechanical free-play of a complete suspension. Not all ball-joints will have the same amount of mechanical free-play, it is dependent on the applied forces and load-cycle. For gaining a better insight on the effects of suspension play, in Figure 6.8 and 6.9 the effect on the camber and toe compliance, as result of increase in mechanical free-play of the ball-joints is illustrated. It is assumed that all ball-joints have the same mechanical free-play. For zero mechanical free-play it can be seen that the camber and toe change already have some initial offset, this is a result of the implemented rod stiffness. The suspension free-play shows mostly a linear relation with the increase of ball-joint free-play.



(a) Camber compliance and play. **(b)** Toe compliance and play.

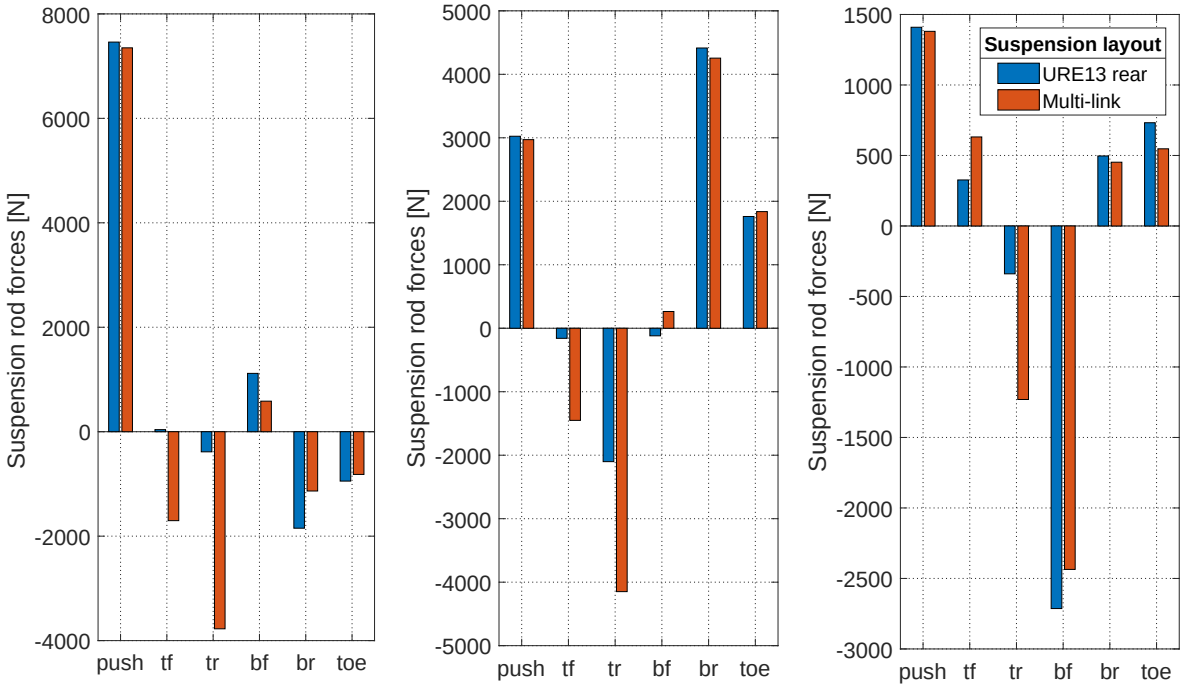
Figure 6.8: Rear suspension compliance and play increase over increased mechanical free-play.



(a) Camber compliance and play. **(b)** Toe compliance and play.

Figure 6.9: Front suspension compliance and play increase over increased mechanical free-play.

The suspension loads are determined by using the defined load cases. The axial forces in the suspension rods are measured using force sensors in the SIMSCAPE simulation model. The forces are measured after the suspension has settled at a steady state position. The resulting forces are given in Appendix I. In Figure 6.10 an example of the suspension loads for the rear suspension are shown. Based on the measured forces it can be concluded that the forces are in general slightly reduced for the multi-link suspension. However, the top front and rear suspension rods are loaded with a higher force. This can be related to fact that a double wishbone consist of a A-arm construction with a bracket for mounting the push-rod. For the rear multi-link design the push rod is connected directly to the upright. As result of decoupling the upper suspension rods the load will increase. For the front multi-link design the push-rod is mounted to the top front rod, which also will increase the loaded force in the top rear suspension rod.



(a) Bump **(b)** Cornering **(c)** Braking

Figure 6.10: Rear suspension forces for three load cases (positive is compression and negative is tensile)

6.5 Steering moment analysis

A steady state roll axis vehicle model is used for analyzing the steering torque. Based on the model and all relevant vehicle parameters, the load transfer during lateral acceleration is determined. The change in lateral and vertical forces acting on the tire contact patches are determined for increasing lateral acceleration. For this the load transfer model is used as discussed in Appendix J. The load transfer results are showed in Figure 6.11. Assumed that the longitudinal forces in x-direction, as result of tire friction, driving torque and braking forces can be neglected, only steady state cornering is assumed. At around 17.5 m/s^2 it can be noticed that the rear left tire is lifted from the ground, as the vertical forces becomes zero. At this point the vertical force for the rear axle is completely concentrated at the outer rear tire. The remaining lateral load transfer is now accommodated by the front axle, which can be noticed in Figure 6.11 by the increase in vertical force for the front left and right tire between 17.5 and 18.5 m/s^2. From 18.5 m/s^2 both left front and rear tires are lifted, at this point the vehicle is at the point of rolling over. Because, during a Formula Student event a maximal lateral acceleration of around 16-17 m/s^2 is achieved due to tire slip limitations, the steering wheel torque is analyzed up to a lateral acceleration of 17.5 m/s^2.

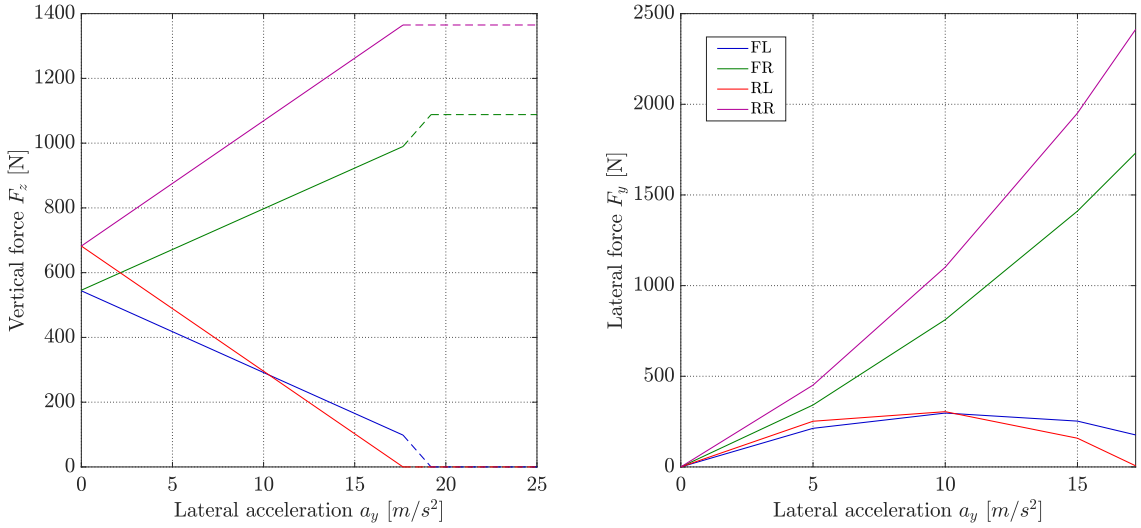
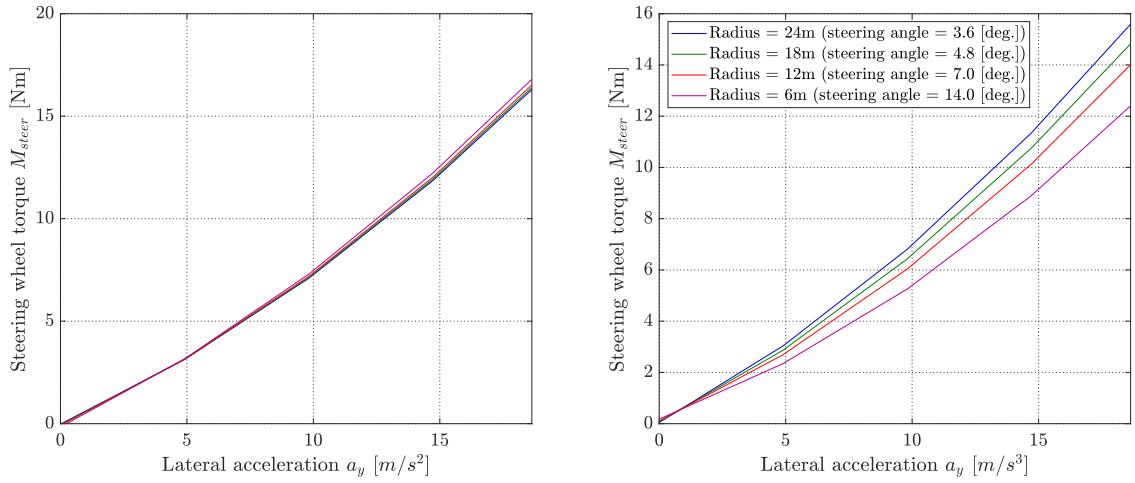


Figure 6.11: Lateral load transfer

The multi-body suspension model is used to calculate the force at the steering rack. Both left and right tire are loaded with the lateral and vertical forces. The resulting steering moment as function of lateral acceleration and steering angle is plotted in Figure 6.12. It can be noticed that the double wishbone suspension has the same steering torque for different cornering radius when the lateral acceleration is increased, see Figure 6.12a. The multi-link suspension shows a decrease in steering wheel torque for a smaller cornering radius, see Figure 6.12b. The decrease of steering wheel torque is related to change in suspension kinematics with steering angle for a multi-link suspension layout. The new steering torque is however of the same magnitude as for the URE13. For the multi-link suspension the steering rack has to be adapted since the ratio between steering rack motion and steering wheel angle is changed. This change is already taken into account for the calculated steering wheel moment as plotted in Figure 6.12.



(a) Double wishbone URE13 **(b)** Multi-link

Figure 6.12: Driver feedback comparison

The steering moment around the king pin axis M_{kp} can be quantified by three contribution, as mentioned in Section 4.4. The moment is induced by the longitudinal, lateral and vertical forces applied at the tire contact point. In figure 6.13 the induced moment around the king pin axis by applying 1 N in longitudinal, lateral and vertical direction is shown. The momentum arm for generating a moment around the kingpin axis for the longitudinal force is the scrub radius r_s, for lateral forces a combination of the mechanical trail n and pneumatic trail t_p, and the vertical force by the wheel load lever arm p.

It can be observed that the king pin moment for longitudinal and vertical forces depends on the steering angle in contrast to the double wishbone suspension. For the URE13 these moments are more constant of the steering range. The moment induced by the lateral force shows a similar trend for both suspension layouts. The steering wheel moment for zero lateral acceleration, in Figure 6.12b, can be clarified by the increase in king pin moment for vertical force F_z. This moment is a result of the wheel load lever arm p. The wheel load lever arm influences the steering wheel torque at low speed, which can be noticed as the slight offset of the steering torque for zero lateral acceleration.

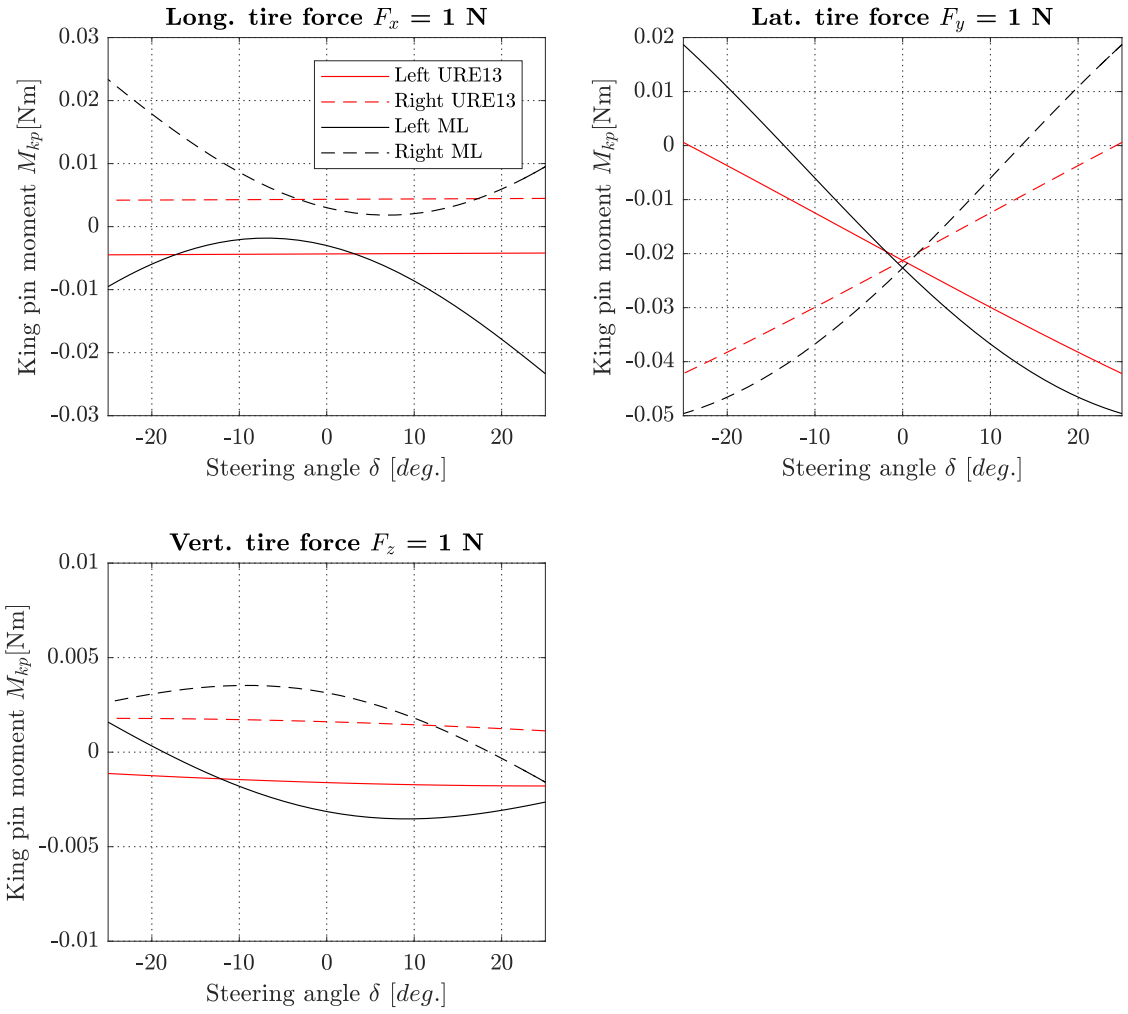


Figure 6.13: Left and right tire king pin steering moment comparison

6.6 Summary

The multi-link suspension is compared with the existing double wishbone layout. During the design process some kinematic characteristics are changed, to improve suspension layout or compliance effects. The main difference is the variable steering axis with steering motion. The variable steering axis is taken in consideration when analyzing cornering performance and driver feedback. The main advantage of using a multi-link suspension is reduction of suspension free-play. The suspension play is analyzed by applying a load at the tire contact patch. Three load-case are evaluated: bump, cornering and braking. For the rear suspension, toe compliance is almost eliminated, whereas camber compliance is reduced. In respect of the front suspension, toe compliance is reduced during bump and even eliminated for braking. The camber compliance is slightly increased by using a multi-link suspension at the front. The suspension forces are analyzed. In general the forces in the suspension links are reduced.

Although, the reaction forces in the upper two suspension rods are increased. These increased forces are related to the position and connection of the push-rod. Furthermore, no collision will occur for the complete requirement motion range. Using the collision check in the simulation model, instead of visual inspection in a 3D CAD program, the overall development time will be reduced. Due to changes in the kinematics of the front suspension, the steering torque is affected. For this the steering torque for lateral acceleration and various steering angles are analyzed. The resulting steering wheel torque for the new multi-link suspension is kept identical with the steering torque of the URE13.

7. Conclusions and recommendations

In this chapter the conclusions and recommendations for future research on the design and analysis of a multi-link suspension are described.

7.1 Conclusions

The main objective of this research is to design a multi-link suspension for a Formula Student race-car and study the advantages and disadvantages in relation to a double wishbone. Based on a literature review, acquired knowledge, problems encountered during the design process and kinematic analysis, the following advantages and disadvantages of a multi-link can be noted. These are based on a comparison between the double wishbone of the current URE racecar and the purposed multi-link design.

Advantages Multi-link vs. Double wishbone

Optimization possibilities
A multi-link suspension has more design freedom and overall less compromises have to be made. This increased design freedom, will result in more possibilities for optimization of the suspension kinematics. Because of the additional kinematic freedom, the multi-link layout is more flexible and the extra hard-points could be comprehensive adjusted to fine-tune the suspension geometry. The multi-link layout offers much more options for adjusting the suspension kinematics compared to double wishbones.

Packaging and space efficiency
For a double wishbone layout, the kinematics are to a large extend restricted by the upright geometry. Based on the current in-wheel packaging of the URE13 double wishbone layout, the suspension kinematics cannot be changed much. For example the kingpin inclination angle can not be smaller than 11 $deg.$, based on the current scrub radius and packaging requirements. Because the steering axis of a multi-link suspension is imaginary, it can be located more freely with respect to the wheel. By using a multi-link the king pin inclination angle has been reduced to 5.9 [$deg.$] for the front suspension.

Positive effect on compliance
A reduction of the effects on toe and camber angle due to suspension play and compliance is achieved. As can be seen in the results of chapter 6.4, the toe compliance is almost reduced to zero, whereas the camber compliance is reduced by 35 % for the rear suspension. The toe compliance at the front suspension is reduced by 40 % and during braking even eliminated. However, by decreasing the toe compliance at the front, the camber compliance is unfortunately increased marginally. A compromise is made between the reduction of toe compliance and increase of camber compliance. Although, the toe compliance is much more sensitive with respect to the overall handling performance. By this, the slightly increase in camber compliance is taken for granted.

No bending moments in suspension rods
Because the suspension rods of a multi-link suspension can freely rotate around the attached points and they are not connected to each other, there are no bending moments present in the connection rods. This is validated with the measured forces using the simulation model. However, there is one exception for the front upper suspension rod of the front suspension. This rod is used to attached the push-rod, which result in a bending moment in this rod.

Production and assembly
For the multi-link layout no complex brackets have to be produced for attaching the push-rod on the upper suspension A-arm. At the rear the push-rod will be directly connected to the upright. At the front the upper front suspension rod will have a more complex design, by the connection of the push-rod. However, the rest of the suspension rods will only differ in length from each other and are straight rods with ball-joints at both ends. This will simplify the production of the suspension rods. Only the upright design will be more complex because of the presence of five or six connection points.

Disadvantages Multi-link vs. Double wishbone

More complex upright design
A disadvantage of the multi-link layout is the slightly more complex design of the upright. The hard-points will be placed further outside the rim. As a result, the brackets for connecting the rods to the upright will have an increased length and are located further away from the wheel center. Also an extra attachment point is needed for connection of the push rod at the rear. As result, the upright will be slightly heavier than for a double wishbone suspension. However, it is expected that the weight will be almost the same because of the absence of the A-arm bracket on the upper 2 suspension links.

Hard-point location sensitivity
Another disadvantage is the sensitivity of incorrect position of the hard-points. An improper design can result in a leverage effect or a collapsing suspension during steering motion. This may increase the forces in the suspension links and ball-joints. Also by improper position of the hard-points the unwanted compliance could increase. The multi-link suspension is therefore more sensitive to position tolerances of the hard-points. However, due to manufacturing tolerances the theoretical finely tuned hard-points may not be realized.

Rotational stiffness
The rotational stiffness around the longitudinal and lateral axis will slightly reduce. The overall stiffness depends on the compliance of the complete suspension layout. However, by placing the connection point further apart from each other and longer arms will result in a higher stiffness. Unfortunately, due to the space limitations and desired kinematics, the resulting stiffness is limited.

7.2 Recommendations

During the execution of the thesis a lot of knowledge is acquired on the design of a multi-link suspension. Based on this knowledge a new multi-link suspension design is proposed. The new suspension design is evaluated using the developed suspension model. However, the current design and simulation model can still be improved and further redefined. Based on the current results and findings the following recommendations can be made.

Simulation model improvements

The developed simulation models can still be improved to give even more useful and realistic suspension characteristics. Possible improvements of these models are:

Tire modelling
More accurate tire data is needed. The current tire models could be improved by performing new measurements. Based on the new tire data a complete new study should be performed in determination of the suspension characteristics. This will result in optimal use of the tires and improve of overall performance. Using the improved tire model, the optimal position and orientation of the tire could be estimated. This has to be validated since the orientation of the tire is defined by the kinematic characteristics of the suspension. In last formula student events it was noticed that the tire shows a lot of lateral deformation during cornering. By taking this deformation and camber stiffness into account the optimal camber variation over steer and bump motion could be estimated. This will result in a increase of the lateral performance.

Steering geometry
For finalizing the steering geometry, Ackermann steering should be further investigated. Ackerman geometry defines the relation between the left and right steering angle during cornering. During the design evaluation, Ackermann steering is not discussed. Because of time limitation and the optimal amount of Ackermann is not studied in detail, the choice is made to neglect this part of the suspension kinematic. However, the simulation model is capable of plotting both left and right steering angle over steering rack displacement. Using this data and by repositioning of the steering hard-point on the upright, the required Ackermann setting could be adjusted and optimized.

Compliance model
The suspension free-play and compliance are modeled by implementing mechanical free-play at the ball-joints and axial stiffness of the suspension rods. Based on the ball-joint motion and deformation of the suspension rods, the suspension kinematics will change. Assumed is that the chassis and upright are rigid and cannot deform. However, in a real vehicle the complete chassis will deform as result of the suspension forces applied to the chassis. To implement this effect in the suspension model, each connection point on the chassis and upright should consists of a stiffness matrix. Also the complete chassis stiffness should be modeled. The deformation of the chassis will effect the kinematic characteristics of all four individual suspensions. However, the exact values for these stiffnesses are hard the determine analytically, but could be partially quantified using compliance measurements for existing suspension layouts.

Steering torque
A simplified steady-state roll axis vehicle model is used to simulate lateral load transfer and determine forces acting on the tire contact patch during cornering. A more advanced analyses of the steering torque could be achieved by using the full vehicle multi-body circuit simulation model. This model should then be extended to include a multi-link suspension layout. Various driving maneuvers and also vehicle parameters like, spring settings, aerodynamics, power-train parameters can be analyzed. Additionally, this model could be expanded by implementing suspension compliance and mechanical free-play.

Numerical optimization
The kinematic suspension model is able of calculating, and plotting all suspension kinematics and characteristics, based on set of input parameters and suspension hard-points. Currently the model is partially automated and different suspension layouts could be analyzed simultaneously during one simulation. This procedure can be further automated using numerical optimization techniques. The optimal suspension design could be obtained, based on a set of design targets, available space and vehicle geometric layout restrictions.

Vehicle testing

There is an opportunity to test the multi-link suspension on the current URE13 vehicle. New uprights and suspension links should be designed. The same hard-points on the chassis should be used, these can not be relocated because of the current composite layup of the monocoque. In order to further develop the multi-link suspension for a future vehicle it is advised to perform extensive tests on a test track. These test can be used to gain insight into the performance increase of using a multi-link suspension. The following list gives an indication of measurements which can be performed:

1. Test of different initial suspension setups using the setup table and analyze of the vehicle handling and driver feedback.

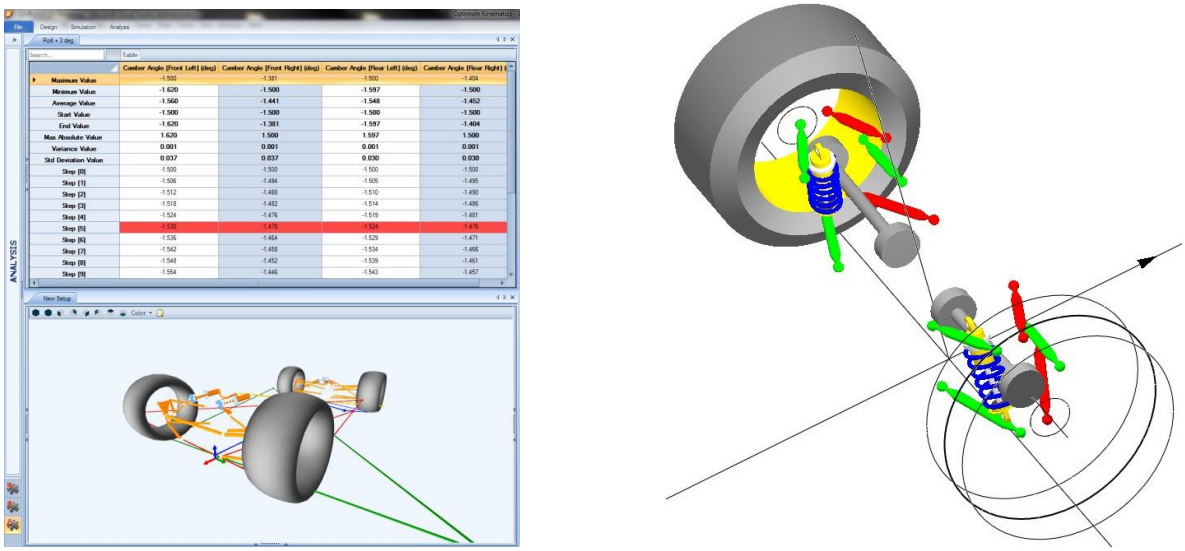
2. Performing compliance and free-play measurements of the suspension and compare these measurements with the results of the suspension simulation model.

3. For a complete quantification of the suspension characteristics, kinematics and compliance (K&C) testing could be performed [43]. There are different companies which are specialized in performing this type of measurements for passenger or race cars. During the K&C measurement, the suspension characteristics are measured by applying individual or combined motions and forces to the vehicle body and tires. The vehicle body is fixed or displaced in heave, roll or pitch motion to simulation different driving maneuvers. Simultaneously cornering, braking or traction forces are applied. The actual suspension characteristics are determined and the results can be used to validate and further improve the suspension model or to redefine the suspension targets.

A. Suspension Analysis Software

Different software packages are available for designing and analyzing suspension systems. The most common are ADAMS Car [32], OptimumKinematics [33] and Susprog3D [34]. The suspension design software packages provide 3D visualization of suspension geometry and motion, see Figure A.1. They enable the user to change hard-points and corresponding suspension geometry. Advanced commercial software packages are also capable of automatically calculate hard-points based on a set of suspension targets and parameters.

These software packages are very basic and calculate only the kinematic behavior of a suspension for a given steer or bump motion. A few software packages are more comprehensive and are also capable of simulating vehicle dynamics using a multi-body model, including tire models and non-linear compliance characteristics.



(a) OptimumKinematics [35] **(b)** Susprog3D [36]

Figure A.1: Commercially available suspension design software packages

However, the problem with these packages is that they only show the user interface and are based on input and output data without providing any knowledge of its internal calculations. The mathematical equations and calculation of additional kinematics are not given and cannot be retrieved from the software package. Furthermore, these commercial suspension packages are not adaptable and force the user to work in a certain way. Some software packages do even not support a multi-link suspension and also the possibility of analyzing suspension play is not implemented. In conclusion, the available software packages are limiting the user in designing a suspension. The packages are a black box with no insight in the exact calculation of the suspension kinematic characteristics.

B. SimScape versus SimMechanics

SIMSCAPE provides a multibody simulation environment for 3D mechanical systems, such as motion based constructions, robots, and vehicle suspensions, see Figure B.1. A multibody system can be created by connecting blocks representing bodies, joints, constraints, force elements, and sensors. SIMSCAPE is able of formulating and solving the equations of motion for the modeled mechanical system. Also, complete CAD files and assemblies can be import, including all masses, inertias, joints, constraints and geometry. Motion animations can be generated to visualize the complete dynamics of the mechanical system. The SIMSCAPE models can be parameterized conveniently using MATLAB variables and expressions, accessed from a MATLAB main file or directly in the model. Additionally, other components of MATLAB SIMULINK can be integrated in the model [37].

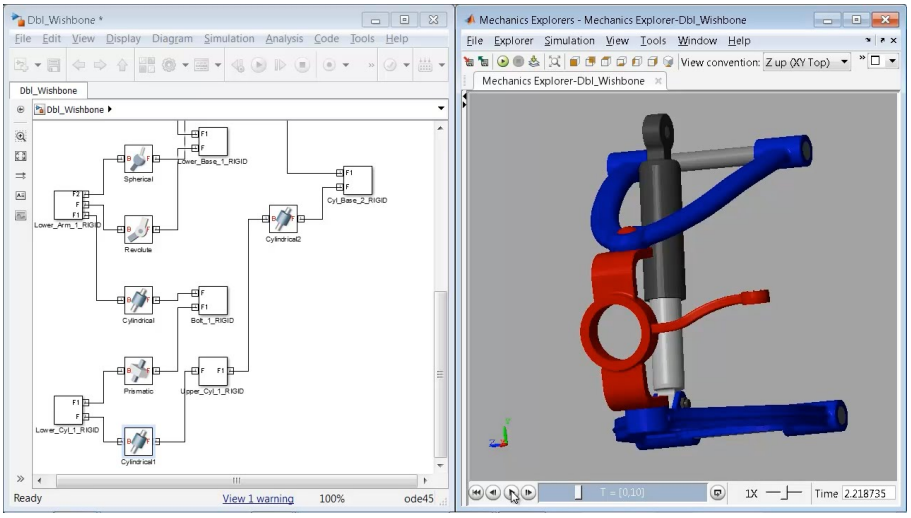


Figure B.1: MATLAB SIMSCAPE [38]

The SIMSCAPE software contains two versions, the first generation and second generation [39]. First generation SIMMECHANICS includes a block library and visualization possibilities found in standard SIMULINK library. From MATLAB version 2012a the second generation SIMSCAPE is introduced. The second generation comes with a simpler modeling approach and new block library. Also the computational, visualization options and integration with other SIMULINK products is improved. From MATLAB version 2018a, the first generation will be no longer supported and the library will be omitted. Table B.1 provides a comparison between first and second generation.

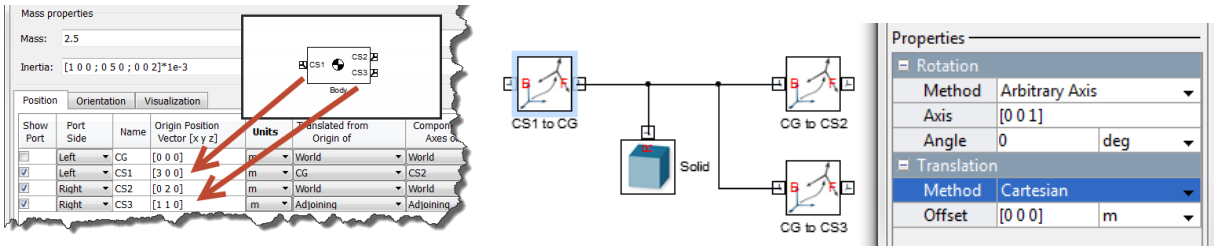
Table B.1: Differences first and second-generation SIMMECHANICS

Feature	1st generation Simmechanics	2nd generation Simscape
Mass/Inertia Calculation	Manual only	Automatic or manual
Solid Geometry	No	Yes
Animation Replay	No	Yes
3-D Model Exploration	Limited	Yes
Initial State Targets	Limited	Yes
Simscape Logging	No	Yes
Code Generation	Yes	Yes
CAD Import	Yes	Yes
Motion Actuation	Yes	Yes
Force/Torque Sensing	Yes	Yes
Complex Constraints	Yes	Gears only
Variable Mass/Gravity	Yes	No

To gain experience with SIMSCAPE and investigate the differences between both generations, a simple

multi-link suspension model created in the first generation of SIMMECHANICS and is converted to the second generation SIMSCAPE. This model is described at the bottom of this page.

The main difference can be found in the usage of body frames and motion constraints. In the first generation, the coordinate systems of each rigid body is defined in the corresponding body element block. Coordinate systems can be defined relative to different frames, including a local or global world frame, see Figure B.2a. In the second generation, each coordinate frame is defined locally by using a rigid translation or rotation transformation block. All connections lines in the second generation diagram correspond to coordinate frames, see Figure B.2b.



(a) SIMMECHANICS (Generation 1) [40] **(b)** SIMSCAPE (Generation 2) [41]

Figure B.2: Definition body element in SIMMECHANICS 1st and 2nd generation

Another difference can be found in the motion constraints and joint connection between bodies. In the second generation, the constraints and joints have a standard convention for its degree of freedom that is always defined locally by the body it connects. The primitive axis of the joints or motion constraint are related to the standardized axis of the connected body, following the standards used in robotic systems. For example the standardized axis of a simple rod is related to the axial z-axis over the length of the rod. This local definition makes it complex to specify a motion constrains to a certain axis of the mechanical system. A series of rigid body rotations or translations is needed to rotate the standardized axis of the body to the desired motion constrained or joint axis.

An improvement compared to the first generation is that initial condition of joints can be specified in the block diagram, stiffness and damping can be directly specified in the joint blocks, and joint blocks includes sensors for measuring displacements and forces.

Example suspension model

In Figure B.3 the modeled multi-link suspension using MATLAB SIMMECHANICS, is shown. Exercise 4 out of advanced vehicle dynamics course of 2014 [42] is used as example. The complete layout is reproduced with all prescribed dimensions and parameters. The shape and position of the bodies are based on the input of the suspension hard-points. The hard-points related to the chassis are joined together to one body, which is fixed to the world. The chassis consist of six connection points: five for the suspension links and one for connection of the spring/damper. The upright is constructed in the same way, consisting of six connection points: five for the suspension links and one for connection of the tire. The spring damper is connected between the chassis and front bottom rod. However, the damper/spring assembly is not visualized in figure B.3. The tire is modeled as a thin square plate and is connected to the upright by a revolute joint. The lowest bottom point of this plate is specified as the tire contact point.

For the connection of the suspension links to the upright spherical joints are used. Although, it seem logical to also use a spherical joint to connect the link with the chassis. However, this will lead to numerical problems, because of the rotation DOF around the axial axes of the links. Due to this extra rotation DOF of the suspension links, the system is under constrained. Therefore, a universal joint is introduced, this joint has 2 rotational degrees of freedom, instead of 3. To define the rotational degrees of freedom for this universal joint, the vector connecting both endpoints of the link is computed and used. A body sensor is connected to the tire center and tire contact point to have numerical values for the position of the upright and tire. The output of the body sensors are saved to the MATLAB work-space and are used to calculate all relevant suspension kinematics.

The chassis is fixed in space, whereas the vertical motion of the tire contact point should have the freedom to move in the longitudinal direction, lateral and yaw direction. It is assumed that the tire cannot slide with respect to the ground surface and is allowed to have a inclination angle. The distance between the wheel center and the tire contact point remains constant and equals the tire radius. The model is extend by a combination of bodies and joints to prescribe these degrees of freedom for the tire contact point. The vertical motion is controlled by applying a vertical force at the tire contact point.

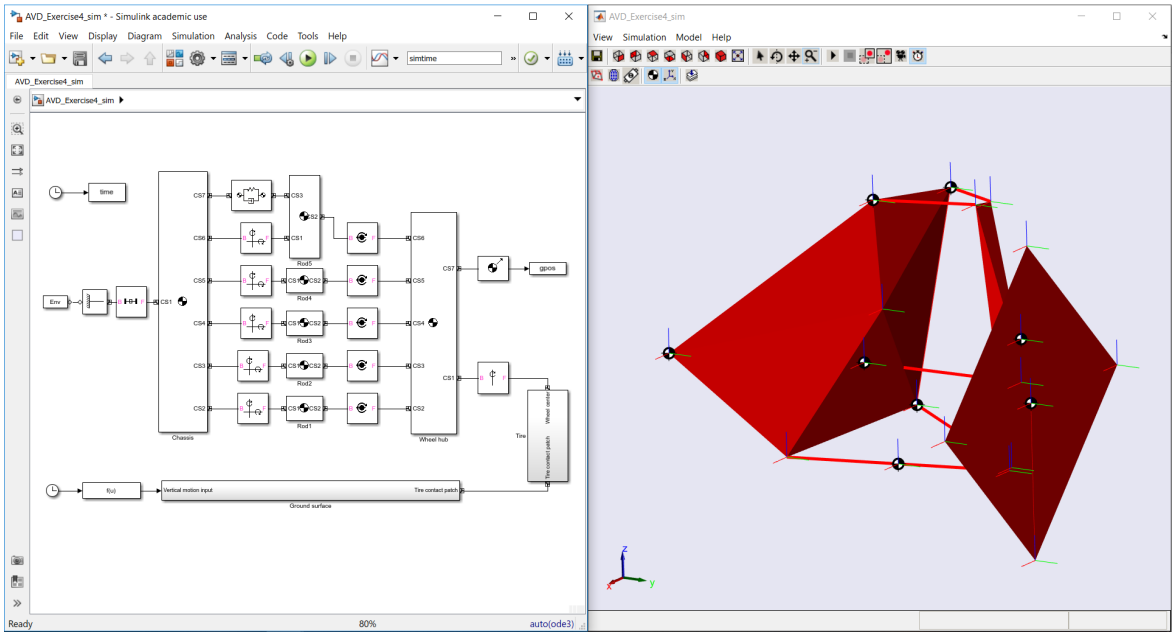


Figure B.3: Multi-link suspension model in SIMMECHANICS

The model create in the second generation of MATLAB SIMMECHANICS, named SIMSCAPE, is constructed in the same way, shown in Figure B.4. All prescribed dimensions and values are kept the same. The main difference can be found in the visualization and shapes of the body elements. The shape of each body element can be freely specified, whereas for the first generation of SIMMECHANICS the bodies are constructed by interconnection of the various hard-points. Also the the primitive axis of the joints are related to the standardized axis of the connected body, which result in some extra difficulties.

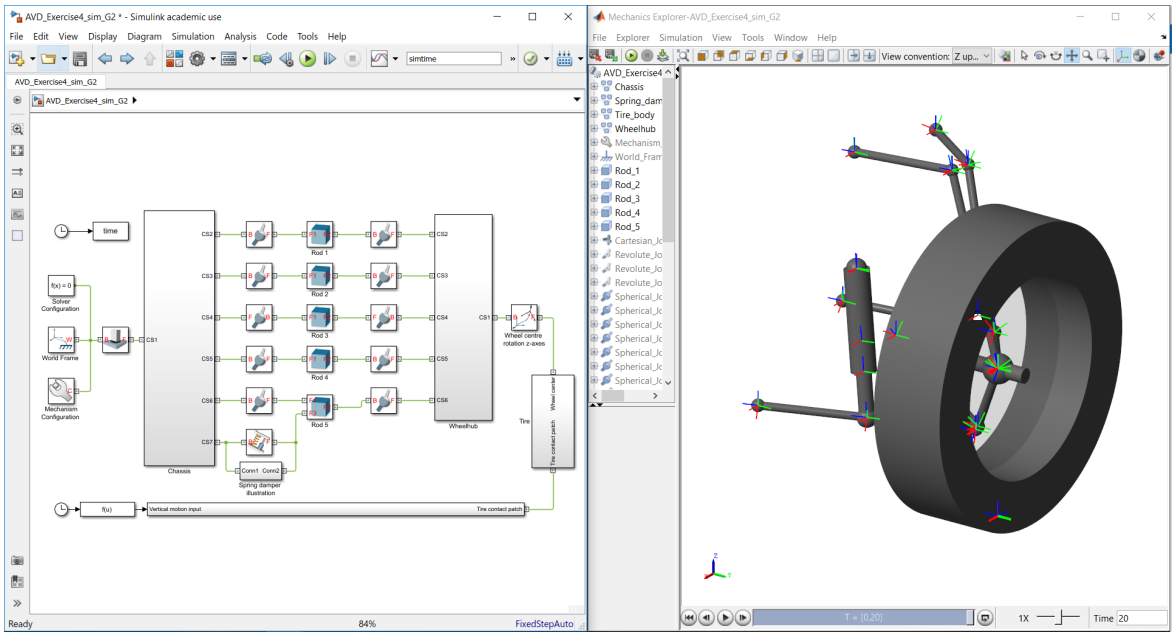


Figure B.4: Multi-link suspension model in SIMSCAPE

The kinematics result of both the model in MATLAB SIMMECHANICS and SIMSCAPE are shown in Figure B.5 below. The wheel alignment, longitudinal and lateral kinematics are plotted over vertical displacement of the tire contact patch z_{cp}. The results of the first and second generation are compared with each other. There were only small differences noticeable, in order of 0.22 %. These differences are marginal and are not further investigated.

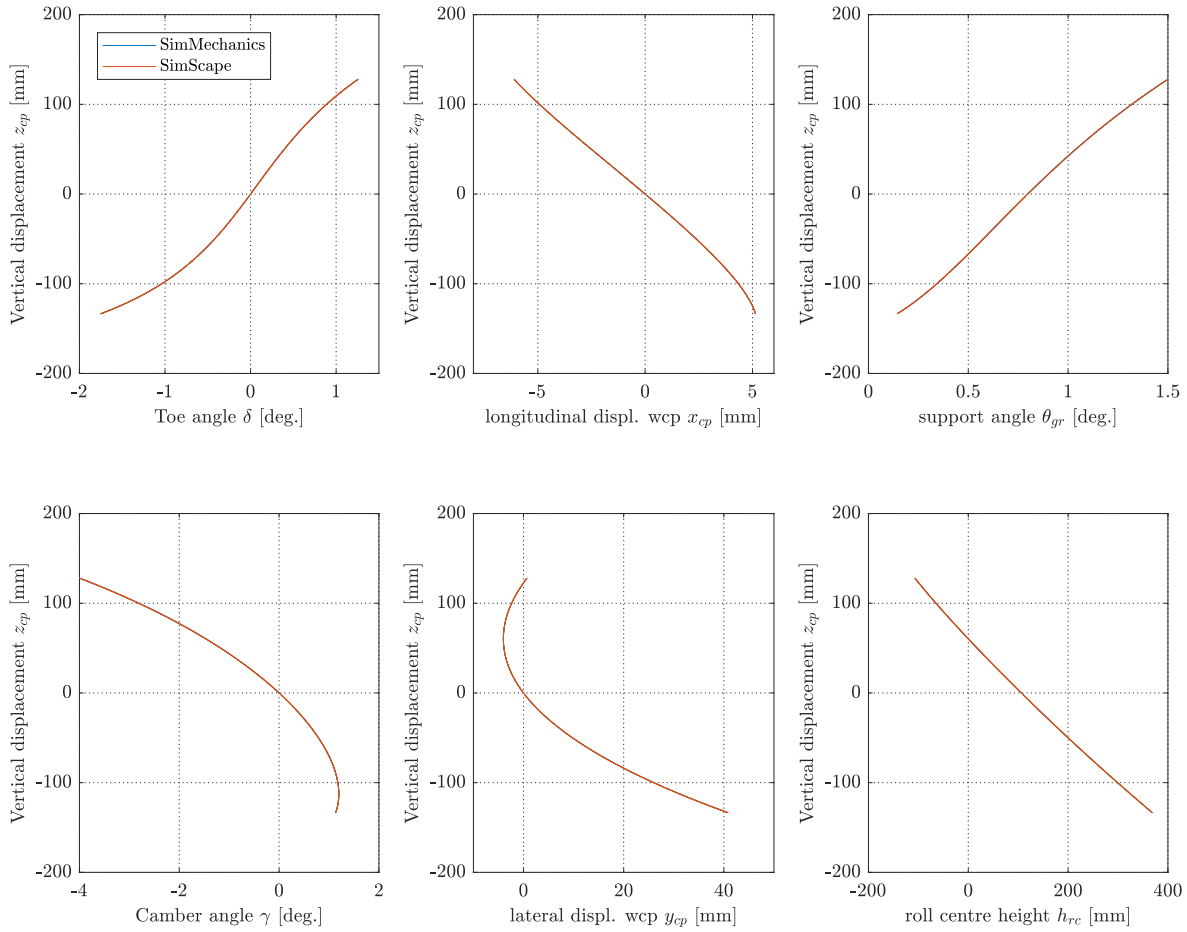


Figure B.5: Multi-link wheel alignment, longitudinal and lateral kinematics using SIMMECHANICS generation 1 and SIMSCAPE.

C. Suspension design tool

In Figure C.1 and C.2 a screen shot of the SIMSCAPE suspension simulation models are shown. Two models are created for analyzing both the steer and bump characteristics. Because of the different motion constrains, it is not feasible to merge both models. Each model has there own constrains and actuation options. The model in Figure C.1 is actuated by applying a steering motion, for analyses of the steering kinematics. In this model the spring/damper is modeled as in-compressible and the tire contact patch can freely move in lateral, longitudinal and vertical direction.

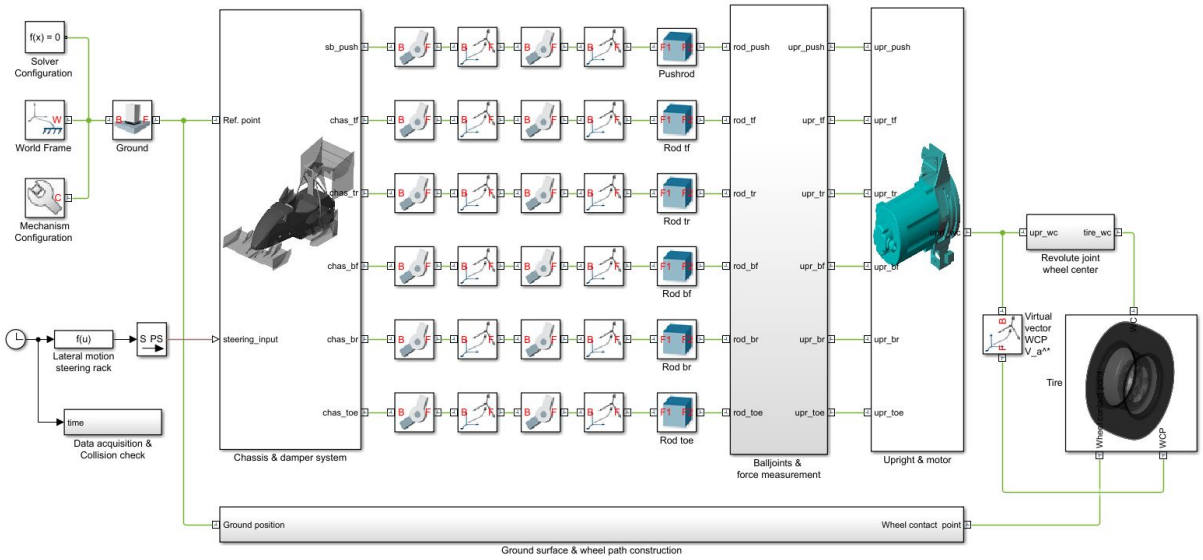


Figure C.1: Suspension model SimMechanics steer motion input.

The model in Figure C.2 is used to apply a vertical motion in order to analyses the bump kinematics. The vertical motion is applied to the tire contact patch. Whereas, the spring/damper is able to extend or to be compressed.

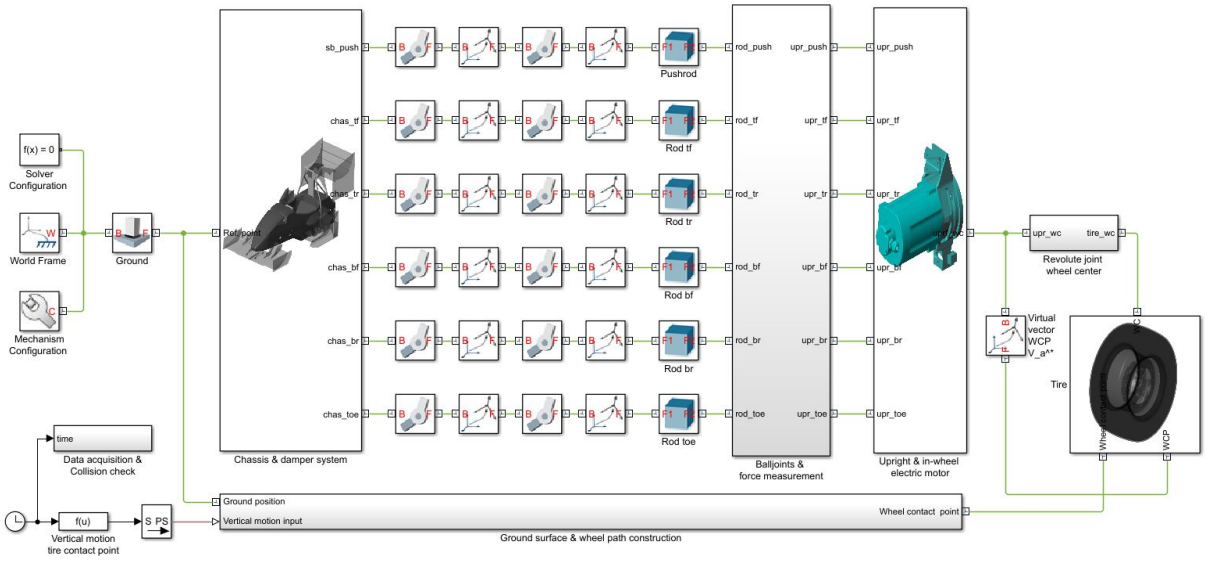


Figure C.2: Suspension model SimMechanics bump motion input.

In both models the chassis is fixed to the world, so rotational and translational motions are restricted. The chassis includes a rocker system, which transfers the motion of the pushrod to a translational motion of the spring/damper.

The rocker system consists of a motion ratio which describes the ratio between velocity of the wheel contact patch in z-direction and the axial velocity of the spring/damper. The chassis has a steering rack system, which is used to impose a lateral motion in y-direction of the toe hard-point at the chassis of the steering link.

Depending on the simulation motion for analyzing steer respectively bump kinematics, either the steering rack system or the damper system is unconstrained, resulting in one degree of freedom.

The four suspension links, steer or toe link and push rod are connected to the chassis. It seems obvious to use spherical joints for this connection. Although, the unconstrained rotation around the z-axis of the links, by using spherical joints, could lead to numerical problems and increase the simulation time. The spherical joints which connect the suspension links to the chassis are therefore constructed using two revolute joints in series. Between and behind the revolute joints a rigid transformation is positioned which prescribes a rotation of the frame by 90 deg respectively -90 deg around the lateral y-axis. By combination of these joints and transformations, the suspension links are able of rotate around there x and y axis, though the rotation around the axial z-axis is restricted.

The six suspension links are connected to the upright by six spherical joints. These spherical joints include of a force sensor to collect the axial rod forces of each suspension link. The upright is modeled as a rigid body. This rigid body represents the upright itself and the electric motor.

The rigid body representing the tire, is connected to the upright by a revolute joint. The lowest point of the tire is defined as the wheel contact point and is connected to the ground by a wheel pad construction. The wheel pad is constructed by using a combination of bodies and joints that restrict the motion of tire. In addition to the vertical motion, the wheel pad should have the freedom to move in the longitudinal, lateral and yaw direction. Furthermore, it is assumed that the tire cannot slide with respect to the wheel pad and is allowed to have a inclination angle. The distance between the wheel center and the contact point remains constant and equals the tire radius r_{tire}.

The inertia of all rigid bodies is assumed to be almost zero by selecting a inertia of $1e^{-6}$ kgm^2. The unsprung mass $m_{unsprung}$ of the complete suspension is taken to be 14 kg and is implemented as a point mass, positioned at the wheel center.

To measure all relevant positions and velocities each rigid body is connected to a body sensor. The body sensors collect position, and lateral and angular velocity. The numerical output of these body sensors is fed into the MATLAB work space and are used to calculate all relevant suspension kinematics and characteristics.

D. Mechanical free-play models

The mechanical free-play in the suspension simulation model is implemented by three possible models. The first model which has been implemented is shown in the Figure D.1. In this model it is assumed that the ball-joint center can move in the axial-direction of the suspension rod. Between the region from $-\varepsilon/2$ till $\varepsilon/2$ there is no contact between ball-joint center and end stops.

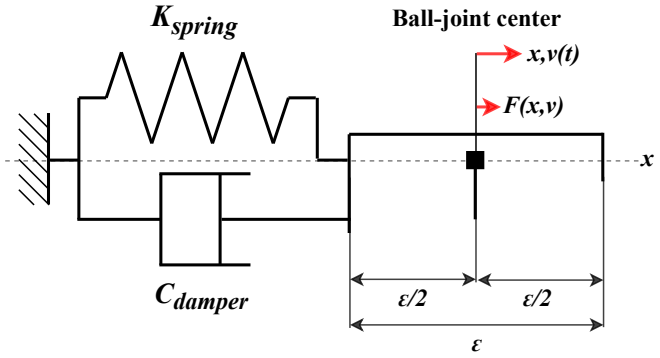


Figure D.1: Schematic representation of the freeplay model with hard stops.

The reaction force $F_{react.}$ is described by:

$$F_{react.}(x,t) = \begin{cases} K_{spring} \cdot (x - \varepsilon/2) + C_{damper} v & \text{if } x \leq \varepsilon/2 \\ 0 & \text{if } \varepsilon/2 < x < \varepsilon/2 \\ K_{spring} \cdot (x + \varepsilon/2) + C_{damper} v & \text{if } x \geq \varepsilon/2 \end{cases}$$

where, ε describes the mechanical free-play in the ball-joint, x is the displacement of the ball-joint center due to mechanical free-play, v is the velocity of the ball-joint center, D_{damper} corresponds to the damper constant of the ball-joint, and K_{spring} corresponds to the stiffness of the suspension rod and ball-joint.

When the suspension is moving the ball-joint center will travel to one of the hard stops. This will take place at high speeds because in the free-play region ($\varepsilon/2 < x < \varepsilon/2$) there is no friction or damping. When the ball-joint center hits one of the hard stops the force will increase rapidly because of the high velocity and reaction force at that point. Due to this abrupt increase in force there is a overshoot in the opposite direction and the simulation model cannot handle these high forces and starts oscillating.

Solving this effect, a second free-play model has been created where friction is added. The friction is a combination of static and viscous friction. The resulting model is showed in Figure D.2.

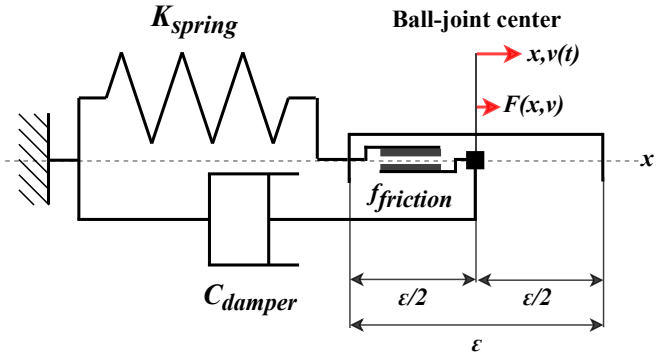


Figure D.2: Schematic representation of the freeplay model with hard stops and translational friction.

The reaction force $F_{react.}$ is now described by:

$$F_{react.}(x,t) = \begin{cases} K_{spring}(x - \varepsilon/2) + C_{damper}v + F_{fric} & \text{if } x \leq \varepsilon/2 \\ F_{fric} & \text{if } \varepsilon/2 < x < \varepsilon/2 \\ K_{spring}(x + \varepsilon/2) + C_{damper}v + F_{fric} & \text{if } x \geq \varepsilon/2 \end{cases}$$

where the friction force F_{fric} can be divided into static $F_{stat,fric}$ and viscous friction $f_{visc,fric}$. Static friction is constant over the complete motion region and the viscous depends on the velocity:

$$F_{fric} = f_{visc,fric}v + F_{stat,fric}. \tag{D.1}$$

Still the problem was not solved. Instead of friction in the free-play region ($\varepsilon/2 < x < \varepsilon/2$) a damper is added. This is currently implemented free-play model as discussed in Section 4.3.1.

E. Simulations mechanical free-play

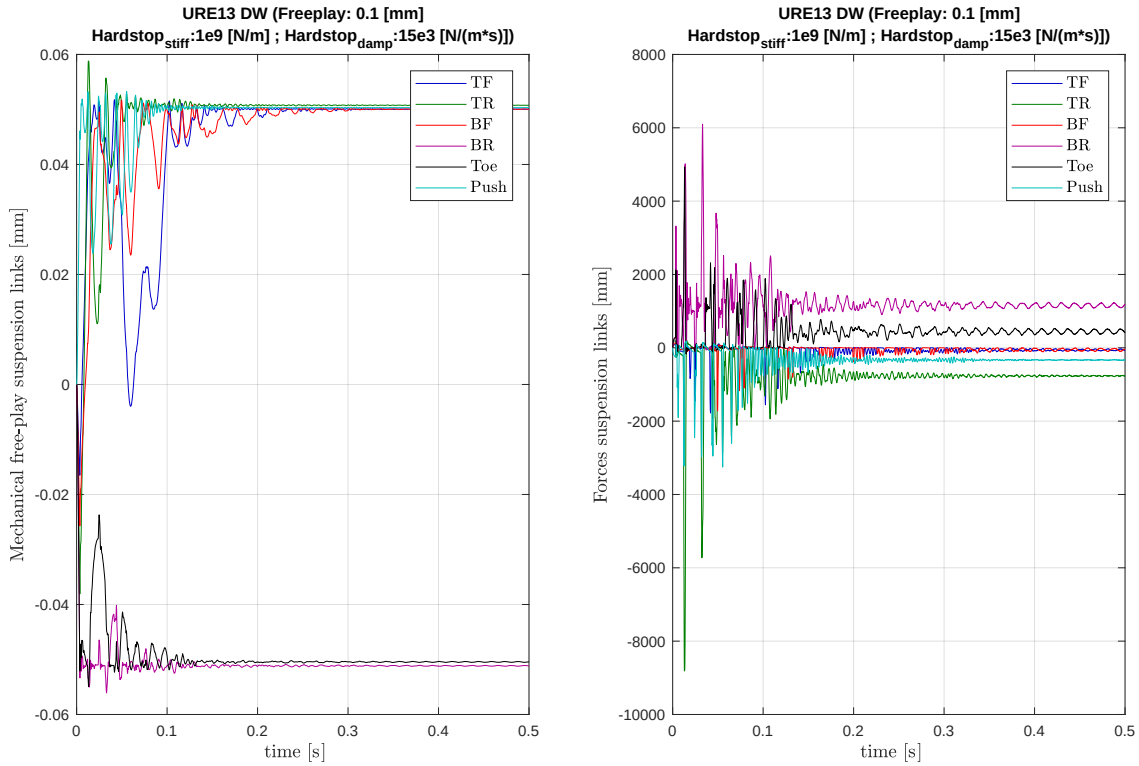


Figure E.1: Suspension mechanical free-play and rod forces,
force applied on the tire contact point is: $F_{wcp} = [0N \quad -1000N \quad 500N]$,
$\varepsilon = 0.1e^{-3}$ $[m]$, $K_{spring} = 1e9$ $[N/m]$ and $C_{damper} = 15e3$ $[N/(ms)]$

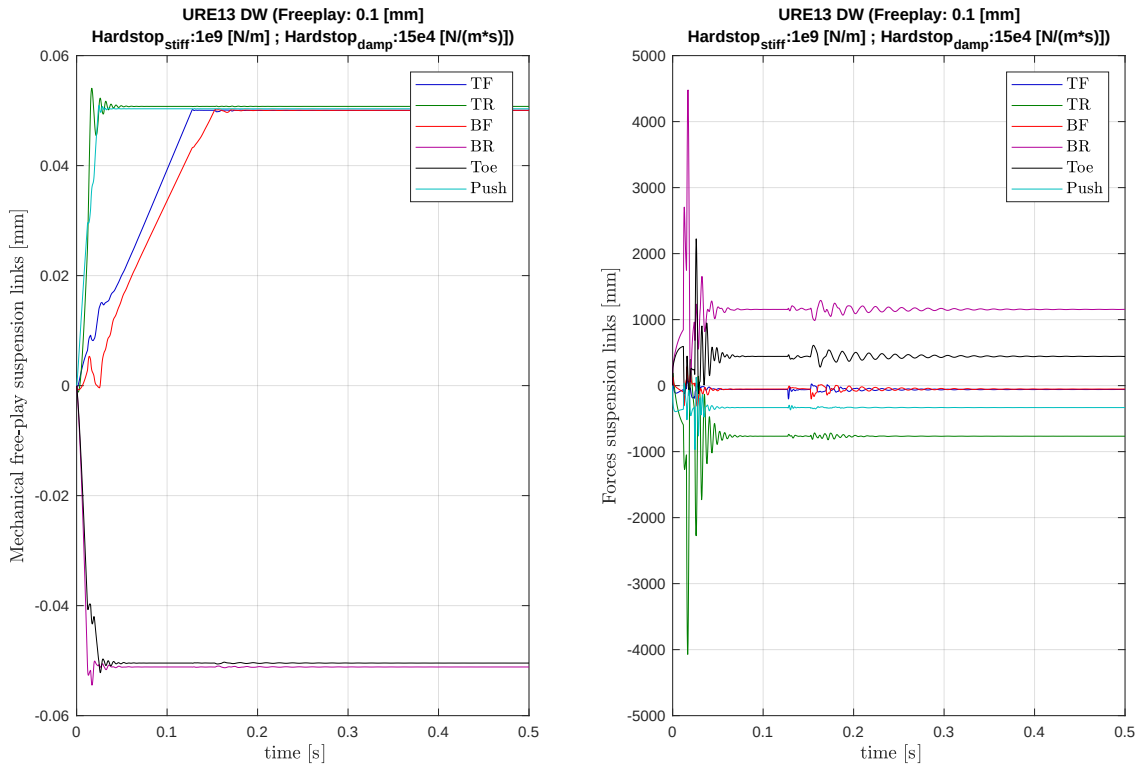


Figure E.2: Suspension mechanical free-play and rod forces,
force applied on the tire contact point is: $F_{wcp} = [0N \quad -1000N \quad 500N]$,
$\varepsilon = 0.1e^{-3}$ $[m]$, $K_{spring} = 1e9$ $[N/m]$ and $C_{damper} = 15e4$ $[N/(ms)]$

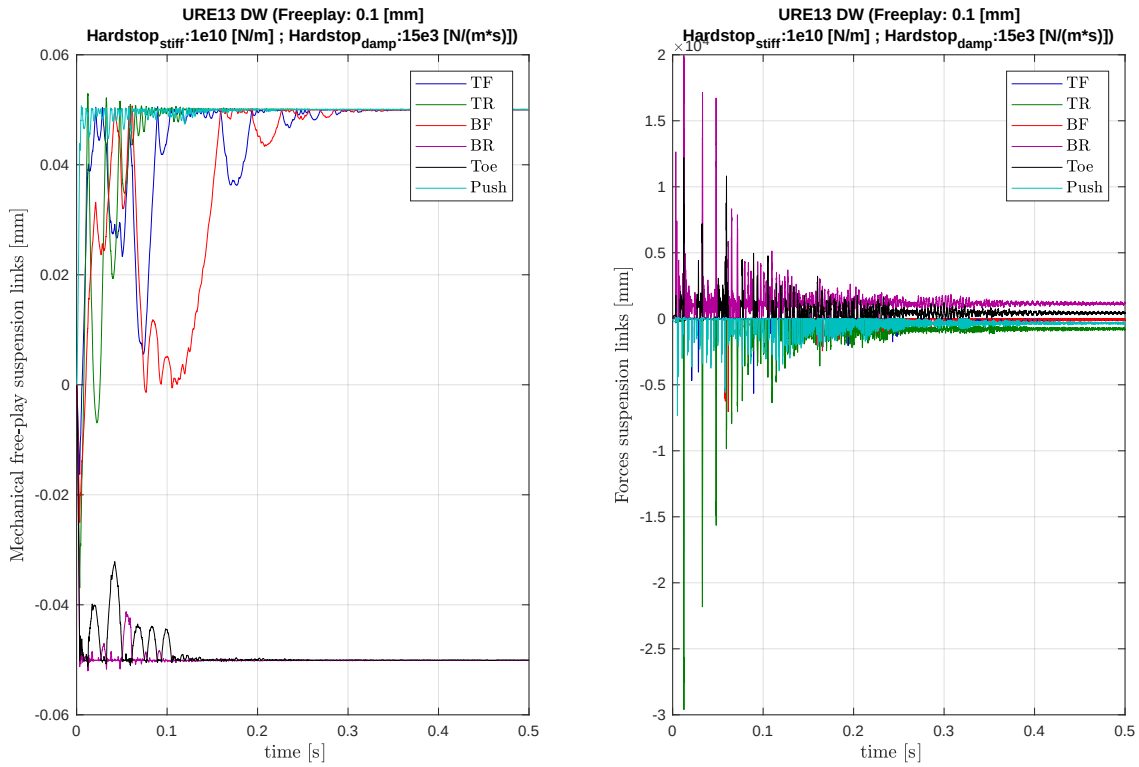


Figure E.3: Suspension mechanical free-play and rod forces,
force applied on the tire contact point is: $F_{wcp} = [0N \quad -1000N \quad 500N]$,
$\varepsilon = 0.1e^{-3}$ $[m]$, $K_{spring} = 1e10$ $[N/m]$ and $C_{damper} = 15e3$ $[N/(ms)]$

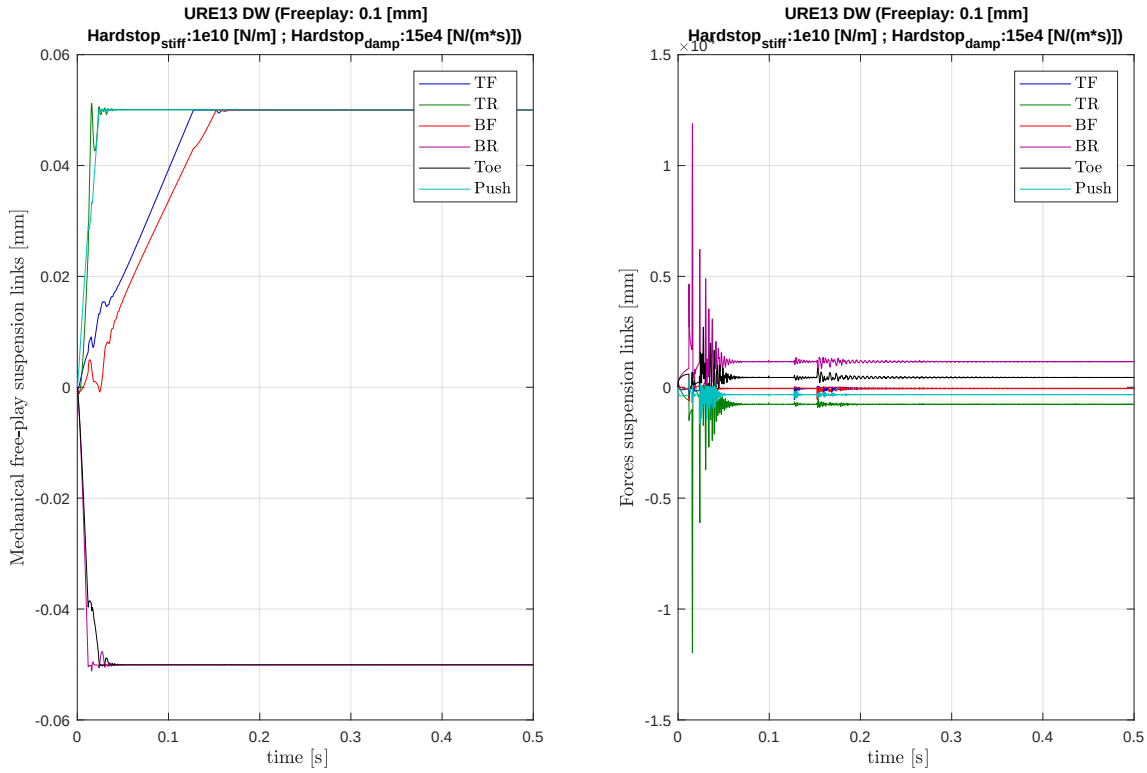


Figure E.4: Suspension mechanical free-play and rod forces,
force applied on the tire contact point is: $F_{wcp} = [0N \quad -1000N \quad 500N]$,
$\varepsilon = 0.1e^{-3}$ $[m]$, $K_{spring} = 1e10$ $[N/m]$ and $C_{damper} = 15e4$ $[N/(ms)]$

F. Setup table

In Figure F.1 the setup tables for static camber and toe of the rear multi-link suspension are given. Based on the number of shims, the static camber and toe angle can be adjusted. The static camber angle is adjusted by changing the distance between the upper two hard-points on the upright and the tire longitudinal plane in x and$z$-direction. This distance is changed by adding or removing shims between the suspension brackets, for connecting the suspension links to the upright, and the upright itself. The static toe angle is adjusted by changing the length of the toe or steering rod. This is also achieved by using shims. A correct combination of the shims result in the desired suspension setup. Using the table in Figure F.1 a quick and easy overview of the effects on number of shims is given. The vertical rows are related to the needed change in shims for the camber adjustment. Whereas, The horizontal columns are related to the needed change in shims for the toe adjustment.

Adjustments of camber and toe are slightly depended of each other. Adjusting the camber angle will induce a minor change of the toe angle or vice versa. Thus a specific procedure of setup is required. First the initial camber should be adjusted by the upper setup table. After the camber angle is set, the desired toe angle has to be adjusted using the bottom setup table. However, as can be noticed, for a specific number of camber shims the camber angle will vary over the used toe shims. This can be observed by slight change in camber angle from left to right in each row of the upper table. By this the upper table should again be used to readjusted the camber angle for the second time. Eventually a proper combination of shims is obtained which result in the desired suspension setup.

Camber [deg.]	Number of shims toe/steering rod (1shim = 0.2mm thick)																			
Number of shims camber bracket (1shim = 1mm thick)	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
0	-2,8	-2,8	-2,8	-2,8	-2,9	-2,9	-2,9	-2,9	-3,0	-3,0	-3,0	-3,0	-3,0	-3,1	-3,1	-3,1	-3,1	-3,2	-3,2	-3,2
1	-2,5	-2,5	-2,5	-2,5	-2,6	-2,6	-2,6	-2,6	-2,7	-2,7	-2,7	-2,7	-2,7	-2,8	-2,8	-2,8	-2,8	-2,9	-2,9	-2,9
2	-2,2	-2,2	-2,2	-2,2	-2,3	-2,3	-2,3	-2,3	-2,4	-2,4	-2,4	-2,4	-2,4	-2,5	-2,5	-2,5	-2,5	-2,6	-2,6	-2,6
3	-1,9	-1,9	-1,9	-1,9	-2,0	-2,0	-2,0	-2,0	-2,1	-2,1	-2,1	-2,1	-2,1	-2,2	-2,2	-2,2	-2,2	-2,3	-2,3	-2,3
4	-1,6	-1,6	-1,6	-1,6	-1,7	-1,7	-1,7	-1,7	-1,7	-1,8	-1,8	-1,8	-1,8	-1,9	-1,9	-1,9	-1,9	-2,0	-2,0	-2,0
5	-1,3	-1,3	-1,3	-1,3	-1,4	-1,4	-1,4	-1,4	-1,4	-1,5	-1,5	-1,5	-1,5	-1,6	-1,6	-1,6	-1,6	-1,7	-1,7	-1,7
6	-1,0	-1,0	-1,0	-1,0	-1,1	-1,1	-1,1	-1,1	-1,1	-1,2	-1,2	-1,2	-1,2	-1,3	-1,3	-1,3	-1,3	-1,4	-1,4	-1,4
7	-0,7	-0,7	-0,7	-0,7	-0,8	-0,8	-0,8	-0,8	-0,8	-0,9	-0,9	-0,9	-0,9	-1,0	-1,0	-1,0	-1,0	-1,1	-1,1	-1,1
8	-0,4	-0,4	-0,4	-0,4	-0,5	-0,5	-0,5	-0,5	-0,6	-0,6	-0,6	-0,6	-0,6	-0,7	-0,7	-0,7	-0,7	-0,8	-0,8	-0,8
9	-0,1	-0,1	-0,1	-0,1	-0,2	-0,2	-0,2	-0,2	-0,3	-0,3	-0,3	-0,3	-0,3	-0,4	-0,4	-0,4	-0,4	-0,5	-0,5	-0,5
10	0,2	0,2	0,2	0,2	0,1	0,1	0,1	0,1	0,0	0,0	0,0	0,0	0,0	-0,1	-0,1	-0,1	-0,1	-0,2	-0,2	-0,2
11	0,5	0,5	0,5	0,5	0,4	0,4	0,4	0,4	0,3	0,3	0,3	0,3	0,3	0,2	0,2	0,2	0,2	0,1	0,1	0,1
12	0,8	0,8	0,8	0,8	0,7	0,7	0,7	0,7	0,6	0,6	0,6	0,6	0,5	0,5	0,5	0,5	0,5	0,4	0,4	0,4
13	1,1	1,1	1,1	1,1	1,0	1,0	1,0	1,0	0,9	0,9	0,9	0,9	0,8	0,8	0,8	0,8	0,8	0,7	0,7	0,7
14	1,4	1,4	1,4	1,3	1,3	1,3	1,3	1,3	1,2	1,2	1,2	1,2	1,1	1,1	1,1	1,1	1,0	1,0	1,0	1,0
15	1,7	1,7	1,7	1,6	1,6	1,6	1,6	1,5	1,5	1,5	1,5	1,5	1,4	1,4	1,4	1,4	1,3	1,3	1,3	1,3
16	2,0	2,0	2,0	1,9	1,9	1,9	1,9	1,8	1,8	1,8	1,8	1,8	1,7	1,7	1,7	1,7	1,6	1,6	1,6	1,6
17	2,3	2,3	2,3	2,2	2,2	2,2	2,2	2,1	2,1	2,1	2,1	2,0	2,0	2,0	2,0	2,0	1,9	1,9	1,9	1,9
18	2,6	2,6	2,5	2,5	2,5	2,5	2,5	2,4	2,4	2,4	2,4	2,3	2,3	2,3	2,3	2,2	2,2	2,2	2,2	2,2
19	2,9	2,9	2,8	2,8	2,8	2,8	2,7	2,7	2,7	2,7	2,7	2,6	2,6	2,6	2,6	2,5	2,5	2,5	2,5	2,4

Toe [deg.]	Number of shims toe/steering rod (1shim = 0.2mm thick)																			
Number of shims camber bracket (1shim = 1mm thick)	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
0	1,5	1,3	1,2	1,1	0,9	0,8	0,6	0,5	0,4	0,2	0,1	-0,1	-0,2	-0,4	-0,5	-0,6	-0,8	-0,9	-1,1	-1,2
1	1,5	1,3	1,2	1,1	0,9	0,8	0,6	0,5	0,4	0,2	0,1	-0,1	-0,2	-0,4	-0,5	-0,6	-0,8	-0,9	-1,1	-1,2
2	1,5	1,3	1,2	1,1	0,9	0,8	0,6	0,5	0,3	0,2	0,1	-0,1	-0,2	-0,4	-0,5	-0,6	-0,8	-0,9	-1,1	-1,2
3	1,5	1,3	1,2	1,0	0,9	0,8	0,6	0,5	0,3	0,2	0,1	-0,1	-0,2	-0,4	-0,5	-0,7	-0,8	-0,9	-1,1	-1,2
4	1,5	1,3	1,2	1,0	0,9	0,8	0,6	0,5	0,3	0,2	0,1	-0,1	-0,2	-0,4	-0,5	-0,7	-0,8	-0,9	-1,1	-1,2
5	1,5	1,3	1,2	1,0	0,9	0,7	0,6	0,5	0,3	0,2	0,0	-0,1	-0,2	-0,4	-0,5	-0,7	-0,8	-0,9	-1,1	-1,2
6	1,4	1,3	1,2	1,0	0,9	0,7	0,6	0,5	0,3	0,2	0,0	-0,1	-0,2	-0,4	-0,5	-0,7	-0,8	-1,0	-1,1	-1,2
7	1,4	1,3	1,2	1,0	0,9	0,7	0,6	0,5	0,3	0,2	0,0	-0,1	-0,3	-0,4	-0,5	-0,7	-0,8	-1,0	-1,1	-1,2
8	1,4	1,3	1,1	1,0	0,9	0,7	0,6	0,4	0,3	0,2	0,0	-0,1	-0,3	-0,4	-0,5	-0,7	-0,8	-1,0	-1,1	-1,3
9	1,4	1,3	1,1	1,0	0,9	0,7	0,6	0,4	0,3	0,2	0,0	-0,1	-0,3	-0,4	-0,6	-0,7	-0,8	-1,0	-1,1	-1,3
10	1,4	1,3	1,1	1,0	0,8	0,7	0,6	0,4	0,3	0,1	0,0	-0,1	-0,3	-0,4	-0,6	-0,7	-0,8	-1,0	-1,1	-1,3
11	1,4	1,2	1,1	1,0	0,8	0,7	0,5	0,4	0,3	0,1	0,0	-0,2	-0,3	-0,4	-0,6	-0,7	-0,9	-1,0	-1,1	-1,3
12	1,4	1,2	1,1	1,0	0,8	0,7	0,5	0,4	0,3	0,1	0,0	-0,2	-0,3	-0,4	-0,6	-0,7	-0,9	-1,0	-1,1	-1,3
13	1,4	1,2	1,1	0,9	0,8	0,7	0,5	0,4	0,2	0,1	0,0	-0,2	-0,3	-0,5	-0,6	-0,7	-0,9	-1,0	-1,2	-1,3
14	1,3	1,2	1,1	0,9	0,8	0,7	0,5	0,4	0,2	0,1	0,0	-0,2	-0,3	-0,5	-0,6	-0,7	-0,9	-1,0	-1,2	-1,3
15	1,3	1,2	1,1	0,9	0,8	0,6	0,5	0,4	0,2	0,1	-0,1	-0,2	-0,3	-0,5	-0,6	-0,8	-0,9	-1,0	-1,2	-1,3
16	1,3	1,2	1,0	0,9	0,8	0,6	0,5	0,3	0,2	0,1	-0,1	-0,2	-0,3	-0,5	-0,6	-0,8	-0,9	-1,0	-1,2	-1,3
17	1,3	1,2	1,0	0,9	0,8	0,6	0,5	0,3	0,2	0,1	-0,1	-0,2	-0,4	-0,5	-0,6	-0,8	-0,9	-1,1	-1,2	-1,3
18	1,3	1,2	1,0	0,9	0,7	0,6	0,5	0,3	0,2	0,0	-0,1	-0,2	-0,4	-0,5	-0,7	-0,8	-0,9	-1,1	-1,2	-1,4
19	1,3	1,1	1,0	0,9	0,7	0,6	0,4	0,3	0,2	0,0	-0,1	-0,2	-0,4	-0,5	-0,7	-0,8	-0,9	-1,1	-1,2	-1,4

Figure F.1: Setup table for static camber and toe for the rear multi-link layout

G. Kinematics comparison rear

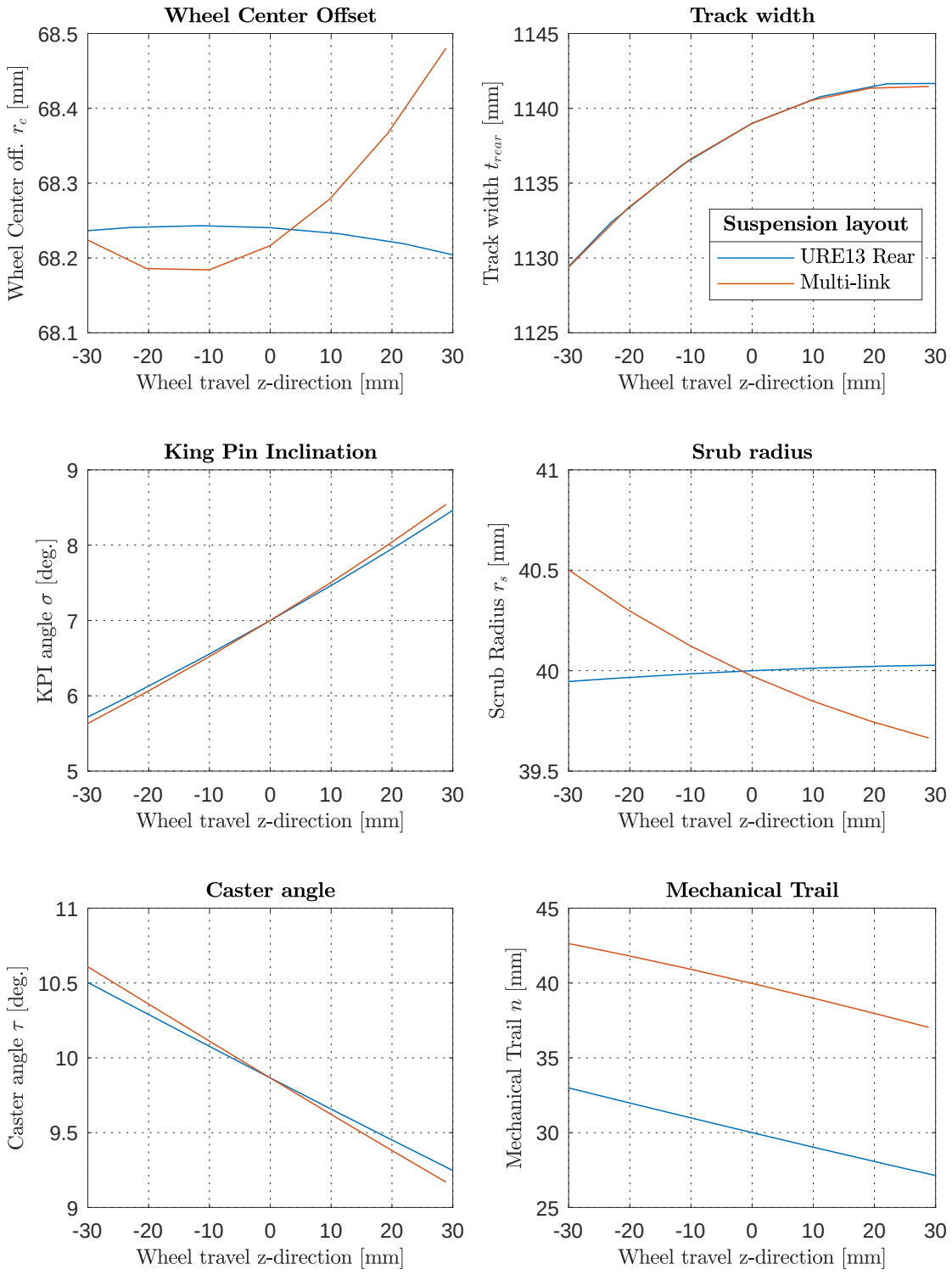


Figure G.1: Steering axis kinematics over wheel travel motion

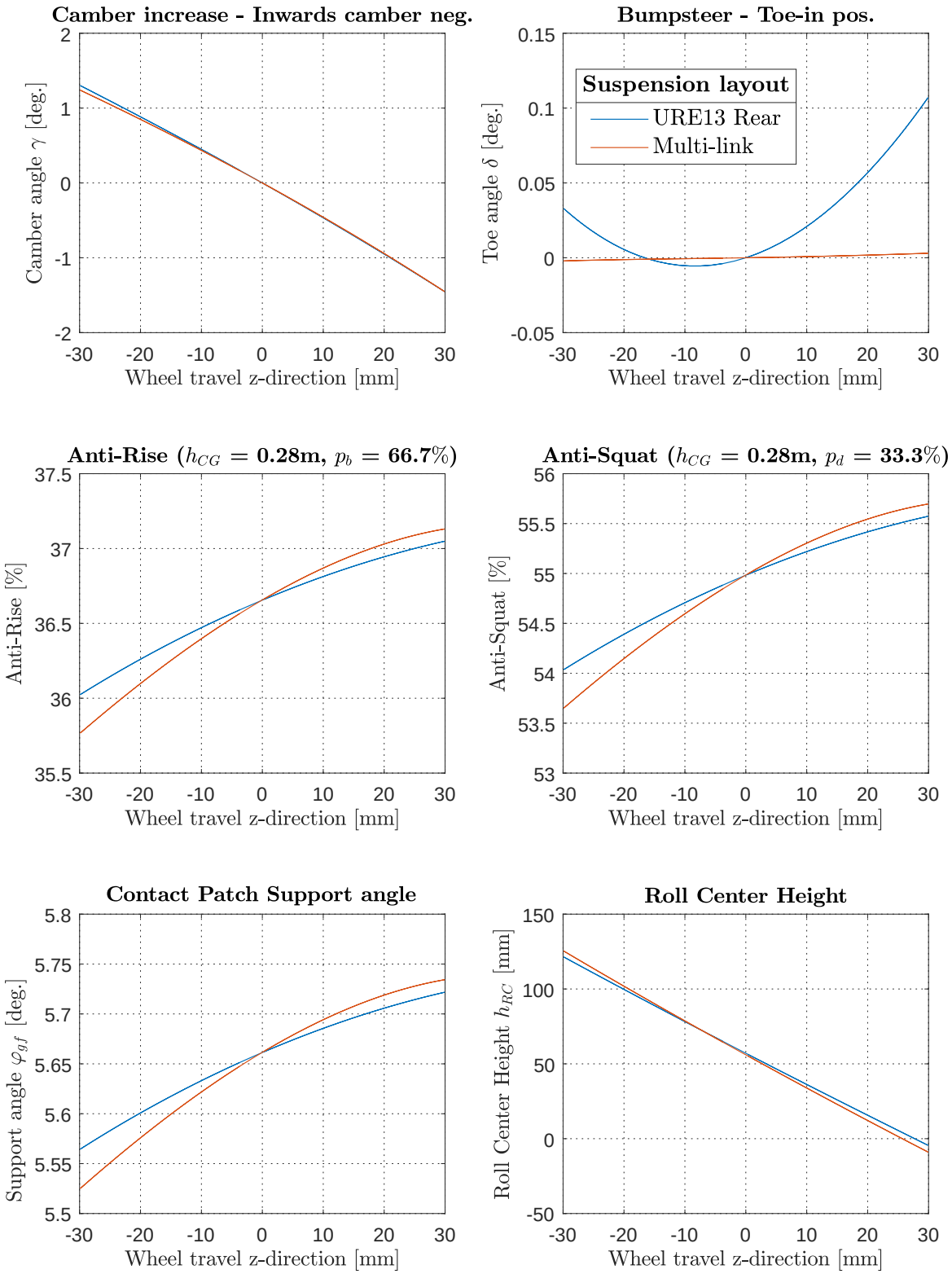


Figure G.2: Bump kinematics over wheel travel motion

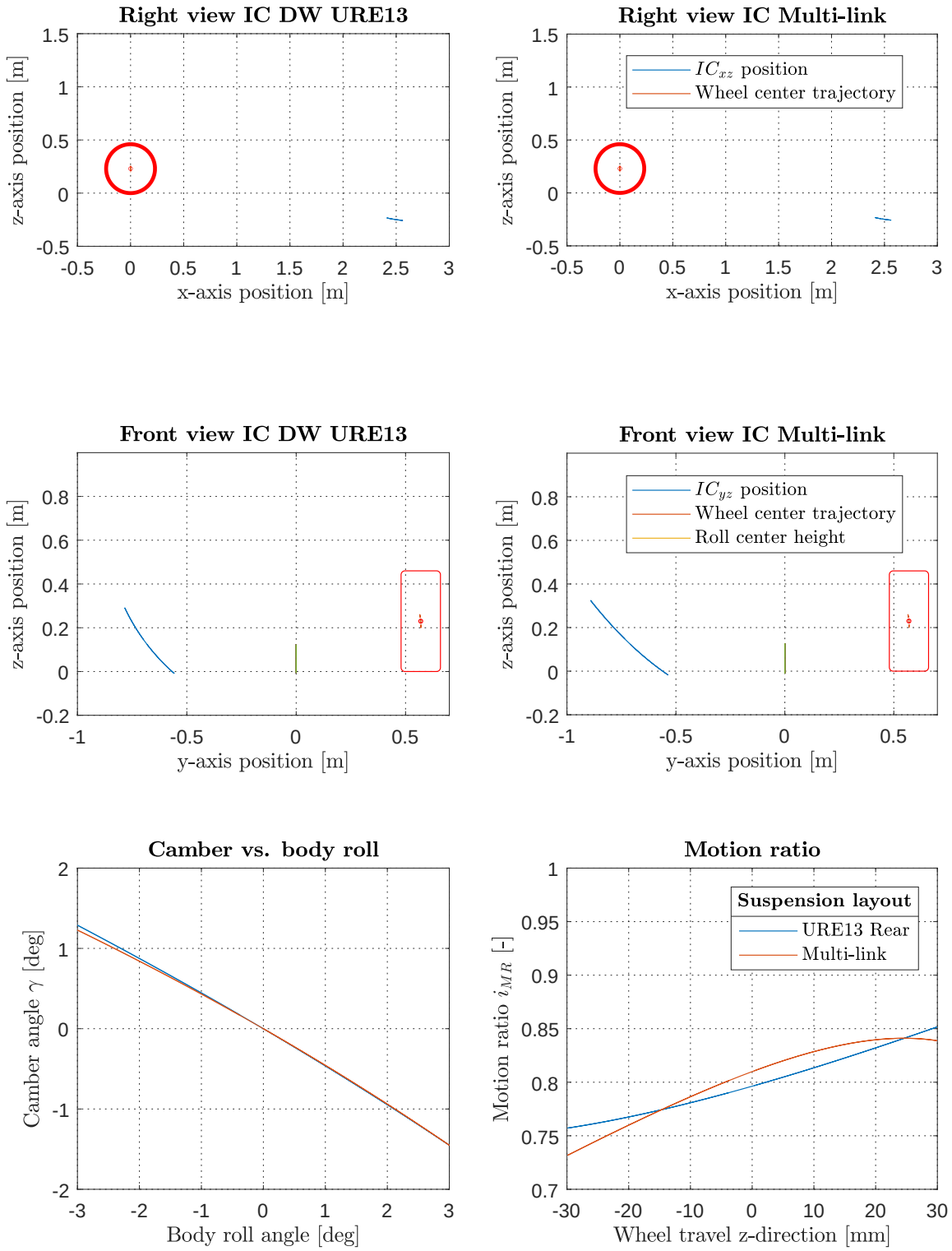


Figure G.3: Bump kinematics over wheel travel motion

H. Kinematics comparison front

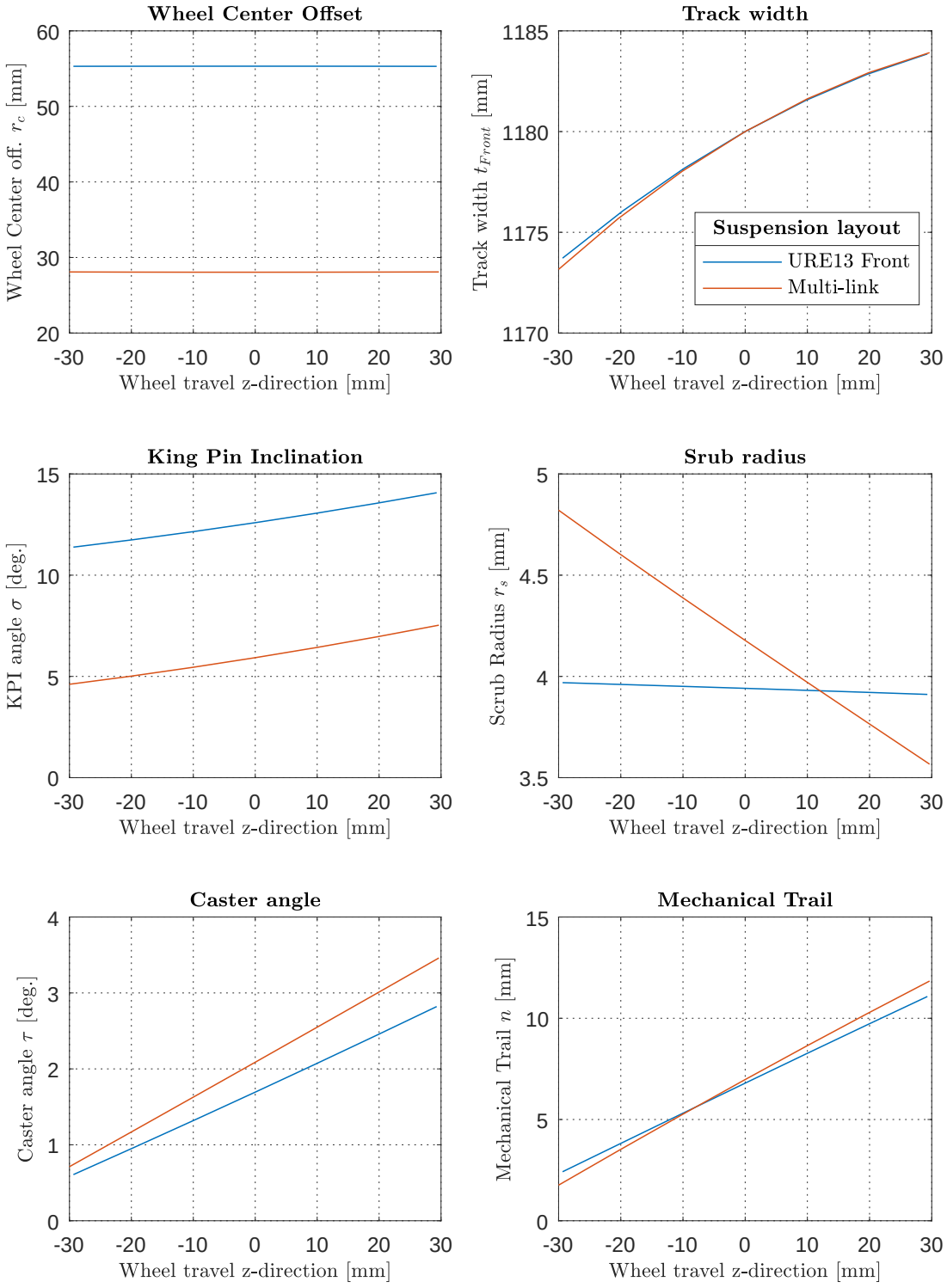


Figure H.1: Steering axis kinematics over wheel travel motion

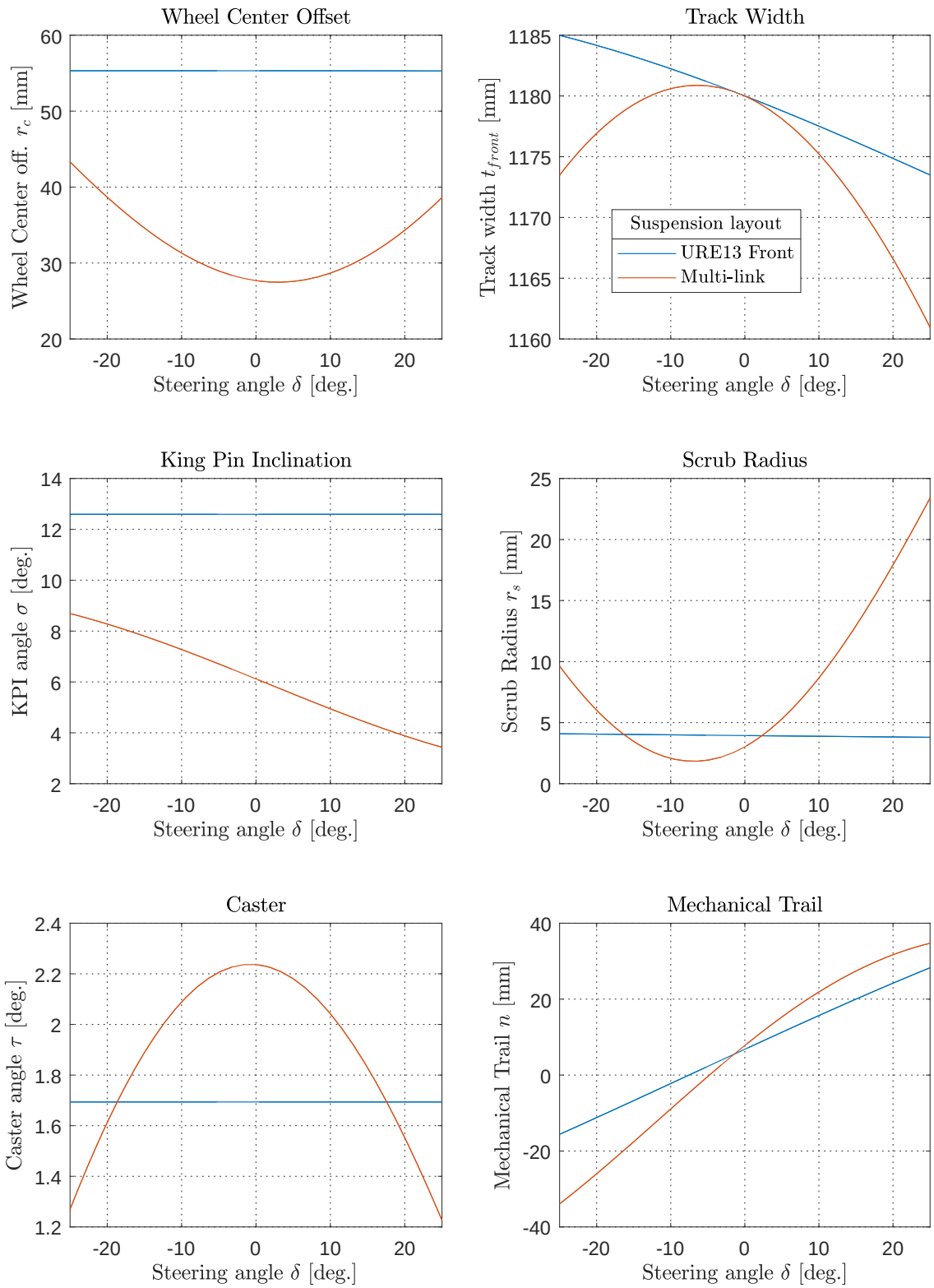


Figure H.2: Steering axis kinematics over steering motion

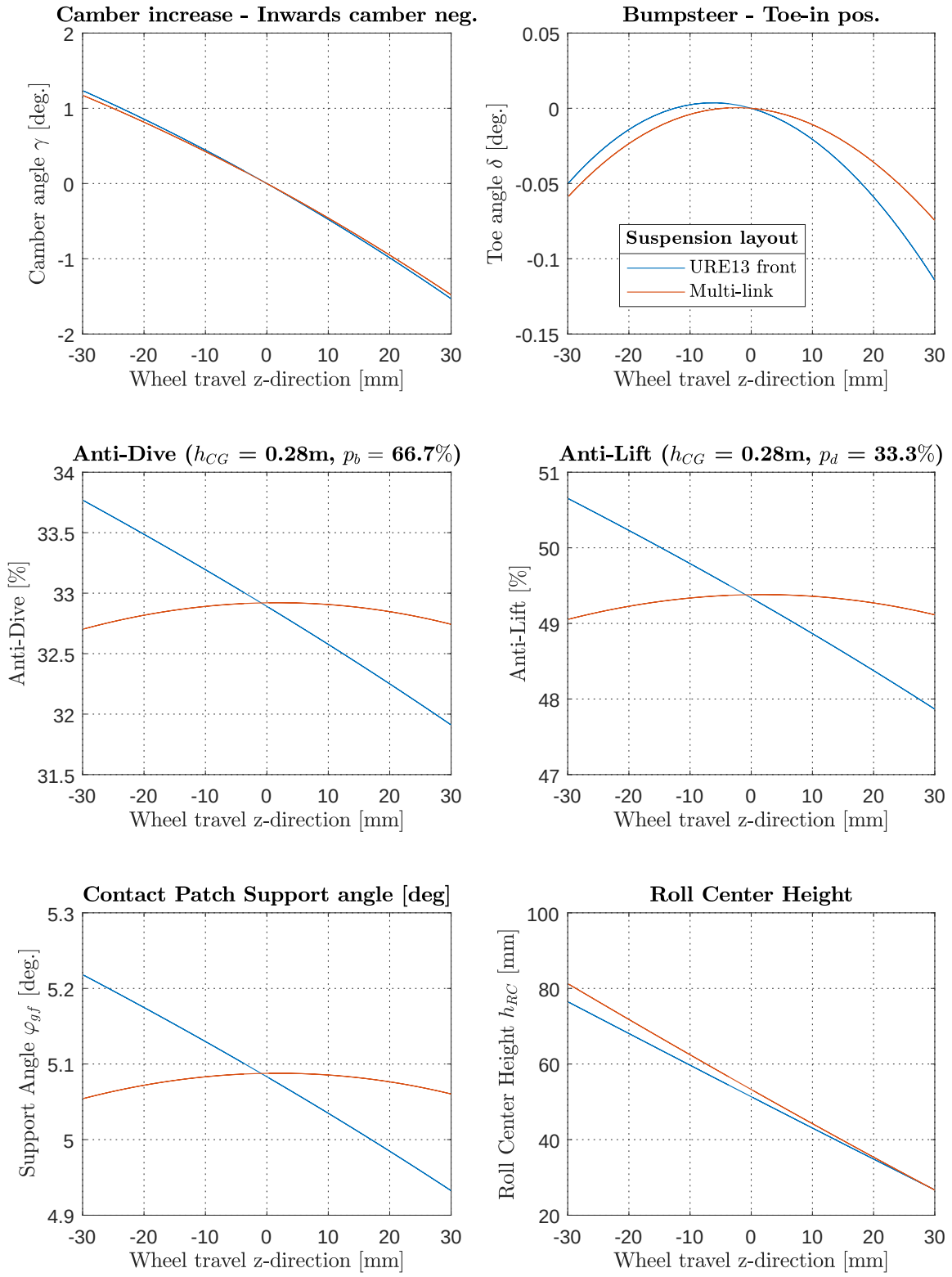


Figure H.3: Bump kinematics over wheel travel motion

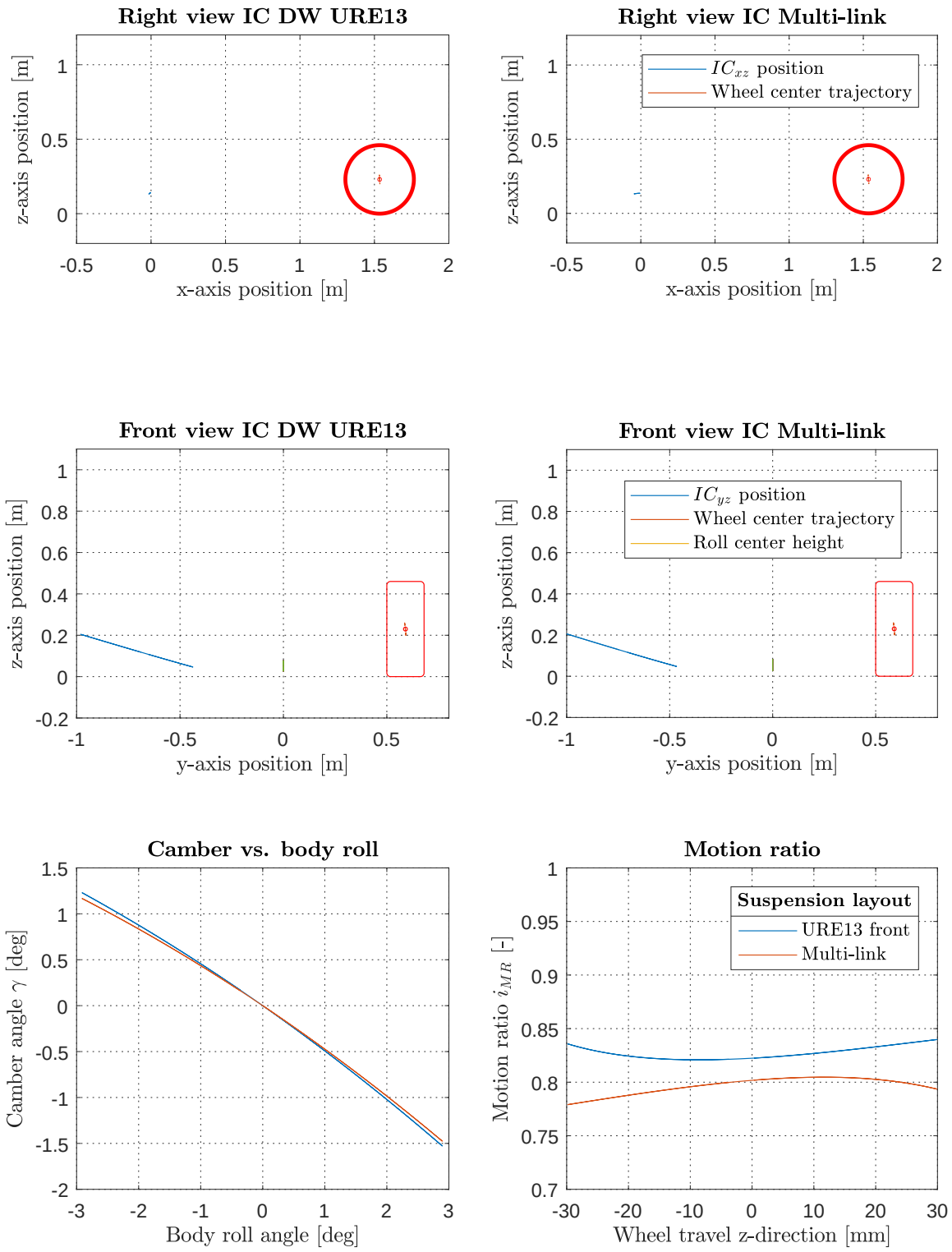


Figure H.4: Bump kinematics over wheel travel motion

I. Suspension compliance and forces

In Figure I.1 the results of the compliance simulation for the rear suspension are shown. Three load case are applied and the change in toe and camber angle are plotted as function of time.

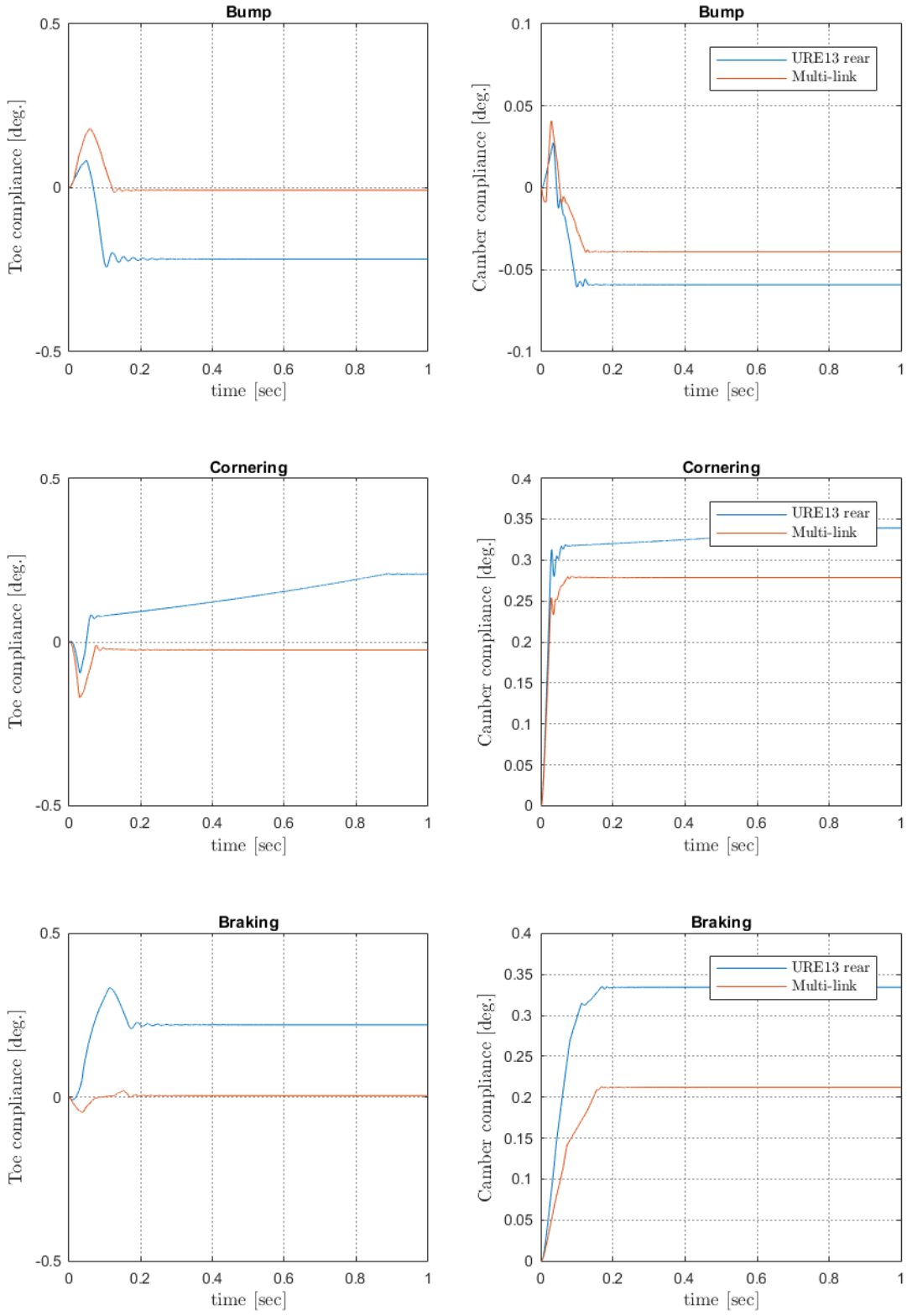


Figure I.1: Compliance and free-play rear suspension

In Figure I.2 the results of the compliance simulation for the rear suspension are shown. Three load case are applied and the change in toe and camber angle are plotted as function of time.

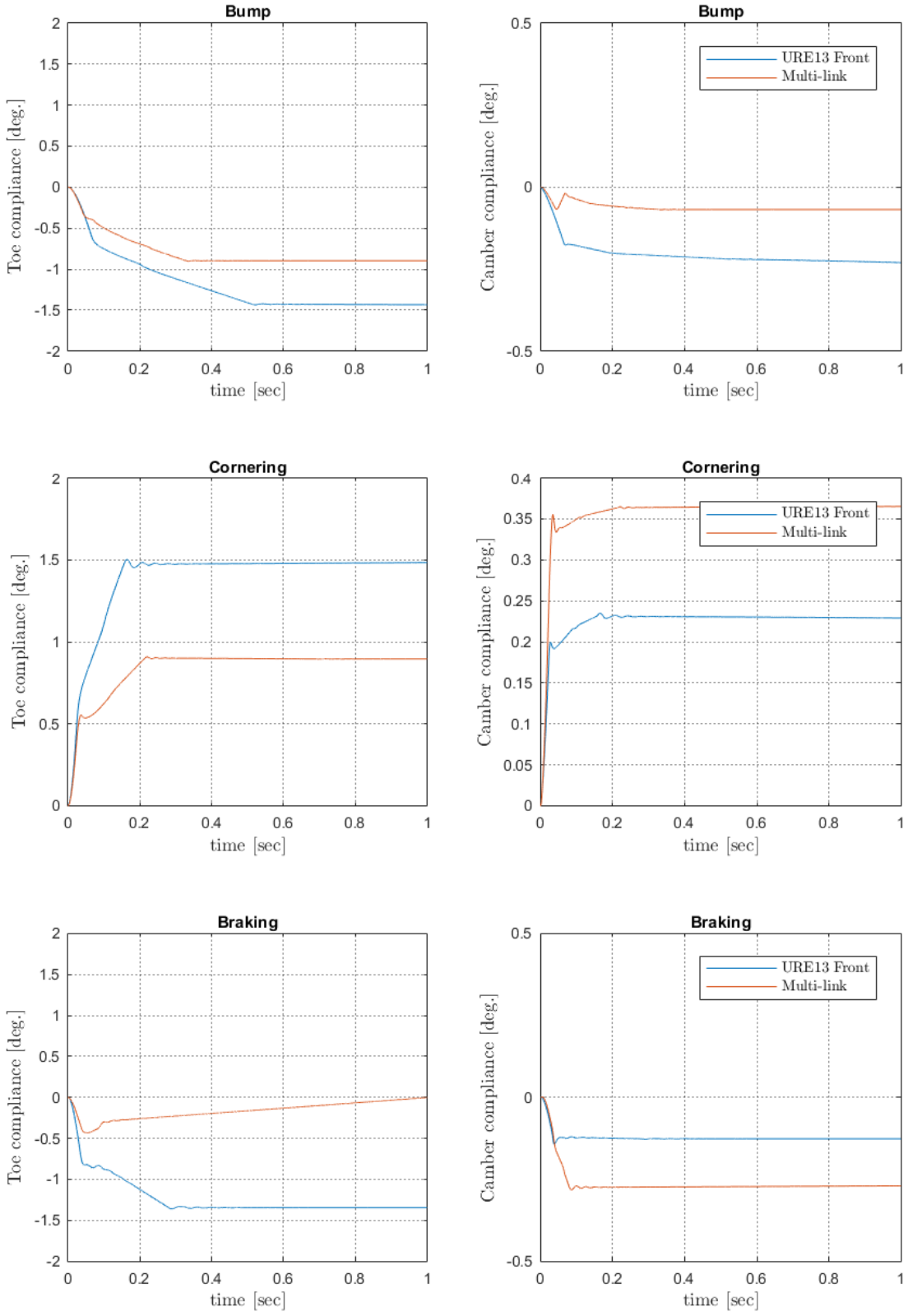


Figure I.2: Compliance and free-play front suspension

In Figure I.3 the resulting suspension load for three load cases are shown. These forces are measured for the rear suspension.

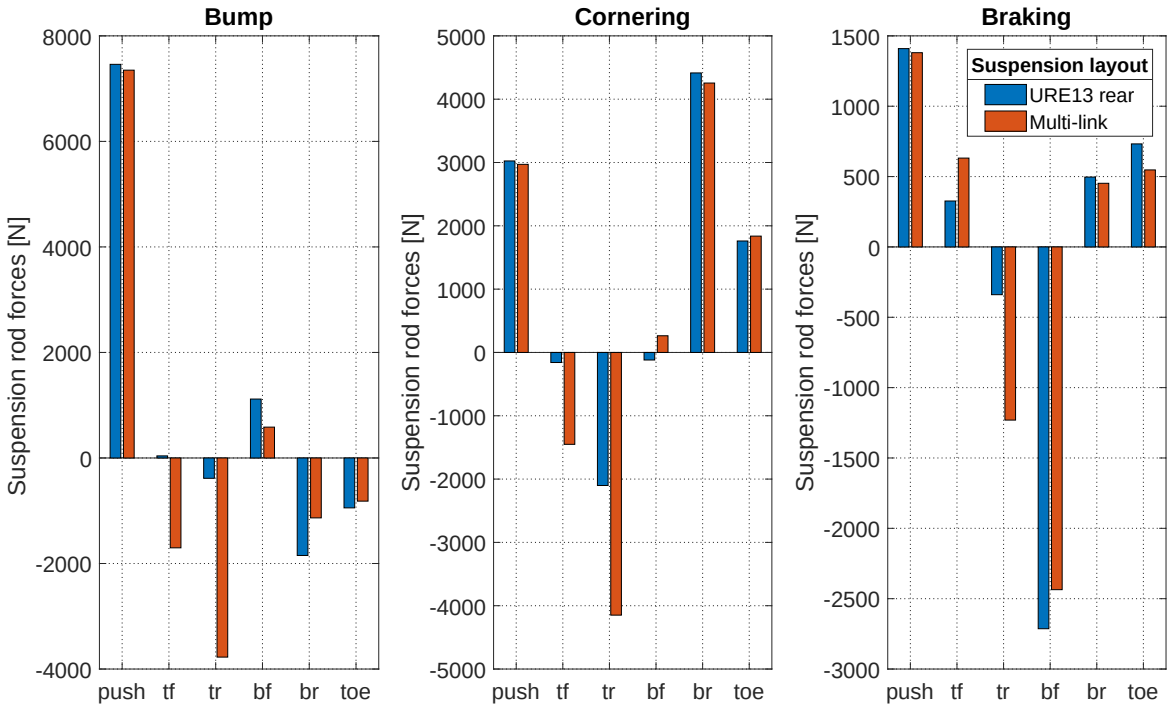


Figure I.3: Suspension loads rear

In Figure I.4 the resulting suspension load for three load cases are shown. These forces are measured for the front suspension.

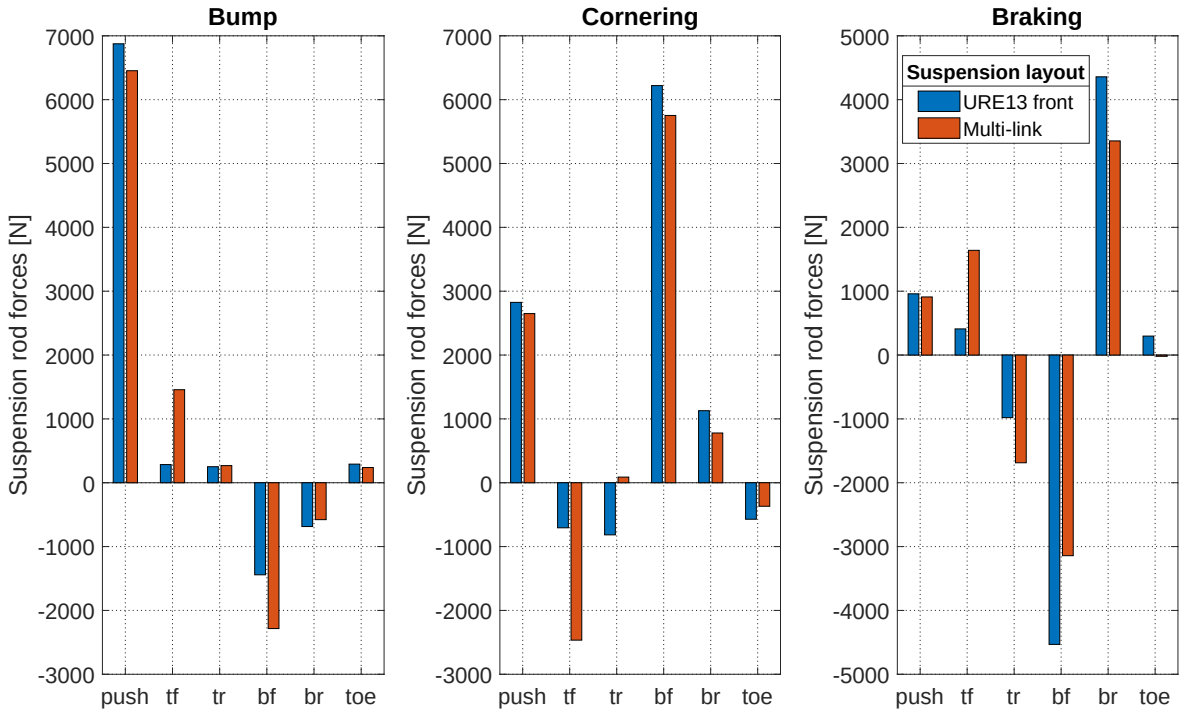


Figure I.4: Suspension loads front

J. Load transfer

Load transfer refers to changes in vertical tire forces during longitudinal and lateral acceleration. The forces which accelerate a vehicle are applied at the tire contact patch. These tire contact patch forces are not directed in a plane through the vehicle center of gravity, causing change of the vertical forces of the four tires. Lowering the center of gravity will reduce the load transfer in both longitudinal or lateral direction. Also changing the wheelbase or track-width will effect the load transfer. Increasing the vehicles wheelbase will reduce longitudinal load transfer whereas, increasing the track-width will reduce the lateral load transfer.

The method and roll axis vehicle model as described in the lectures notes of the course vehicle dynamics [11] are used for estimation of the load transfer. In Figure J.1 a schematic representation of this roll axis vehicle model is shown.

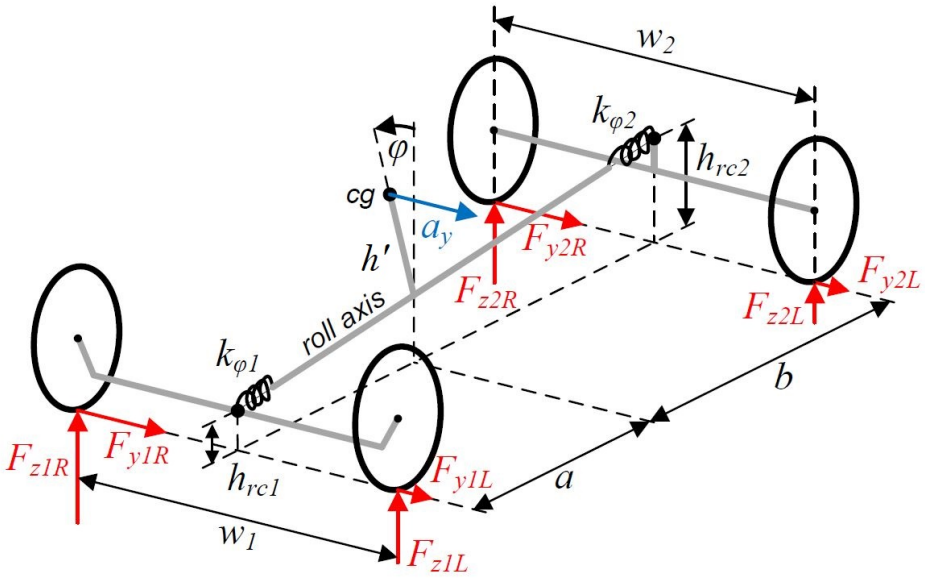


Figure J.1: Roll axis vehicle model.

The lateral load transfer is described by the change of the vertical forces of the tires due to the lateral acceleration, when cornering. It is the amount of load transfer between the left and right tires. The vertical force on the outer tires is increased, whereas the force of the inner tires is reduced. The total lateral load transfer of the vehicle can be calculated using the free body diagram of Figure J.1. The vehicle is illustrated from a angled point of view from the front. Values indicated by subscript 1 are related to the front axle and values indicated by subscript 2 are related to the rear axle.

The load distribution between the left and right tire is derived by taking the moment equilibrium about the roll axis. A extensive explination is described in the lectures notes of the course vehicle dynamics [11]. After substitution and rearranging, the vertical tire loads are formulated by the following equations:

$$F_{z1L} = \frac{bmg}{2(a+b)} - \left(\frac{k_{\varphi 1} h'}{k_{\varphi 1} + k_{\varphi 2} - mgh'} + \frac{bh_{rc1}}{(a+b)} \right) \frac{ma_y}{w_1} \tag{J.1}$$

$$F_{z1R} = \frac{bmg}{2(a+b)} + \left(\frac{k_{\varphi 1} h'}{k_{\varphi 1} + k_{\varphi 2} - mgh'} + \frac{bh_{rc1}}{(a+b)} \right) \frac{ma_y}{w_1} \tag{J.2}$$

$$F_{z2L} = \frac{amg}{2(a+b)} - \left(\frac{k_{\varphi 2} h'}{k_{\varphi 1} + k_{\varphi 2} - mgh'} + \frac{bh_{rc2}}{(a+b)} \right) \frac{ma_y}{w_2} \tag{J.3}$$

$$F_{z2R} = \frac{amg}{2(a+b)} + \left(\frac{k_{\varphi 2} h'}{k_{\varphi 1} + k_{\varphi 2} - mgh'} + \frac{bh_{rc2}}{(a+b)} \right) \frac{ma_y}{w_2} \tag{J.4}$$

where, a and b as the horizontal distance from the center of gravity position to front respectively rear axle. l is the wheelbase. w_1 and w_2 are the track-width for front and rear axle. m is the mass of the vehicle including a driver. The front and rear roll center height are noted by h_{rc1} and h_{rc2}. h' is the distance from the center of gravity to the roll axis. The roll stiffness for front and rear are described by $k_{\varphi 1}$ and $k_{\varphi 2}$.

The amount of roll stiffness for front or rear axle, are a combination of the shock absorber and the anti-roll bar stiffness. Combination of both will result in the roll stiffness for front $k_{\varphi 1}$ and rear $k_{\varphi 2}$. The roll stiffness at the front is calculated by:

$$k_{\varphi 1} = k_{w,s1} + k_{w,arb1} \tag{J.5}$$

where, $k_{w,s1}$ is the roll stiffness at the front induced by the shock absorbers and $k_{w,arb1}$ is the roll stiffness at the front induced by the anti-roll bar system. The same is done for the rear axle.

$$k_{\varphi 2} = k_{w,s2} + k_{w,arb2} \tag{J.6}$$

Finally, the distribution of lateral tire forces are calculated by the using the distribution in vertical loads between front and rear. The total lateral forces for front F_{y1} and rear F_{y2} axle are calculated by:

$$F_{y1} = \frac{b}{(a+b)} m a_y, \tag{J.7}$$

$$F_{y2} = \frac{a}{(a+b)} m a_y. \tag{J.8}$$

Using the vertical load distribution for left and right tires the lateral forces are divided over all four tires by using the following equations:

$$F_{y1L} = F_{y1} \frac{F_{z1L}}{(F_{z1L} + F_{z1R})} \tag{J.9}$$

$$F_{y1R} = F_{y1} \frac{F_{z1R}}{(F_{z1L} + F_{z1R})} \tag{J.10}$$

$$F_{y2L} = F_{y2} \frac{F_{z2L}}{(F_{z1L} + F_{z1R})} \tag{J.11}$$

$$F_{y2R} = F_{y2} \frac{F_{z2R}}{(F_{z1L} + F_{z1R})} \tag{J.12}$$

The parameters of the URE13 vehicle are used for the calculation of the load transfer in the simulation model. The used values are shown in Table J.1.

Table J.1: Roll axis model vehicle parameters URE13

Parameter:	Value:	Unit:
m	250	$[kg]$
a	0.854	$[m]$
b	0.681	$[m]$
l	1.535	$[m]$
w_1	1.18	$[m]$
w_2	1.14	$[m]$
h'	0.25	$[m]$
h_{rc1}	0.055	$[m]$
h_{rc2}	0.052	$[m]$
$k_{\varphi 1}$	43.26	$[kNm/rad]$
$k_{\varphi 2}$	68.56	$[kNm/rad]$
$k_{w,s1}$	30.76	$[kNm/rad]$
$k_{w,arb1}$	48.66	$[kNm/rad]$
$k_{w,s1}$	12.5	$[kNm/rad]$
$k_{w,arb1}$	19.9	$[kNm/rad]$

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